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Applied Economics and Data Science

# Master Thesis

**Title**

# Improving Reliability of Vision-Language Reasoning in Medical VQA through Uncertainty Calibration

submitted by

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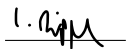
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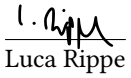
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Luca Rippe

[1] https://cursor.com
[2] https://claude.com

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# Abstract

Medical visual question answering (Medical VQA) is usually scored by answer accuracy, yet a fluent rationale and a correct letter do not show that a model is uncertain when it is wrong, or that it uses the image for visual evidence. This thesis evaluates explanation-based Medical VQA as a multiple-choice letter prediction task on VQA-RAD, Med-VQA-2019, and SLAKE. Answer-only generation is compared with zeroshot JSON chain-of-thought, in answer-first, rationale-first, and two-stage order, and with supervised fine-tuning on text-verified teacher rationales, across LLaVA-1.5, Qwen2.5-VL, and Qwen3-VL. Uncertainty is measured at the answer token by entropy, confidence, and sample agreement. Selective prediction is reported as risk-coverage curves and area under the risk-coverage curve. Visual use is probed by replacing the image with a black or shuffled counterpart.

Prompt-only explanation does not replace in-domain letter supervision. Fine-tuning on verified rationales matches letter-only accuracy to within a few items and does not produce the large chain-of-thought dividend often claimed in accuracy-only Medical VQA. Generation order matters more than the presence of an explanation. Rationale-first emission saturates confidence and collapses entropy while worsening ranking quality, so the scores then track consistency with the model's own trace rather than residual doubt about the letter. Answer-first emission keeps a usable letter distribution. After the image is removed, residual accuracy remains far above chance, especially on Med-VQA-2019, so language priors are reduced rather than eliminated. The same scores support selective answering only when that residual variation remains. These findings identify which generation contracts could later enter a workload or oversight analysis.



**Keywords:** Medical Visual Question Answering, Vision–Language Models, Uncertainty Calibration, Selective Prediction, Chain-of-Thought Explanation, Health Economics

# Contents


**1 Introduction** **1**
- 1.1 Motivation and Problem Statement . . . 1
  - 1.1.1 The Medical VQA Task . . . 1
  - 1.1.2 Reliability Challenges and Current Limitations . . . 2
  - 1.1.3 Explanation-Based Medical VQA . . . 3
  - 1.1.4 Health-Economic Relevance . . . 4
  - 1.1.5 Research Questions . . . 4
- 1.2 Research Contributions . . . 5
- 1.3 Thesis Outline . . . 5

**2 Related Work** **7**
- 2.1 Visual Question Answering Foundations . . . 7
- 2.2 Medical Visual Question Answering . . . 8
- 2.3 Uncertainty Measurement and Calibration . . . 9
- 2.4 Datasets and Evaluation . . . 10
- 2.5 Health-Economic Perspective on Medical AI . . . 11

**3 Technical Background** **13**
- 3.1 Vision-Language Models . . . 13
  - 3.1.1 Vision-Encoder and Language-Decoder . . . 13
  - 3.1.2 Evaluated model families . . . 14
- 3.2 Adaptation . . . 16
  - 3.2.1 Supervised fine-tuning . . . 16
- 3.3 Decoding . . . 17
- 3.4 Natural-Language and Structured Explanations . . . 17
- 3.5 Distillation and Verification . . . 18
- 3.6 Uncertainty and Calibration . . . 19
  - 3.6.1 Calibration and overconfidence . . . 19
  - 3.6.2 Answer-token scores . . . 19
- 3.7 Selective Prediction . . . 21
- 3.8 Visual Grounding . . . 23

**4 Methodology** **25**
- 4.1 Experimental Workflow Overview . . . 25
- 4.2 Student and Baseline Checkpoints . . . 26
- 4.3 Datasets . . . 26
  - 4.3.1 Source corpora . . . 26
  - 4.3.2 Dataset modification and preparation . . . 30
- 4.4 Shared Reporting Protocol . . . 32
  - 4.4.1 Evaluation metrics . . . 32
  - 4.4.2 Image ablations . . . 33
  - 4.4.3 Format error analysis . . . 33

4.5 Experimental Regimes . . . 33
4.5.1 Experiment 1: Answer-only VQA baseline . . . 34
4.5.2 Experiment 2: Answer+explanation . . . 34
4.5.3 Qualitative SHAP overlays . . . 37
4.5.4 Selective prediction . . . 38

**5 Results** **39**
5.1 VQA-RAD . . . 40
5.1.1 Risk-coverage analysis . . . 43
5.2 Med-VQA-2019 . . . 45
5.2.1 Risk-coverage analysis . . . 48
5.3 SLAKE . . . 50
5.3.1 Risk-coverage analysis . . . 53
5.4 Mixed-order RF+AF . . . 55
5.5 Medically pretrained models . . . 55
5.6 Certainty drop under shuffled images . . . 57
5.7 SHAP analysis . . . 59

**6 Discussion** **62**
6.1 Technical Discussion . . . 62
6.1.1 Experiment 1: Answer-only VQA baseline . . . 62
6.1.2 Experiment 2: Zeroshot explanation . . . 63
6.1.3 Experiment 2: Explanation-based SFT . . . 64
6.1.4 Selective prediction . . . 66
6.1.5 Medically pretrained models . . . 67
6.1.6 Visual grounding . . . 68
6.1.7 Cross-protocol synthesis . . . 69
6.2 Health-Economic Implications . . . 70
6.2.1 Clinical workload . . . 70
6.2.2 Trust and oversight costs . . . 71
6.2.3 Conditions for economically sustainable deployment . . . 72

**7 Conclusion** **74**
7.1 Summary of Contributions . . . 74
7.2 Limitations . . . 75
7.3 Future Work . . . 76

**A Supplementary Material** **78**
A.1 Qwen3-VL architecture diagram . . . 78
A.2 SLAKE question content types . . . 79
A.3 Qualitative VQA-RAD example . . . 79
A.4 Format adherence: Answer-only . . . 80
A.5 Format adherence: Zeroshot JSON . . . 81
A.6 Additional zeroshot test: 2S-L . . . 81
A.7 Format adherence: Explanation-based SFT . . . 82
A.8 Original-image confidence and agreement . . . 83
A.9 Explanation SFT: original-image letter and explanation scores . . . 84
A.10 Closed and open ablation splits . . . 86

# List of Acronyms

**ACE** Adaptive Calibration Error.

**AI** Artificial Intelligence.

**AURC** Area Under the Risk-Coverage Curve.

**AUROC** Area Under the Receiver Operating Characteristic Curve.

**BAN** Bilinear Attention Network.

**BLEU** Bilingual Evaluation Understudy.

**CAD** Computer-Aided Detection.

**CDSS** Clinical Decision Support System.

**CHEERS** Consolidated Health Economic Evaluation Reporting Standards.

**CI** Confidence Interval.

**CLIP** Contrastive Language–Image Pre-training.

**CNN** Convolutional Neural Network.

**CoT** Chain-of-Thought.

**CT** Computed Tomography.

**CV** Computer Vision.

**DALY** Disability-Adjusted Life Year.

**ECE** Expected Calibration Error.

**EHR** Electronic Health Record.

**EU** European Union.

**F1** F1 Score.

**GradCAM** Gradient-weighted Class Activation Mapping.

**GRPO** Group Relative Policy Optimization.

**HTA** Health Technology Assessment.

**ICER** Incremental Cost-Effectiveness Ratio.

**IQWiG** Institute for Quality and Efficiency in Health Care.

**LDCT** Low-Dose Computed Tomography.

**LLM** Large Language Model.

**LSTM** Long Short-Term Memory Network.

**LVLM** Large Vision-Language Model.

**MAC** Memory, Attention, and Composition.

**MASAI** Mammography Screening with Artificial Intelligence.

**MC** Multiple Choice.

**MCB** Multimodal Compact Bilinear Pooling.

**MCE** Maximum Calibration Error.

**Medical VQA** Medical Visual Question Answering.

**MEVF** Mixture of Enhanced Visual Features.

**MLLM** Multimodal Large Language Model.

**MLP** Multilayer Perceptron.

**MoE** Mixture of Experts.

**MRI** Magnetic Resonance Imaging.

**NCE** Net Calibration Error.

**NICE** National Institute for Health and Care Excellence.

**NLL** Negative Log-Likelihood.

**NLP** Natural Language Processing.

**NMN** Neural Module Network.

**OR** Odds Ratio.

**PRISMA** Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

**QALY** Quality-Adjusted Life Year.

**R-CNN** Region-based Convolutional Neural Network.

**RL** Reinforcement Learning.

**RLHF** Reinforcement Learning from Human Feedback.

**RNN** Recurrent Neural Network.

**ROI** Region of Interest.

**SFT** Supervised Fine-Tuning.

**SHAP** SHapley Additive exPlanations.

**UQ** Uncertainty Quantification.

**ViT** Vision Transformer.

**VLM** Vision-Language Model.

**VLP** Vision-Language Pretraining.

**VQA** Visual Question Answering.

**VRAM** Video Random Access Memory.

# List of Figures


1.1 Examples from the VQA-RAD Dataset [1]. . . . . 1

2.1 Schematic of a risk-coverage curve (RCC). Selective risk $R$ is plotted against coverage, the area under the curve (AURC) summarises ranking quality between an oracle and an uninformative baseline. . . . . 10

3.1 LLaVA-Med curriculum adaptation from LLaVA: Stage 1 medical concept alignment (1 epoch on 600K samples) and Stage 2 medical instruction tuning (3 epochs on 60K samples), followed by downstream medical VQA and visual chat [2]. . . . . 15

3.2 Qwen2.5-VL architecture with native-resolution vision encoding, variable visual token counts, and a Qwen2.5 language-model decoder [3]. . . . . 15

4.1 Overview of the experimental pipeline: source datasets are converted to multiple-choice letter prediction and evaluated under two setups (answer-only and answer+explanation). Selective prediction is co-reported for both. . . . . 26

5.1 Answer-only risk–coverage curves on VQA-RAD (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL SFT (0.116). . . . . 43

5.2 Zeroshot JSON risk–coverage curves on VQA-RAD (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL RF (0.116). . . . . 44

5.3 Explanation SFT risk–coverage curves on VQA-RAD for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL AF (0.124). . . . . 44

5.4 Answer-only risk–coverage curves on Med-VQA-2019 (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL SFT (0.080). The closed split comprises 64 items. . . . . 48

5.5 Zeroshot JSON risk–coverage curves on Med-VQA-2019 (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.119). . . . . 49
5.6 Explanation SFT risk–coverage curves on Med-VQA-2019 for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.055). . . . . 49
5.7 Answer-only risk–coverage curves on SLAKE (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL SFT (0.032). . . . . 53
5.8 Zeroshot JSON risk–coverage curves on SLAKE (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.080). . . . . 54
5.9 Explanation SFT risk–coverage curves on SLAKE for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL RF+AF merged (0.036). . . . . 54
5.10 Med-VQA-2019 shuffle pairs for Qwen2.5-VL mixed-order explanation SFT, evaluated answer-first. Question, options, and gold letter stay on the original sample. Only the image is replaced by another test image. Both items are letter-correct on the original image and letter-wrong after the shuffle (incorrect letters in red). Scores under each image are letter entropy and letter confidence. Traces are the greedy answer-first `REGION/OBSERVATION/EXPLANATION` blocks. These panels are qualitative illustrations on individual items and are not used to draw dataset-level conclusions. . . . . 58
5.11 SLAKE SHAP overlays for the MR-weighting item under the original image, a shuffled image, and a black image. Protocols are answer-only LLaVA-1.5 SFT, zeroshot Qwen2.5-VL AF, and explanation SFT (Qwen2.5 AF+RF). Incorrect letters are shown in red. Warm patches support the explained scalar and cool patches oppose it: predicted answer-token probability for answer-only, sequence log-probability for zeroshot and explanation SFT. Colour scales are panel-specific and not comparable across cells. Black-image SHAP maps are spatially flat. These maps are qualitative illustrations on individual items and are not used to draw dataset-level or clinical-faithfulness conclusions. 60

5.12 SLAKE SHAP overlays for the closed T2 item under the original image, a shuffled image, and a black image. Protocols are answer-only LLaVA-1.5 SFT, zeroshot Qwen2.5-VL AF, and explanation SFT (Qwen3 AF). Incorrect letters are shown in red. Warm patches support the explained scalar and cool patches oppose it: predicted answer-token probability for answer-only, sequence log-probability for zeroshot and explanation SFT. Colour scales are panel-specific and not comparable across cells. Black-image SHAP maps are spatially flat. These maps are qualitative illustrations on individual items and are not used to draw dataset-level or clinical-faithfulness conclusions. 61

A.1 Qwen3-VL architecture with native-resolution inputs, interleaved vision-text tokens, and DeepStack fusion into the Qwen3 decoder [4]. Deferred from Chapter 3 because the layout largely mirrors Figure 3.2. 78
A.2 Qualitative example from VQA-RAD: one open-ended test item. . . . 79

# List of Tables


3.1 Comparative overview of the evaluated model families. Vision towers are abbreviated (CLIP-L/14 is CLIP ViT-L/14, DStack is DeepStack). Among the five checkpoints, only MedVLM-R1 continues a Qwen2-VL-2B stack under Group Relative Policy Optimization (GRPO) instead of letter-level Supervised Fine-Tuning (SFT), and only it must be scored through the paper's <think> then <answer> wrapper [5]. . 16

4.1 Student and baseline checkpoints used in this thesis. . . . . 26
4.2 Overview of the three radiology Medical Visual Question Answering (Medical VQA) corpora used in this thesis. Paper versus public counts are shown where they differ. SLAKE is released with organ and disease segmentation masks and bounding boxes. Those annotations are never loaded or used in this thesis. . . . . 27
4.3 Official splits. VQA-RAD has no validation partition. SLAKE counts follow Liu et al. [6], a public snapshot redistributes 9,840 / 2,100 / 2,090 pairs with the same total of 14,028. Med-VQA-2019 training pairs are 12,792 in the challenge report [7], some redistributions list 12,800 training rows. VQA-RAD train and test image counts are not disjoint (see text). . . . . 27
4.4 VQA-RAD image coverage and elicitation variants as reported by Lau et al. [1]. The public release used in this thesis has 314 images and 2,244 unique pairs (see text). . . . . 28
4.5 SLAKE image coverage by body region and modality [6]. . . . . 29
4.6 Closed versus open question types in the splits used for training and evaluation after multiple-choice conversion. Closed items have a yes/no gold answer, open items have any other gold answer. SLAKE is the English subset. Percentages are within each split. . . . . 31
4.7 Text-only verification of pseudo-explanations by teacher and dataset (verifier: Qwen2.5-VL-7B, greedy). *Generated* counts all sampled rationales before filtering. *Retained* counts letter-matching traces used for SFT. . . . . 32
4.8 Default hyperparameters for the reported experimental pipeline (shared across both setups unless stated otherwise). . . . . 34

5.1 VQA-RAD stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema. . . . . 40
5.2 VQA-RAD stacked overall ablations for all generative protocols. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as −. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.11 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 42
5.3 Med-VQA-2019 stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema. . . . . 45
5.4 Med-VQA-2019 stacked overall ablations for all generative protocols. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as −. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.12 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 48

5.5 SLAKE stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema. . . . . 50
5.6 SLAKE stacked overall ablations for all generative protocols. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as −. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.13 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 52
5.7 Mixed-order RF+AF explanation SFT: Qwen2.5-VL-7B trained on the union of verified Qwen3-VL-8B RF and AF traces. This additional test is omitted from Tables 5.1–5.5 so that those tables compare matched single-order RF/AF. Datasets are shown left to right. Acc.: accuracy. Ent.: entropy. . . . . 55
5.8 Medically pretrained models on the original-image test splits. LLaVA-Med v1.5 settings are answer-only Base/SFT and zeroshot JSON AF/RF/2S. MedVLM-R1 is the off-the-shelf medical explanation baseline (native tag schema. Uncertainty and format columns undefined). Acc.: accuracy. Format: parse rate (%). Best Acc. and lowest defined AURC in each split are shown in bold. . . . . 57

A.1 SLAKE question content types by official split [6]. Counts are bilingual (English and Chinese). . . . . 79
A.3 Qualitative examples from VQA-RAD. . . . . 80
A.4 Answer-only format adherence: percentage of samples for which the predicted answer letter could be parsed correctly. Adaptations match the answer-only block of Table 5.1. . . . . 80
A.5 Zeroshot JSON format adherence: percentage of samples for which the structured output could be parsed correctly. Modes match the zeroshot block of Table 5.1 (AF: answer-first. RF: rationale-first. 2S: two-stage). . . . . 81
A.6 Additional zeroshot test: two-stage letter-only (2S-L). Stage 1 generates a rationale without a letter. Stage 2 answers with a letter-only multiple-choice response. Acc.: accuracy. Format: parse rate of the structured zeroshot output (%). Best Acc. and lowest AURC in each split are shown in bold. . . . . 81

A.7 Explanation-based SFT format adherence: percentage of samples for which the structured JSON output could be parsed correctly. Teachers and orders match the explanation SFT block of Table 5.1 (RF: rationale-first. AF: answer-first). . . . . 82
A.8 Answer-only SFT on original images: answer-token confidence (Conf.) and sample agreement (Agr.) per split. Accuracy, entropy, and AURC for the same rows are in Tables 5.1–5.5. . . . . 83
A.9 Zeroshot JSON on original images: answer-token confidence (Conf.) and sample agreement (Agr.) per split (AF: answer-first. RF: rationale-first. 2S: two-stage). Accuracy, entropy, and AURC for the same rows are in Tables 5.1–5.5. . . . . 84
A.10 Original-image explanation SFT for Qwen3-teacher students. Companion to the explanation-SFT blocks of Tables 5.2–5.6. RF+AF is the mixed-order additional test of section 5.4. Acc.: accuracy. Ent.L / Conf.L: answer-letter entropy and confidence. Ent.X / Conf.X: explanation-span entropy and confidence. Bold Acc. and AURC are within-dataset extrema. . . . . 86
A.11 VQA-RAD stacked closed and open ablations, companion to the overall stack of Table 5.2. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 87
A.12 Med-VQA-2019 stacked closed and open ablations, companion to the overall stack of Table 5.4. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 88
A.13 SLAKE stacked closed and open ablations, companion to the overall stack of Table 5.6. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema. . . . . 89

# 1 Introduction

This chapter introduces Medical VQA as a research direction and motivates the need to evaluate such systems not only by answer accuracy but also by reliability. It outlines the clinical motivation for the uncertainty calibration of Medical VQA models and identifies reliability challenges that arise in explanation-based generation. In addition, the health-economic relevance of trustworthy uncertainty estimates for clinical decision support and resource allocation is discussed. Further on, research gaps are addressed and the main contributions of this thesis are summarized. Finally, an overview of the overall thesis structure is provided.

## 1.1 Motivation and Problem Statement

### 1.1.1 The Medical VQA Task

Artificial Intelligence (AI) systems are increasingly being integrated into clinical and healthcare-related workflows [8]. Within this context, Medical VQA has emerged as a promising research direction. By combining methods from Natural Language Processing (NLP) and Computer Vision (CV), Medical VQA aims to develop multimodality models that are capable of generating natural language answers given a medical image such as a radiograph, Computed Tomography (CT) scan, or Magnetic Resonance Imaging (MRI) data and an according question [9]. See the examples in Figure 1.1.

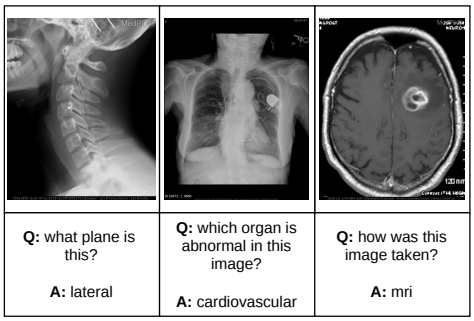



Figure 1.1: Examples from the VQA-RAD Dataset [1].

Tasks can include for example identifying pathological findings, describing anatomical structures, or answering diagnostic questions based on imaging evidence. Potential use cases of Medical VQA include assisting diagnostic workflows and clinical decision-making [10], supporting radiologists and clinicians under high workload [9], and enabling patients to better understand and interact with their own medical data [11].

### 1.1.2 Reliability Challenges and Current Limitations

Particularly in high-risk environments such as the medical domain, the reliability of model answers is crucial. Machine learning models inevitably make errors, and even state-of-the-art VQA systems still exhibit non-negligible rates of incorrect predictions [12], which can limit their practical usability. With the growing interest in deploying VQA systems in real-world settings, the ability of such models to provide reliable and trustworthy assistance becomes increasingly important [13]. For this reason, it is useful to optimize for uncertainty calibration alongside accuracy itself. Several limitations in Medical VQA still undermine these goals.

Models frequently generate fluent answers that contradict or ignore source evidence, often because localization is inadequate [14, 15]. Rather than inspecting pathological regions, they exploit textual and visual shortcuts: disease terms co-occur with anatomical mentions, so a system may answer *yes* to a localization question without using visual information from the given image [14]. Even when responses contain medical visual claims, a substantial fraction remains ungrounded. Accuracy can therefore improve without actual visual dependence, and some models retain high performance in ablation studies even with blank or mismatched images [16]. The same apparent competence is brittle: answers fluctuate strongly under semantically equivalent rephrasings, with reported recall drops of around 40-50% relative to original benchmarks [17]. Under simple probing or adversarial settings, even strong multimodal models can fall below random guessing on diagnostic questions, with average decreases of nearly 28% across tested systems [18], and sanity baselines that ignore the image can still perform surprisingly well [19].

These failures are amplified by the benchmarks themselves. Widely used datasets such as VQA-RAD [1], SLAKE [6], and PathVQA [20] exhibit strong language priors and unbalanced answer distributions, especially for yes/no questions. Text-only settings still yield non-trivial accuracy on VQA-RAD and SLAKE, indicating exploitable linguistic shortcuts [21], while bias-sensitive splits show pronounced degradation under out-of-distribution conditions [22]. A classic pattern is to favour the majority answer *no* on yes/no questions even when the image suggests otherwise [23]. Scale and construction compound the effect: VQA-RAD contains only 315 images and 3,515 QA pairs [1], SLAKE 642 images [6], and PathVQA is partly generated via a semi-automated caption-to-QA pipeline [20], which can embed additional textual regularities such as annotation artefacts [24]. Exact-match accuracy remains the dominant score, yet it is overly strict for open-ended answers and does not account for semantically equivalent paraphrases [25]. More critically, accuracy alone does not measure visual dependence, so models may score well while relying on language priors rather than the image [16, 19].

Reliability is further limited by overconfidence, weak explanations, and poor transfer. Medical vision-language models are frequently poorly calibrated, which increases the risk of misdiagnoses when confidence is treated as a trust signal [26, 27]. Overconfidence persists across model families and is not resolved by scaling or prompting

strategies such as Chain-of-Thought (CoT) or verbalized confidence. Simple post-hoc calibration (e.g. Platt scaling) often reduces calibration error more effectively [28]. Attention maps are commonly offered as visual explanations, but their usefulness is context-dependent and they do not consistently provide the faithful insights required for medical decision-making [29]. Although they can outperform Grad-CAM in some settings, they are often surpassed by transformer-specific interpretability methods and should not be treated as reliable evidence of what the model actually used [29]. Performance is also highly modality-sensitive and degrades under distribution shift across datasets, acquisition protocols, and question subjects [19, 30], which limits transfer from curated benchmarks to heterogeneous clinical environments.

### 1.1.3 Explanation-Based Medical VQA

Recent Medical VQA research has moved from answer-only generation toward explanation-based methods inspired by CoT [31], on the assumption that intermediate steps improve accuracy, interpretability, and trust. MedVLM-R1 raises diagnostic accuracy with limited data via reinforcement learning [5], MedLVR preserves diagnostically relevant visual evidence during decoding [30], and localize-before-answer as well as structured visual CoT reduce shortcut learning [14, 32]. These gains do not automatically yield visual dependence or calibrated confidence. Accuracy-oriented rewards can leave models accurate under blank or mismatched images while still producing ungrounded visual claims [16]. Because many VLMs encode the image once as static context, subsequent explanation can drift toward language priors [30], and fluent traces may be post-hoc justifications rather than causal evidence of the decision [5].

Liang et al. isolate this gap with generation order: if explanations were causal, rationale-first and answer-first decoding should agree [33]. Order, scale, and explanation-supervised fine-tuning jointly shape accuracy and faithfulness, and larger models are a prerequisite for reliable rationale-first explanation. This thesis transfers that diagnostic to Medical VQA and additionally asks whether order-conditioned explanation improves calibrated uncertainty, not only answer accuracy. The same questions underpin the Amenable agenda on interactive learning and trustworthy explanation generation, which combines expert-in-the-loop representation learning with consistency-aware entailment [34]. The present protocols are preliminary experimental groundwork for that agenda.

Explanation can move calibration in opposite directions, depending on the explanation style and the confidence estimator. CoT often raises task accuracy while worsening token-likelihood uncertainty, because decoding becomes conditioned on the model's own trace rather than on residual doubt about correctness [35]. Slower, search-like generation can instead improve expressed confidence [36]. In Medical VQA, naive single-shot CoT and verbalized confidence do not resolve overconfidence [28], whereas grounding- and verification-based procedures can reduce calibration error [37, 38]. Under explanation, agreement and multi-path verification are more reliable signals than raw token probabilities [35, 39], self-verification alone is not [40]. Dedicated explanation systems nevertheless report accuracy and explainability far more often than calibration jointly, so whether explanation makes Medical VQA better calibrated, not only more capable, remains insufficiently established.

### 1.1.4 Health-Economic Relevance

Beyond technical performance, the reliability of Medical VQA systems has direct implications for the allocation of scarce clinical resources. Imaging AI is increasingly discussed as a response to rising workloads and workforce shortages in radiology. Medical VQA may reduce interpretation and communication burden, support decision-making under time pressure, and could improve access in resource-constrained settings [9]. Randomized evidence from AI-supported mammography screening illustrates that such efficiency gains are attainable in related imaging tasks, including substantial reductions in reading workload without compromising detection safety [41]. At the same time, recent systematic reviews show that rigorous health-economic evaluations of radiology AI remain rare and highly context-dependent: economic value emerges mainly in high-volume, resource-intensive settings when accuracy meets clinical need, rather than as a universal dividend [42]. These considerations make calibration and uncertainty estimation economically relevant. Diagnostic error already imposes large clinical and societal costs through preventable harm, delayed treatment, and unnecessary follow-up [43]. Overconfident yet incorrect model outputs can amplify automation bias and liability risk, whereas poorly trusted systems may erase intended productivity gains by forcing extensive manual verification [26, 28]. Well-calibrated confidence estimates, by contrast, enable selective prediction and human oversight to be targeted where risk is highest, which is a prerequisite for translating technical improvements into safer and more efficient clinical workflows. Importantly, this bridge between Medical VQA and health economics is still largely missing. After systematic literature search (as of July 2026), no peer-reviewed study or recognized preprint was identified that combines a Medical VQA system with a formal health-economic evaluation, such as a cost-effectiveness, cost-utility, or budget-impact analysis. Even recent work on clinical integration barriers [44] and on confidence calibration in Medical VQA [38] does not quantify downstream economic consequences. This thesis therefore treats calibration not only as a reliability criterion, but also as a condition for assessing if Medical VQA can be deployed in an economically sustainable way.

### 1.1.5 Research Questions

Based on the outlined context and the stated challenges this thesis addresses the following research questions:

**RQ1: Reliability of Explanation-Based Medical VQA**
*Can explanation-based vision-language reasoning improve the reliability of VQA in the medical domain?*
This question is examined through three related aspects:

- **Uncertainty measurement.** Comparable quantification of predictive uncertainty and overconfidence across answer-only and explanation-based settings, using scores such as answer-token entropy, confidence, and sample agreement.
- **Calibration.** Whether those scores track empirical correctness rather than self-consistency with a generated explanation, and how generation order and explanation-based training affect that relationship.
- **Selective prediction.** Whether the resulting uncertainty estimates improve selective answering by concentrating errors among abstained items.

**RQ2: Health-Economic Implications**
*What health-economic implications follow from the empirical findings on accuracy, calibration, and selective prediction, particularly regarding clinical workload, trust and oversight costs, and the conditions for economically sustainable deployment of explanation-based Medical VQA?*

## 1.2 Research Contributions

This thesis makes the following contributions toward a reliability-centred evaluation of explanation-based Medical VQA:

1. **Matched evaluation protocol.** It operationalises Medical VQA as multiple-choice letter prediction with closed, open, and overall splits on complementary radiology benchmarks (VQA-RAD, Med-VQA-2019, and SLAKE), so that answer-only and explanation-based regimes can be compared under a shared accuracy and calibration definition.

2. **Reliability evaluation of explanation-based Medical VQA.** It systematically contrasts answer-only generation with explanation-based strategies: supervised fine-tuning on text-verified teacher rationales (GPT-4o-mini vs. Qwen3-VL-8B as teachers and explanation-first vs. answer-first order) and zeroshot JSON CoT variants, across generalist and medically adapted student models (LLaVA-1.5, LLaVA-Med, Qwen2.5-VL, Qwen3-VL) and an off-the-shelf explanation baseline (MedVLM-R1). That comparison is evaluated jointly along three aspects: uncertainty measurement (answer-token entropy, confidence, and sample agreement under a shared decoding and sampling protocol), calibration (whether those scores track correctness rather than trace consistency), and selective prediction (risk-coverage analysis and Area Under the Risk-Coverage Curve (AURC), including coverage at fixed selective risk).

3. **Visual-dependence diagnostics.** It complements accuracy with optional grounding analyses (counterfactual image ablations) so that apparent performance gains can be inspected for genuine image use versus language priors.

4. **Health-economic interpretation.** It links empirical findings on accuracy, calibration, and selective prediction to clinical workload, trust and oversight costs, and conditions for economically sustainable deployment, without claiming a formal health-economic evaluation of a Medical VQA system.

## 1.3 Thesis Outline

The remainder of this thesis is structured as follows.

**Chapter 2: Related Work** reviews related work on Medical VQA covering the task formulation, datasets and benchmarks, model families, evaluation metrics, and the still limited treatment of reliability and calibration, and concludes with the health-economic literature on imaging AI.

**Chapter 3: Technical Background** provides the technical background of the concepts used throughout this work. It introduces the evaluated vision-language models

and formalises adaptation (supervised fine-tuning), decoding, chain-of-thought and structured explanations, teacher-student verification, uncertainty and calibration, selective prediction, and visual-grounding concepts.

**Chapter 4: Methodology** presents the methodology: workflow overview, datasets and their adaptations, the shared reporting protocol (metrics, ablations, format errors), and the two experimental setups (answer-only, and answer+explanation as zeroshot JSON plus explanation SFT), each with its prompt contracts and training defaults, aligned with the research questions. Selective prediction is co-reported for both setups and presented per dataset in the results chapter.

**Chapter 5: Results** presents quantitative findings by dataset for accuracy, uncertainty ranking, and selective prediction under the two experimental setups, with selected shuffle pairs that drop letter confidence and qualitative SHAP overlays on selected SLAKE items.

**Chapter 6: Discussion** evaluates the findings and places them in a theoretical context. The results are interpreted with respect to RQ1 and the health-economic implications of the empirical evidence (RQ2) are developed, including clinical workload, trust, oversight costs, and conditions for economically sustainable deployment of explanation-based Medical VQA.

**Chapter 7: Conclusion** summarises the main contributions and limitations and identifies directions for future work on reliable, calibrated, and economically informed Medical VQA.

**Appendix A: Supplementary Material** reports the qualitative VQA-RAD output panel, original-image letter and explanation scores for explanation SFT students (Table A.10), closed and open ablation splits (Tables A.11–A.13), and format-adherence matrices. Overall stacked ablations are in Chapter 5.

Together, these chapters move from positioning the problem in the literature, through formal definitions and experimental design, to empirical results and their technical as well as health-economic interpretation, culminating in a concise synthesis of what this thesis contributes and leaves open.

# 2 | Related Work

This chapter positions the thesis in the technical and health-economic literature that surrounds reliable Medical VQA. Section 2.1 traces the shift from classification-based Visual Question Answering (VQA) to generative multimodal models of the families evaluated later. Section 2.2 reviews medical adaptations, with emphasis on LLaVA-Med and explanation-order diagnostics. Section 2.3 surveys calibration and selective prediction insofar as they inform answer-token scores and AURC. Section 2.4 restricts dataset and metric discussion to the three radiology benchmarks and the letter-prediction protocol used in this thesis. Finally, Section 2.5 reconstructs health-economic implications from imaging AI, because no formal health-economic evaluation of a Medical VQA system was identified.

## 2.1 Visual Question Answering Foundations

This section sketches only the architectural path that leads to the generative Multimodal Large Language Models (MLLMs) evaluated in later chapters.

**From Classification to Vision-Language Pretraining.** Early neural VQA treated answering as classification over a fixed vocabulary: a Convolutional Neural Network (CNN) encoded the image, a recurrent or bag-of-words encoder encoded the question, and a fused vector scored answer classes [45, 46, 47]. Even simple language features already approached recurrent baselines, which established language priors as a persistent confound rather than a solved visual-understanding problem [47, 48]. Transformer Vision-Language Pretraining (VLP) then replaced hand-designed fusion with dual- or single-stream attention over region or patch tokens [49, 50, 51, 52, 53], but still typically decoded from a closed answer set.

**From Classification to Generative Multimodal Models.** The models used in this thesis belong to a later shift: VQA as open-ended text generation. Contrastive image-text pretraining, notably Contrastive Language–Image Pre-training (CLIP) [54], supplied the vision encoders of most current systems. Instruction-tuned MLLMs connect a vision encoder to an Large Language Model (LLM) through a learned projector. LLaVA popularises a linear (later Multilayer Perceptron (MLP)) connector with visual instruction tuning [55, 56]. Qwen2-VL / Qwen2.5-VL emphasise native-resolution inputs and stronger vision-language alignment at open weights [57, 3]. Those two families, together with their medical adaptations, are the student and baseline backbones of the experiments. Broader catalogues of BLIP-style querying Transformers or mixture-of-experts generalists are not evaluated here.

## 2.2 Medical Visual Question Answering

Medical VQA transfers the generative paradigm to radiology images (CT, MRI, radiographs) under data scarcity, domain shift from natural images, and a specialised vocabulary [58, 59]. The experimental chapters use three such corpora (Section 2.4). This section reviews the model line that is actually compared.

**Challenge Origins and Core Benchmarks.** Organised radiology VQA began with ImageCLEF VQA-Med, VQA-Med 2019 remains the most cited competition split (modality, plane, organ, abnormality) [60, 7]. Clinician-authored VQA-RAD [1] and bilingual SLAKE [6] supply the other two evaluation sets of this thesis. Open-ended generation scores (token recall, token-F1 Score (F1), Bilingual Evaluation Understudy (BLEU)) and closed exact-match accuracy are not interchangeable, which is why later chapters convert all three corpora to the same multiple-choice letter protocol [58, 59].

**From transfer learning to generative medical models.** Before instruction-tuned MLLMs, medical systems transferred ImageNet CNNs and BERT question encoders under tiny labelled sets, later adding medical contrastive VLP on radiographs and reports [61, 62, 63]. Those pipelines are historical context for the accuracy numbers still quoted on VQA-RAD and SLAKE, they are not re-implemented here.
From 2023 onward, generative medical MLLMs treat Medical VQA as text generation. The biomedical checkpoint used later is LLaVA-Med: it continues from a general LLaVA model, first matching PMC figure captions and then instruction-tuning on GPT-4 biomedical dialogues [2]. Lee et al. place that family in a captioning survey in which MultiMedQA-tuned medical language models already match clinician scores on board-style items, and in which later systems stop treating a medical encoder and a medical decoder as two independent stacks [64]. PMC-VQA / MedVInT attach a vision encoder to an LLM and report large open-ended gains on VQA-RAD and SLAKE relative to earlier encoder-decoder baselines [10]. Among explanation-oriented systems, MedVLM-R1 applies GRPO to a Qwen2-VL-2B backbone and is included later as an off-the-shelf medical explanation baseline [5]. Accuracy-first leaderboards for other generalists (Med-PaLM M, RadFM, Med-Gemini, and similar) are not part of the matched letter-prediction comparison.

**Adaptation, order, and open challenges.** Adapting general MLLMs to Medical VQA typically uses supervised fine-tuning on in-domain VQA pairs. Zero-shot generalists remain fragile on specialised diagnostic questions: ProbMed reports mean accuracy drops of 27.78 % under adversarial pairing, including GPT-4o and Gemini Pro [18]. The specialist-generalist boundary is therefore as much a function of data and adaptation as of architecture [59].
A complementary line of work asks whether natural-language explanations are causally involved in the model's decision rather than attached post hoc. Liang et al. isolate generation order (rationale-first versus answer-first) as a minimal intervention and show that order, scale, and explanation-supervised fine-tuning jointly shape accuracy and cross-order consistency [33]. This thesis adopts that order diagnostic in a medical multiple-choice setting and extends it with uncertainty and selective prediction.

## 2.3 Uncertainty Measurement and Calibration

Beyond predictive accuracy, a safety-critical question is whether reported confidence tracks empirical correctness [65]. Formal score definitions used later are given in Chapter 3. This section reviews only the literature that motivates those scores and the choice of ranking (AURC) over bin-wise calibration as the headline reliability metric.

**Uncertainty Quantification and Classical Calibration.** Uncertainty Quantification (UQ) distinguishes aleatoric from epistemic uncertainty [66]. Selective prediction / abstention uses a scalar score to refuse answers under high uncertainty. Conformal prediction is a related but unused guarantee in the experiments [67]. Deep networks are systematically overconfident [65]. The usual summary is Expected Calibration Error (ECE) from confidence binning, together with reliability diagrams. Those estimators are sensitive to binning and sample size [68, 69], and post-hoc temperature scaling that works in-distribution can fail under shift [70]. The experiments therefore treat bin-wise ECE as background, not as the primary comparison: RQ1 asks, under selective prediction, whether uncertainty *ranks* errors for abstention.

**Calibration in Large Language Models and VLMs.** Vision-Language Model (VLM) practice inherits two competing confidence channels from LLMs. Kadavath et al. and the GPT-4 report already document that answer-token likelihoods can track correctness in a pretrained model and become over-sharp once the same model is aligned [71, 72]. Prompted percentages can then look better calibrated than those likelihoods [73, 74], while remaining too high across prompting recipes [75]. Sanz-Guerrero et al. recast that literature for chat templates: after SFT or Reinforcement Learning from Human Feedback (RLHF), assistants over-trust a completion they just produced relative to the identical string shown as a user turn, and they shrink the gap at inference by pasting the generated answer back as user input before asking for confidence [76]. Neither verbalised self-evaluation nor that recasting is used below. Generative MLLMs in the LLaVA and Qwen-VL families remain overconfident on multimodal benchmarks and prefer answering over admitting uncertainty [77]. Free-text ECE additionally depends on sequence likelihoods, verbalised scores, or an LLM-as-judge, which is a further reason to evaluate closed letter prediction. This thesis measures answer-token entropy, confidence, and sample agreement because the letter-prediction protocol supplies a single answer token whose likelihood is comparable across answer-only and explanation regimes.

**Calibration in Medical AI and Med-VQA.** In clinical imaging, poorly calibrated probabilities distort decision thresholds and risk communication [78]. For Medical VQA, accuracy itself is brittle: ProbMed drops often occur without a commensurate drop in confidence [18]. Byun et al. find persistent overconfidence across Qwen3-VL, InternVL3, and LLaVA-NeXT on VQA-RAD, SLAKE-EN, and VQA-Med 2019, unsolved by scale, CoT, or verbalised prompting alone, although Platt scaling can reduce verbalised ECE [28]. Du et al. report that medical SFT can *worsen* calibration (LLaVA-1.5-med-7B vanilla ECE 44.28 %) [38]. The open gap that this thesis addresses is the joint reporting of explanation-regime accuracy with ranking-based selective prediction, rather than accuracy or ECE in isolation.

**Selective Prediction.** For discriminative VQA, Whitehead et al. formulate reliable answering as preferring abstention to incorrect answers and introduce risk-coverage analysis [12]. Figure 2.1 illustrates the geometry: as coverage increases from answering only the most confident cases to answering all cases, selective risk $R$ rises toward the full-set risk $R(1)$. An oracle ranking that answers easy cases first yields a low curve and a small AURC, an uninformative ranking stays close to $R(1)$. Despite accuracies above 71 % on VQA v2, softmax thresholding yields under 8 % coverage at 1 % risk, and a learned multimodal selector raises that coverage 2.4×.

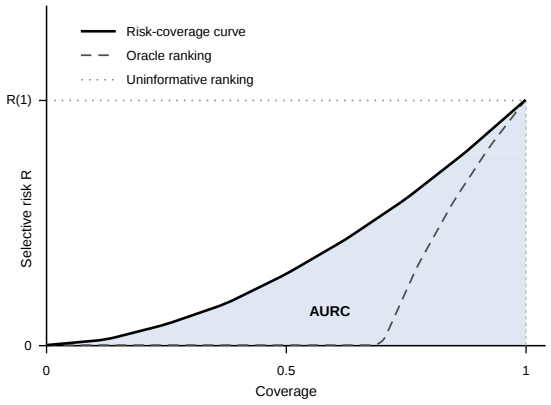



Figure 2.1: Schematic of a risk-coverage curve (RCC). Selective risk $R$ is plotted against coverage, the area under the curve (AURC) summarises ranking quality between an oracle and an uninformative baseline.

## 2.4 Datasets and Evaluation

This section covers the three radiology corpora used later and the scoring choices that follow from them. Calibration metrics remain in Section 2.3.

**Dataset Construction and Benchmark Families.** VQA-RAD comprises 315 MedPix radiology images and about 3,515 clinician-authored questions, of which the free-form subset is 58 % closed (92 % of those yes/no) and 42 % open-ended, with a canonical 451-pair test split [1]. SLAKE provides 642 images (282 CT, 181 MRI, 179 radiographs), 14,028 bilingual pairs, and an official 70/15/15 split [6]. The public release also includes segmentation masks and bounding boxes which are not used in this thesis. VQA-Med 2019 supplies 4,200 MedPix images and 15,292 pattern-generated pairs in modality, plane, organ-system, and abnormality categories, of which only the 500-pair test set was reviewed by a physician and a radiologist [7, 60]. Larger instruction-scale or multi-modality resources exist, but they are not part of the matched letter-prediction evaluation.

**Bias, Contamination, and Annotation Quality.** Language-prior critiques from general VQA [48] transfer directly. Text-only baselines can still achieve non-trivial

accuracy on VQA-RAD and SLAKE by exploiting answer regularities. VQA-RAD additionally suffers train/test image overlap: 202 of 203 test images also appear in training, with only questions (and sometimes not even answers) differing [79]. Clinician-authored sets are small but cleaner. Caption-generated sets such as VQA-Med 2019 buy scale at the cost of limited lexical diversity and rarely reported inter-annotator agreement.

**Evaluation Metrics.** Closed-ended evaluation typically uses exact-match accuracy. Surface n-gram metrics such as BLEU are poorly suited to short clinical answers: Lau et al. already noted that BLEU can favour a constant *no* baseline over longer correct responses [1]. Report-generation entity scores and LLM-as-judge protocols target free-text radiology reports, not a single option letter, and are not used here. The experimental protocol therefore scores letter match on closed, open, and overall splits, and ranks errors with AURC.

**Reproducibility and Benchmark Design.** Leaderboard practice is hindered by heterogeneous splits: VQA-RAD training sizes vary with whether paraphrases are included, while only the 451-pair test set is a stable anchor. High closed accuracy on VQA-RAD and SLAKE can coexist with collapse under adversarial probes [18, 80]. These observations motivate the closed letter-prediction setting, image-ablation controls, and selective-prediction reporting used later in this thesis.

## 2.5 Health-Economic Perspective on Medical AI

Medical VQA combines image-based finding detection with natural-language communication of those findings. Within the literature reviewed for this thesis, neither component is covered by a formal health-economic evaluation of a Medical VQA system. Throughout this thesis, health-economic evaluation denotes analytic forms such as cost-effectiveness analysis, cost-utility analysis, and budget-impact analysis. Quality-Adjusted Life Years (QALYs) are a measure of health-related utility and an input to cost-utility analysis. The Incremental Cost-Effectiveness Ratio (ICER) is the summary result of a cost-effectiveness or cost-utility analysis, not an analysis form. The present work estimates neither quantity and does not conduct those evaluations. The context for RQ2 is therefore reconstructed from imaging AI and from work on report communication, abstention, and trust, the quantities that the technical chapters actually speak to.

**Evidence Base and Determinants of Value.** A systematic review of economic evaluations of AI in diagnostic imaging identified only ten eligible modelling studies and no prospective real-world cost data [81]. A broader radiology-workflow review retained 21 studies with quantified economic endpoints [42]. The two reviews relate to this thesis in complementary ways. From Brin et al., the implication is negative: existing ICER estimates are modelled, concern detection pipelines rather than dialogic Medical VQA, and are not transferred as numerical inputs here. From Molwitz et al., the transferable part is the value condition that RQ2 inherits: for short, high-frequency imaging tasks, economic benefit arises mainly in shortage or high-volume settings when accuracy meets clinical need, per-case savings remain small, and pricing must not erase them. Matching a human reader in mean accuracy is not required

for that condition. A slightly weaker system can still be the relevant comparator if it is more stable over time than a fatigued or shift-varying human reader, or if selective prediction confines auto-answers to a high-coverage, low-risk subset. Medical VQA, as studied here, sits in that task class. The experiments test whether explanation-based generation improves the accuracy and calibrated abstention that such a condition would require, not its cost or effect. Compared with usual care without AI, diagnostic AI is systematically harder to justify than therapeutic AI: it typically adds a layer to an existing pathway rather than displacing a high-cost treatment, while bias reduction, explainability, and implementation are first-order incremental costs [82]. Evaluations of Electronic Health Record (EHR)-based Clinical Decision Support System (CDSS) often omit integration and monitoring outlays, a gap that would dominate any future Medical VQA budget-impact analysis [83].

**Capacity, Workflow, and Prospective Screening.** Imaging demand, limited training capacity, and retirement create an access gap along which AI is discussed as demand management, workflow efficiency, and capacity building [84]. Efficiency gains do not automatically translate into relief. In a survey of 6,726 radiologists, AI use was associated with higher adjusted burnout odds and a dose-response to usage frequency [85]. The methodologically strongest imaging reference that couples detection with workload remains the randomised Mammography Screening with Artificial Intelligence (MASAI) mammography trial: higher detection without additional false positives coincided with a 44.3 % reduction in screen-reading workload [41, 86]. That constellation is the evidence standard that Medical VQA still lacks.

**Report Communication, Abstention, and Trust.** Closest to the functional logic of Medical VQA is work on natural-language simplification of radiology reports. A meta-analysis finds large gains in perceived understandability together with a 7.2 % error rate and clinically relevant errors in 0.9 % of cases [87]. Patient-understandable summaries can improve comprehension and reduce anxiety [88], which is a plausible demand-side benefit only if residual error is costed as work-up, false reassurance, and loss of trust.

Abstention is not economically neutral. In a clinician vignette study, decision accuracy under incorrect AI predictions fell from 66 % to 56 % and recovered to 64 % under selective prediction, yet missed diagnoses and omitted treatments rose when abstention was communicated [89]. Cost-aware conformal deferral treats the abstention threshold as a clinical-cost variable rather than a purely technical knob [90]. Trust and distributional error further condition whether laboratory rankings can become deployment value: opaque systems and misaligned incentives erode care-seeking [91], AI-associated underdiagnosis by sex, ethnicity, and socioeconomic status is documented in chest radiography [92], and liability insurance may reallocate residual harm without replacing those inputs [93].

# 3 | Technical Background

This chapter formalises the concepts that the experimental pipeline uses as primitives and which are therefore referred to in the methodology Chapter 4. It defines the vision-language architecture shared by the evaluated models, the adaptation and decoding mechanisms that produce answers and explanations, and the reliability quantities used to score those outputs. The experimental task (Medical VQA as multiple-choice letter prediction), Hugging Face checkpoints, prompts, and numerical defaults are specified in Chapter 4. Section 3.1 introduces the vision-encoder plus language-decoder paradigm and the five model families. Section 3.2 covers supervised fine-tuning. Section 3.3 defines greedy and stochastic decoding. Section 3.4 distinguishes chain-of-thought families, emission order, single- versus two-pass generation, and faithfulness. Section 3.5 introduces teacher-student distillation and text-only verification. Section 3.6 relates calibration to the uncertainty scores used later. Section 3.7 covers selective prediction and AURC. Section 3.8 summarises visual-dependence diagnostics (concrete protocols follow in Chapter 4).

## 3.1 Vision-Language Models

The models used in this thesis are generative vision-language models (VLMs). They map an image and a natural-language prompt to a token sequence. Five checkpoints in the 2-8 B range serve as students or off-the-shelf baselines. The selection spans two axes so that later comparisons can be attributed, at least in part, to architecture and scale versus domain adaptation:

**1) Architecture generation:** early LLaVA-style models (fixed resolution, MLP connector) versus the Qwen-VL line (native dynamic resolution, multimodal rotary embeddings).

**2) Domain adaptation:** generalist checkpoints (LLaVA-1.5, Qwen2.5-VL, Qwen3-VL) versus medically adapted ones (LLaVA-Med, MedVLM-R1).

Hugging Face identifiers and the evaluation protocol that holds the checkpoints fixed are recorded in Section 4.2.

### 3.1.1 Vision-Encoder and Language-Decoder

All five models follow the dominant vision-encoder plus language-decoder design, in which a vision encoder is attached to a decoder-only language model [55, 56]. A Vision Transformer (ViT) vision encoder maps the image to a sequence of patch features. A lightweight connector (projector or merger) maps those features into the language

embedding space as visual pseudo-tokens. A pretrained LLM decoder then generates text token by token, conditioned on the interleaved visual and textual prefix. If $x_{<t}$ denotes the tokens already produced (including the visual prefix and the user prompt), the model defines

$$p_\theta(x_t \mid x_{<t}) = \mathrm{softmax}\big(z_t\big)_{x_t}, \tag{3.1}$$

where $z_t \in \mathbb{R}^{|V|}$ are the pre-softmax logits over vocabulary $V$. A complete answer is the sampled or greedily decoded continuation $x_{t_0}, \ldots, x_T$. In letter-prediction Medical VQA the scientifically relevant token is the option letter. Longer explanation traces merely change where that token sits in the sequence (Section 3.6).

Pretraining typically comprises an alignment stage (often connector-only) followed by instruction tuning with partially or fully unfrozen components. Instruction tuning teaches the model to follow chat-style prompts. The task-specific supervised fine-tuning of this thesis is a later, narrower adaptation to letter or JSON targets (Section 3.2) and should not be confused with that earlier instruction stage.

Three architectural differences act as confounds in the experiments. First, *resolution*: LLaVA-style encoders resize to a fixed 336×336 grid, whereas Qwen-VL encoders consume native-resolution patches and emit a variable number of visual tokens. Higher, input-dependent token counts preserve fine radiological detail but change memory use. Pixel bounds are therefore an experimental control rather than a modelling detail. Second, *vision pretraining*: LLaVA models use a contrastive CLIP ViT [54]. Qwen3-VL replaces this with a SigLIP-2 tower [4]. CLIP-style encoders are trained to align image and text embeddings. They are not trained to localise small findings, which limits what a later connector can recover. Third, *fusion*: LLaVA injects visual tokens once through an MLP projector. Qwen2.5-VL adds multimodal rotary position embeddings so that spatial and textual positions share one coordinate system [3]. Qwen3-VL further injects intermediate ViT features into selected decoder layers (DeepStack) [4]. These mechanisms change how strongly later tokens can still attend to image evidence, which matters once explanation traces intervene between the image and the answer letter.

### 3.1.2 Evaluated model families

**LLaVA-1.5-7B.** LLaVA-1.5 is the consolidated second iteration of the LLaVA family and serves as the generalist first-generation baseline [56]. It uses a CLIP ViT-L/14 encoder at $336 \times 336$ (fixed 576 visual tokens), a two-layer MLP connector, and Vicuna-1.5-7B as the decoder. Relative to LLaVA-1.0 it replaces the linear projector, raises resolution, and mixes academic VQA data with response-formatting prompts that encourage short answers [55, 56]. The fixed, comparatively low resolution is a known limitation for fine medical detail.

**LLaVA-Med v1.5 (Mistral-7B).** LLaVA-Med transfers the LLaVA recipe to biomedicine [2]. Concept alignment and medical instruction data are derived from PMC figure-caption resources, with GPT-4 dialogues generated from text alone (captions and inline mentions). Curriculum training first aligns the projector on biomedical concepts, then instruction-tunes projector and LLM (Figure 3.1). The checkpoint used here replaces the Vicuna backbone of the v1.0 paper with Mistral-7B-Instruct-v0.2 and follows the LLaVA-1.5 architecture stack. It is a research model and is not intended for clinical decision making.

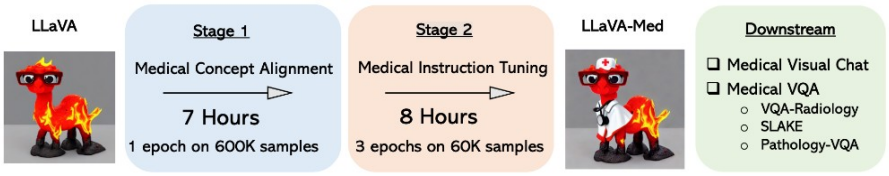



Figure 3.1: LLaVA-Med curriculum adaptation from LLaVA: Stage 1 medical concept alignment (1 epoch on 600K samples) and Stage 2 medical instruction tuning (3 epochs on 60K samples), followed by downstream medical VQA and visual chat [2].

**Qwen2.5-VL-7B-Instruct.** Qwen2.5-VL is the second-generation student architecture used later [3]. The vision tower is a ViT trained from scratch: 2D rotary embeddings and windowed attention let it ingest native-resolution images, which are rounded so that height and width are multiples of 28 and then patched at stride 14. An MLP merger packs each $2 \times 2$ neighbourhood of patch features into one visual token (4:1) before a Qwen2.5 decoder whose 1D RoPE is replaced by multimodal rotary embeddings aligned to absolute time (Figure 3.2). Variable token counts per image are favourable for medical detail but affect memory and throughput.

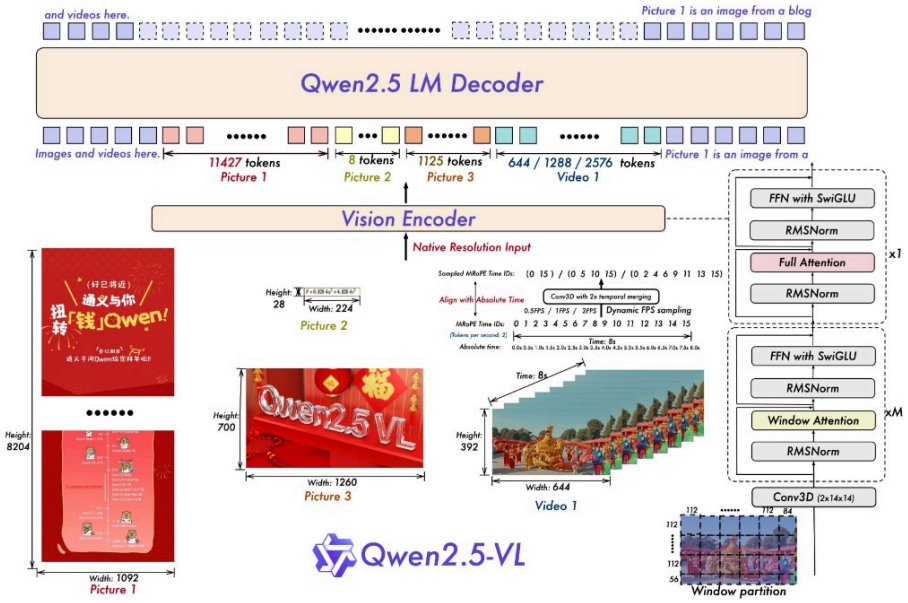



Figure 3.2: Qwen2.5-VL architecture with native-resolution vision encoding, variable visual token counts, and a Qwen2.5 language-model decoder [3].

**Qwen3-VL-8B-Instruct.** Qwen3-VL keeps the three-component schema of Figure 3.2 but updates fusion: a SigLIP-2-based vision encoder, an MLP merger, a dense Qwen3-8B decoder, interleaved multimodal rotary embeddings, and DeepStack

injection of intermediate ViT features into decoder layers [94, 4]. The corresponding architecture diagram is deferred to Figure A.1 (Appendix A) because it is largely redundant with the Qwen2.5-VL layout. The Instruct variant used as a student answers directly. A separate Thinking variant emits an explicit explanation trace and is not the primary student checkpoint.

**MedVLM-R1.** MedVLM-R1 is the off-the-shelf medical explanation baseline, continued from Qwen2-VL-2B [5, 57]. Pan et al. replace extra SFT with GRPO on radiology VQA, arguing that letter-level cross-entropy rewards surface regularities and leaves the decision path unstated. The released checkpoint therefore writes a free-text rationale and then a letter, which they present as a compactness result: a 2B model with a format-plus-accuracy reward can outscore larger SFT systems trained on far more pairs [5]. It is not fine-tuned on the corpora of this thesis. Outputs follow a native `<think>` / `<answer>` tag schema, so letter extraction must follow that contract when the model is scored.

**Comparative overview.** Table 3.1 summarises backbone, vision tower, connector, resolution, domain, and post-training. Two implications follow for later chapters. MedVLM-R1 ($\approx 2$ B) is substantially smaller than the $\approx 7$-8 B students, so accuracy comparisons confound checkpoint family with scale. Prompt and format conventions also differ: LLaVA-1.5 was trained with short-answer formatting, whereas MedVLM-R1 expects a strict tag schema. Model-specific wrappers at identical question content are documented in Section 4.5.

| | LLaVA-1.5 | LLaVA-Med | Qwen2.5-VL | Qwen3-VL | MedVLM-R1 |
|---|---|---|---|---|---|
| LLM | Vicuna-1.5-7B | Mistral-7B | Qwen2.5-7B | Qwen3-8B | Qwen2-1.5B |
| Vision | CLIP-L/14 | CLIP-L/14 | custom ViT | SigLIP-2 | custom ViT |
| Connector | 2-layer MLP | 2-layer MLP | MLP merger | MLP+DStack | MLP merger |
| Resolution | fixed 336 | fixed 336 | native dyn. | native dyn. | native dyn. |
| Domain | generalist | biomedical | generalist | generalist | biomedical |
| Post-training | SFT | SFT (curr.) | SFT | SFT (Instr.) | RL (GRPO) |
| Explicit expl. | no | no | no | Instruct: no | yes (tags) |

Table 3.1: Comparative overview of the evaluated model families. Vision towers are abbreviated (CLIP-L/14 is CLIP ViT-L/14, DStack is DeepStack). Among the five checkpoints, only MedVLM-R1 continues a Qwen2-VL-2B stack under GRPO instead of letter-level SFT, and only it must be scored through the paper's `<think>` then `<answer>` wrapper [5].

## 3.2 Adaptation

### 3.2.1 Supervised fine-tuning

Supervised fine-tuning (SFT) adapts a pretrained multimodal model to letter-prediction and, optionally, explanation targets by minimising token-level cross-entropy on assistant tokens only. Let $x = (x_1, \ldots, x_T)$ be the tokenised training sequence and let $M \subseteq \{1, \ldots, T\}$ be the set of assistant (target) positions. Prompt

tokens, including image tokens and the user multiple-choice block, and padding tokens lie outside $M$ and receive zero loss weight. With model parameters $\theta$,

$$\mathcal{L}_{\mathrm{CE}}(\theta) = -\sum_{t \in M} \log p_\theta(x_t \mid x_{<t}) \,. \tag{3.2}$$

This focuses learning on the answer and, in explanation settings, on the structured explanation, rather than on reconstructing the question text. Explanation-based targets require longer sequence lengths than letter-only targets because the assistant output includes a JSON explanation block.

## 3.3 Decoding

The predictive distribution $p_\theta(\cdot \mid x_{<t})$ does not by itself determine the emitted string. Decoding chooses a token at each step and thereby determines both the reported letter and the logits from which uncertainty is read.

**Greedy decoding.** Greedy decoding selects $\hat{x}_t = \arg\max_v p_\theta(v \mid x_{<t})$ at every step. It is deterministic given the prompt and is the default for the official answer in fine-tuned and letter-only evaluation.

**Temperature sampling.** Stochastic decoding draws $x_t$ from a sharpened or flattened distribution

$$p_T(v \mid x_{<t}) \propto p_\theta(v \mid x_{<t})^{1/T}. \tag{3.3}$$

$T < 1$ concentrates mass on the mode, $T > 1$ increases diversity. Agreement-based uncertainty requires independent samples from this family (Section 3.6).

**Nucleus sampling.** Nucleus, or top-$p$, sampling [95] further restricts the draw to the smallest set of tokens whose cumulative probability is at least $p$, then renormalises. It avoids the long tail of near-zero tokens that temperature sampling alone would still visit. The numerical defaults for $T$, $p$, and the number of samples per item are stated in Section 4.4. All that is needed here is that every evaluated item is decoded several times under this stochastic distribution, and that both the reported answer and the uncertainty scores are derived from that same set of samples.

## 3.4 Natural-Language and Structured Explanations

Explanation-enhanced Medical VQA emits not only an answer letter but also a natural-language justification. Chain-of-thought (CoT) prompting elicits intermediate steps before the final answer and can improve multi-step performance without changing weights [31]. The same idea can be turned into a training target: the assistant sequence then contains a rationale followed, or preceded, by the letter. Five distinctions matter for the later regimes.

**Structured versus free-form.** A structured explanation occupies fixed fields (in this thesis: region, observation, and explanation). A free-form rationale is an unconstrained short paragraph. Structure constrains what the student must imitate and what a verifier can read. It does not by itself make the explanation faithful to the image.

**Training-time versus prompt-only.** Under explanation-based SFT the rationale is part of the supervised target. Under zeroshot JSON the same surface schema is requested only at inference time, with no in-domain rationale fine-tuning. These are different interventions: the first changes weights, the second changes only the decoding contract.

**Emission order.** Let $r$ be the rationale and $a$ the answer letter. *Explanation-first* (rationale-first) generation emits $(r, a)$, *answer-first* generation emits $(a, r)$. This order intervention follows the controlled diagnostic of Liang et al. [33]. Because decoding is autoregressive, explanation-first generation conditions the letter logits on the model's own trace. Token probabilities then increasingly reflect consistency with that trace rather than residual uncertainty about the letter [35]. Answer-first generation writes the letter before the justification, so the letter distribution is not conditioned on $r$. The trailing rationale is then a post-hoc explanation of an already committed decision.

**Single-pass versus two-pass.** The order distinction above assumes one continuation. *Single-pass* generation emits both fields in that continuation. *Two-pass* (two-stage) generation first samples a rationale $\tilde{r}$ with no letter, then runs a second forward pass that conditions on the frozen trace. The second pass requests a structured pair $(\tilde{r}, a)$. The letter distribution of pass 2 is conditioned on a committed rationale, as under explanation-first emission, but $\tilde{r}$ itself was produced without jointly decoding $a$. This isolation is why two-pass prompting is a separate experimental axis from single-pass RF versus AF (Section 4.5.2).

## 3.5 Distillation and Verification

Explanation-based SFT requires rationale targets. This thesis does not use clinician-authored chain-of-thought. It uses a teacher-student construction in the sense of knowledge distillation [96]: a teacher model writes rationales, and a student is trained to imitate the retained traces.

The teacher sees the image and the gold option *text* (not the letter) and writes a structured region-observation-explanation block. Because the gold answer is revealed, teacher traces are gold-guided explanations rather than independently solved answers. A separate verifier then answers the multiple-choice question from the rationale text and the question alone, without the image. A trace is retained if and only if the verifier recovers the gold letter.

That filter is a test of *linguistic self-consistency*: the written justification must determine the correct letter as text. It is not a test of visual correctness, visual grounding, or faithfulness. A rationale can pass verification while citing a finding that is not in the image, provided the text still entails the gold letter. The student therefore receives distilled, internally consistent explanations, not human-validated visual proofs. Sampling counts, teacher identities, and pass rates belong to data preparation in Section 4.3.2.

## 3.6 Uncertainty and Calibration

The title of this thesis refers to uncertainty calibration. The two words are related but not interchangeable, and the scores used in the result tables measure ranking quality rather than bin-wise calibration.

### 3.6.1 Calibration and overconfidence

A predictor is perfectly calibrated if, among all items issued at confidence $p$, a fraction $p$ are correct [65]. The expected calibration error (ECE) turns that definition into a size-weighted bin average: items are grouped by reported score, and each bin $B_m$ contributes $|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|$, with $\mathrm{conf}(B_m)$ the mean score in the bin and $\mathrm{acc}(B_m)$ the empirical hit rate [97, 65]. When $\mathrm{conf}(B_m) > \mathrm{acc}(B_m)$, surplus probability sits on errors. That one-sided surplus is the clinically costly direction, so some authors isolate it as an overconfidence error rather than folding it into a two-sided ECE [98]. Related Work surveys ECE and its variants. This thesis reports neither as a headline metric. Letter-prediction test splits are small, especially on VQA-RAD. Binning estimators are unstable in that regime, and generative traces add a further choice of which token's probability to bin. More importantly, RQ1 asks, under selective prediction, whether uncertainty can *rank* items for abstention, not whether a reliability diagram is flat. Risk-coverage analysis and AURC (Section 3.7) answer that ranking question directly. A complementary distinction is aleatoric versus epistemic uncertainty [66]. Aleatoric uncertainty is irreducible noise in the item (ambiguous images, underspecified questions). Epistemic uncertainty is reducible uncertainty about the model. Answer-token entropy mixes both. Agreement and vote entropy, computed over independent samples, track instability of the parsed letter and are closer to an epistemic signal. No claim is made that the two sources are identified separately.

### 3.6.2 Answer-token scores

Uncertainty is derived from the predictive distribution at the *answer token*. Let $z_{t^*} \in \mathbb{R}^{|V|}$ denote the pre-softmax logits at answer-token position $t^*$, and let $p = \mathrm{softmax}(z_{t^*})$. Each item is decoded $K$ times under the stochastic decoding distribution of Section 3.3, and these are the samples from which both the answer and the uncertainty scores are derived. Write $\hat{y}^{(k)}$ for the parsed letter of sample $k$, and $c^\star$ for the plurality letter among valid parses. Operational $K$ and sampling hyperparameters are given in Section 4.4.1.

**Accuracy:** On a split of size $N$, with gold letter $y_i$ and reported prediction $\hat{y}_i$,

$$\mathrm{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i] \,. \tag{3.4}$$

In both letter-only and explanation JSON evaluation, $\hat{y}_i$ is the majority vote over the $K$ sampled completions.

**Answer-token location:** For letter-only generation, $t^*$ is the first generated token. For explanation JSON, $t^*$ is the last generated token whose identity encodes a valid option letter (including common tokeniser variants), searched backwards over the

generated sequence. Under explanation-first emission this token is conditioned on $r$, which is why entropy can collapse even when the letter is wrong (Section 3.4).

**Token entropy:** The Shannon entropy of a single run at $t^*$ (in nats, with zero-mass terms dropped) is

$$H(z_{t^*}) = -\sum_{v:p_v>0} p_v \log p_v. \tag{3.5}$$

Token entropy is a within-generation score: it is low when the softmax at $t^*$ is peaked, and high when probability mass is spread across the vocabulary. The item-level score is the mean over majority-agreeing runs with a finite entropy,

$$H_i^{\text{tok}} = \frac{1}{|K_i^\star|} \sum_{k \in K_i^\star} H\big(z_{t^*}^{(i,k)}\big), \qquad K_i^\star = \big\{k : \hat{y}_i^{(k)} = c_i^\star,\ H\big(z_{t^*}^{(i,k)}\big) < \infty\big\}. \tag{3.6}$$

Tables report the split mean of $H_i^{\text{tok}}$. That column is a descriptive average, not the ranking score of AURC.

**Confidence:**

$$\text{confidence}_{\text{ans}} = \max_v p_v. \tag{3.7}$$

Confidence is a certainty score: large values indicate a dominant letter. The reported item-level score is the mean over all $K$ runs with a finite value,

$$\bar{c}_i = \frac{1}{K} \sum_{k=1}^{K} \text{confidence}_{\text{ans}}^{(i,k)}. \tag{3.8}$$

The two item-level aggregations differ in scope: entropy is averaged over $K_i^\star$ only, confidence over all $K$ runs. When the $K$ votes split, minority runs therefore contribute to $\bar{c}_i$ but not to $H_i^{\text{tok}}$, so the entropy and confidence columns are not strictly comparable item by item. Both are reported as descriptive columns, and ranking for selective prediction uses the combined score defined below.

**Agreement:** Plurality agreement is

$$\text{agreement}_i = \frac{\max_c \#\{k : \hat{y}_i^{(k)} = c\}}{\#\{k : \hat{y}_i^{(k)} \neq \emptyset\}}. \tag{3.9}$$

**Vote entropy:** Vote entropy is the Shannon entropy of the empirical letter distribution over the $K$ parsed samples. Let $n_{i,c} = \#\{k : \hat{y}_i^{(k)} = c\}$ and $n_i = \sum_c n_{i,c}$ (valid parses only). Then

$$H_i^{\text{vote}} = -\sum_{c:n_{i,c}>0} \frac{n_{i,c}}{n_i} \log \frac{n_{i,c}}{n_i}. \tag{3.10}$$

Vote entropy is a between-generation score: it is zero when all valid samples agree on one letter, and it is large when the $K$ letters are split. Unlike binary agreement it distinguishes, for example, a $3/2$ split from $3/1/1$. Vote entropy is not a table column, it enters evaluation only through the mixture of Section 3.7. When explanation JSON is generated, per-token entropies along the sequence can additionally be stored for trajectory analysis.

## 3.7 Selective Prediction

Selective prediction uses a scalar score $s_i$ (uncertainty or confidence) and a per-sample quality indicator $m_i \in [0, 1]$ (default: letter match $m_i = \mathbf{1}[\hat{y}_i = y_i]$) to decide which questions to answer and which to abstain from [12]. A threshold $\tau$ partitions the test set into answered versus abstained sets. Coverage and risk are

$$\text{coverage}(\tau) = \frac{n_{\text{answered}}}{N}, \qquad \text{risk}(\tau) = 1 - \frac{1}{n_{\text{answered}}} \sum_{i \in A(\tau)} m_i. \tag{3.11}$$

For uncertainty scores (token entropy, combined uncertainty), the model answers when $s \leq \tau$. For certainty scores (confidence, agreement), it answers when $s \geq \tau$.
The risk-coverage curve is obtained by sorting thresholds by increasing coverage. The *area under the risk-coverage curve* (AURC) is the trapezoidal integral of risk with respect to coverage. Lower AURC is better: less risk at comparable coverage. Complementary operating points include coverage at target risks (e.g. 1%, 5%, 10%) and risk at target coverages (e.g. 20%, 50%, 80%). Overconfidence is particularly harmful here: if high-confidence errors remain frequent, risk stays elevated even when coverage is restricted to the most confident fraction of the test set, so AURC degrades. This is the sense in which the thesis treats calibration as a reliability property: not as a binned ECE, but as the extent to which confidence or entropy concentrates errors among the deferred cases.

**Combined uncertainty:** Token entropy $H^{\text{tok}}$ (Section 3.6.2) and vote entropy $H^{\text{vote}}$ (Section 3.6.2) measure within- versus between-generation uncertainty. They live on incomparable scales (vocabulary versus at most $K$ letter labels), so both are min-max normalised on the evaluation subset under consideration (closed, open, or overall, separately):

$$\tilde{H}_i^{\text{tok}} = \frac{H_i^{\text{tok}} - \min_j H_j^{\text{tok}}}{\max_j H_j^{\text{tok}} - \min_j H_j^{\text{tok}}}, \qquad \tilde{H}_i^{\text{vote}} = \frac{H_i^{\text{vote}} - \min_j H_j^{\text{vote}}}{\max_j H_j^{\text{vote}} - \min_j H_j^{\text{vote}}}. \tag{3.12}$$

If $\max_j H_j = \min_j H_j$ on a split, that component has no ranking information and is set to 0 after normalisation, so $U_{\text{combined}}$ falls back to the remaining component. This is the usual case when agreement is at or near 1.00 and every run emits the same letter: vote entropy is then constant, and ranking uses token entropy alone. If both components are constant, $U_{\text{combined}}$ is constant and AURC equals full-set risk. The combined ranking score is the convex mixture

$$U_{\text{combined},i} = \alpha \, \tilde{H}_i^{\text{tok}} + (1 - \alpha) \, \tilde{H}_i^{\text{vote}}, \tag{3.13}$$

with default $\alpha = 0.5$ (Table 4.8). Normalisation is eval-set-dependent, and both components are rescaled separately on each dataset-split combination. Rankings, $U_{\text{combined}}$ values, and the AURC computed from them are therefore only comparable within one such combination. AURC magnitudes are accordingly not compared across corpora anywhere in this thesis, and readings that set one split of a corpus against another are indicative rather than quantitative. Within one dataset-split combination the columns do what they are meant to do: they rank rows against each other. The mixture complements a peaky majority softmax when the $K$ votes still split, it does not correct consistently wrong, peaky predictions.

**AURC:** The reported AURC ranks items by $u_i = U_{\text{combined},i}$. A threshold $\tau$ answers the items with $u_i \le \tau$ and abstains on the rest (empty answered sets excluded):

$$A(\tau) = \{i : u_i \le \tau\}, \qquad \text{coverage}(\tau) = \frac{|A(\tau)|}{N}, \qquad \text{risk}(\tau) = 1 - \frac{1}{|A(\tau)|} \sum_{i \in A(\tau)} m_i. \tag{3.14}$$

Sorting the distinct coverage values $c_k$ with associated risks $r_k$ in increasing coverage yields the risk-coverage curve. The reported AURC is the trapezoidal integral of that curve,

$$\text{AURC} = \sum_k \frac{c_{k+1} - c_k}{2} (r_k + r_{k+1}). \tag{3.15}$$

Lower AURC is better: less selective risk at comparable coverage. Evaluation reports this combined-uncertainty AURC on closed, open, and overall splits.

## 3.8 Visual Grounding

Accuracy alone does not establish that a prediction used the paired image. Language priors and shortcut learning allow non-trivial accuracy from the question and option list alone [48, 16]. The diagnostics below probe visual dependence without changing the letter-match definition of correctness. Operational SHAP settings are specified in Section 4.5.3, dataset-level visual dependence uses the image ablations of Section 4.4.2.

**Counterfactual image ablations.** Replacing the image by a solid black field or shuffling image-question pairings removes or mismatches visual evidence while keeping the question fixed. A large drop in accuracy relative to the matched image indicates visual dependence. A small drop suggests that language priors or shortcuts suffice.

**SHAP attribution.** SHapley Additive exPlanations (SHAP) [99] treats each input coordinate as a player in a cooperative game whose payoff is the model output [100]. A local surrogate is additive in binary presence indicators $z' \in \{0,1\}^M$,

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z'_j, \tag{3.16}$$

with baseline $\phi_0 = \mathbb{E}[f(x)]$ and per-feature credits $\phi_j$. Lundberg and Lee show that, among all such additive scores, only the classical Shapley allocation meets local accuracy, missingness, and consistency at once [99]:

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j, \qquad \phi_j = \sum_{S \subseteq \{1,\dots,M\} \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \Big( f_{S\cup\{j\}}(x_{S\cup\{j\}}) - f_S(x_S) \Big), \tag{3.17}$$

where $f_S$ runs the model with coalition $S$ kept and every other coordinate replaced by a baseline. The factorial weights average the marginal gain of feature $j$ over every insertion order into the coalition. That uniqueness result is what lets SHAP sit as a common reference for otherwise unrelated local explainers: methods that break the three axioms can assign credits that move the wrong way when a feature becomes more important. Exact evaluation is exponential in $M$. Practical estimators

include model-agnostic KernelSHAP (weighted least squares with the Shapley kernel, related to LIME [101]), DeepSHAP (DeepLIFT-style backpropagation with SHAP multipliers [102]), GradientSHAP (Integrated Gradients with input noise [103]), and ViT-Shapley for Vision Transformers [104]. On images, features are patches or superpixels rather than single pixels. Region masking yields grid heatmaps in which positive $\phi_j$ support the explained output and negative values oppose it. In medical imaging and clinical decision support, SHAP is a widely used model-agnostic XAI tool for spatial justification of predictions [105]. Operational MedVQA settings (explained scalar, masker, and visualisation) are specified in Section 4.5.3.

# 4 Methodology

This chapter describes the experimental methodology used to address the research questions of this thesis. It operationalises RQ1 from Chapter 1. Health-economic implications (RQ2) are developed in Chapter 6. Section 4.1 gives a workflow overview. Section 4.2 records the Hugging Face checkpoints whose architectures are defined in Section 3.1. Section 4.3 analyses the source corpora and the adaptations applied to them. Section 4.4 fixes the shared reporting protocol (evaluation metrics, image ablations, and format-error analysis). Section 4.5 defines the two experimental setups that are reported by dataset in Chapter 5: answer-only (Experiment 1) and answer+explanation (Experiment 2: zeroshot JSON and explanation SFT), including the prompts, training infrastructure, and default hyperparameters of each protocol. Selective prediction is co-reported per dataset rather than as a separate experiment. Implementation-level engineering details are omitted unless they affect scientific interpretation.

## 4.1 Experimental Workflow Overview

Figure 4.1 summarises the pipeline described in this section. The workflow starts from the three free-text radiology corpora specified in Section 4.3.1 (VQA-RAD, Med-VQA-2019, and SLAKE) and first constructs shared evaluation artefacts (Section 4.3.2). Free-text answers are converted into multiple-choice letter targets. Closed items become two-way yes/no options. Open items receive four GPT-4o-mini distractors under a constrained JSON schema, after which options are shuffled and the gold letter is recorded. For explanation-based training, teacher models (GPT-4o-mini and Qwen3-VL-8B) generate structured `REGION/OBSERVATION/EXPLANATION` pseudo-rationales.
On these artefacts, two experimental setups are run with the student and baseline models of Section 4.2 under the shared reporting protocol of Section 4.4 (Section 4.5). Experiment 1 is answer-only evaluation: letter targets without explanations, comparing base checkpoints with supervised fine-tuning. Experiment 2 is answer+explanation: zeroshot JSON without in-domain fine-tuning (answer-first, rationale-first, and two-stage orderings, plus MedVLM-R1 as an off-the-shelf medical explanation baseline) and supervised fine-tuning on the verified pseudo-explanations, varying teacher model and emission-order regime (explanation-first and answer-first, mixed RF+AF is an additional test). Both setups co-report selective prediction through entropy, sampling agreement, and risk-coverage summaries (AURC) on the same closed, open, and overall splits. Qualitative shuffle pairs and SHAP overlays on selected samples are reported only as illustrative side material in section 5.6 and section 5.7 and are not treated as a standalone experiment.

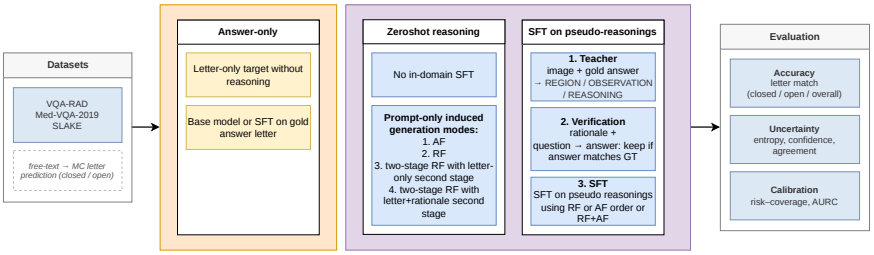



Figure 4.1: Overview of the experimental pipeline: source datasets are converted to multiple-choice letter prediction and evaluated under two setups (answer-only and answer+explanation). Selective prediction is co-reported for both.

## 4.2 Student and Baseline Checkpoints

Architectural properties of used models are defined in Section 3.1. Table 4.1 records the Hugging Face weights used in Experiments 1 and 2. Unless noted otherwise, each fine-tuned checkpoint is compared with its corresponding base weights on the same multiple-choice test splits under the same decoding and uncertainty protocol.

| Short name | Hugging Face ID | Parameters |
|---|---|---|
| LLaVA-1.5-7B | `llava-hf/llava-1.5-7b-hf` | ≈ 7 B |
| LLaVA-Med v1.5 | `microsoft/llava-med-v1.5-mistral-7b` | ≈ 7 B |
| Qwen2.5-VL-7B-Instruct | `Qwen/Qwen2.5-VL-7B-Instruct` | ≈ 7 B LLM + 0.7 B ViT |
| Qwen3-VL-8B-Instruct | `Qwen/Qwen3-VL-8B-Instruct` | ≈ 8 B LLM + ≈ 0.4 B ViT |
| MedVLM-R1 | `JZPeterPan/MedVLM-R1` | ≈ 2 B |

Table 4.1: Student and baseline checkpoints used in this thesis.

VQA-RAD, SLAKE, and related corpora appear in pretraining or adaptation of several of these models (for example LLaVA-Med fine-tuning, and MedVLM-R1 via HuatuoGPT-Vision aggregates).

## 4.3 Datasets

This section first analyses the three source corpora as published (Section 4.3.1), then describes the dataset modifications and preparations that turn those free-text resources into the datasets used for training and evaluation (Section 4.3.2): multiple-choice letter targets and text-verified pseudo-explanations. How these artefacts are consumed is specified in Sections 4.4 and 4.5.

### 4.3.1 Source corpora

Experiments are conducted on three complementary radiology Medical VQA corpora: VQA-RAD [1], ImageCLEF VQA-Med 2019 (hereafter Med-VQA-2019) [7], and SLAKE [6]. This subsection analyses the source corpora as published and publicly distributed. Table 4.2 summarises construction and scale. Table 4.3 records the official

partitions. Tables 4.4 , 4.5 and A.1 display the image and question inventories that the VQA-RAD and SLAKE papers contain in full. The following subsection describes the adaptations and extensions applied to these free-text resources for the experiments of this thesis. For VQA-RAD, no held-out validation file is distributed. Early stopping on a validation split is therefore unavailable unless the test split is reused explicitly, and reported results always use the designated test split.

| | VQA-RAD | Med-VQA-2019 | SLAKE |
|---|---|---|---|
| Year / venue | 2018, *Sci. Data* | 2019, CLEF | 2021, ISBI |
| Image source | MedPix (one image per case) | MedPix (filtered) | Decathlon, ChestX-ray14, CHAOS |
| Images | 315 (paper), 314 (public) | 4,200 | 642 |
| QA pairs | 3,515 (paper), 2,244 (public) | 15,292 | 14,028 |
| Elicitation | Clinician-authored | Template-generated | Physician-curated templates |
| Languages | English | English | English and Chinese |
| Masks / boxes | No | No | Yes (ITK-SNAP, unused) |
| Official split | Train / test | Train / val / test | 70/15/15 by image |

Table 4.2: Overview of the three radiology Medical VQA corpora used in this thesis. Paper versus public counts are shown where they differ. SLAKE is released with organ and disease segmentation masks and bounding boxes. Those annotations are never loaded or used in this thesis.

| | Train | | Validation | | Test | |
|---|---|---|---|---|---|---|
| Corpus | QA | Images | QA | Images | QA | Images |
| VQA-RAD (public) | 1,793 | 313 | – | – | 451 | 203 |
| Med-VQA-2019 | 12,792 | 3,200 | 2,000 | 500 | 500 | 500 |
| SLAKE | 9,849 | 450 | 2,109 | 96 | 2,070 | 96 |

Table 4.3: Official splits. VQA-RAD has no validation partition. SLAKE counts follow Liu et al. [6], a public snapshot redistributes 9,840 / 2,100 / 2,090 pairs with the same total of 14,028. Med-VQA-2019 training pairs are 12,792 in the challenge report [7], some redistributions list 12,800 training rows. VQA-RAD train and test image counts are not disjoint (see text).

### VQA-RAD

VQA-RAD is the first fully clinician-authored radiology VQA dataset [1]. Images are drawn from MedPix, with one image per case so that patients are unique, and images containing graphical overlays such as arrows or circles were excluded. The published collection contains 315 images balanced across three anatomy-modality groups (Table 4.4). Fifteen clinicians produced questions and answers in three variants: free-form

(as if asking a colleague), rephrased by a second person, and framed to a template (Table 4.4). The 3,515 pairs correspond to about ten questions per image. Questions are short (typically five to seven words) and answers have a median length of one word. Eleven clinically motivated types are distinguished: modality, plane, organ system, abnormality, object/condition presence, positional reasoning, colour, size, attribute other than colour/size, counting, and other. Only two free-form type counts are published. Presence is the most frequent ($n$ = 483) and counting the rarest ($n$ = 15), i.e. 31.9 % and 1.0 % of the 1,515 free-form pairs. Among free-form items, 42 % are open-ended ($n$ = 637) and 58 % closed ($n$ = 878). 92 % of the closed items are yes/no (405 yes, 406 no).

| Body region | Imaging | Images | Share |
|---|---|---|---|
| Head | Axial CT/MRI | 104 | 33 % |
| Chest | Radiograph | 107 | 34 % |
| Abdomen | Axial CT | 104 | 33 % |
| **Total** | | 315 | 100 % |

| Elicitation | QA pairs | Share |
|---|---|---|
| Free-form | 1,515 | 43.1 % |
| Rephrased | 733 | 20.9 % |
| Framed | 1,267 | 36.0 % |
| **Total** | 3,515 | 100 % |

Table 4.4: VQA-RAD image coverage and elicitation variants as reported by Lau et al. [1]. The public release used in this thesis has 314 images and 2,244 unique pairs (see text).

The publicly released distribution used in this thesis contains 2,244 unique question-answer pairs on 314 images (train 1,793 pairs / 313 images: test 451 pairs / 203 images), not the 3,515 pairs stated in the paper. The 451-pair test set matches the paper's free-form plus rephrased evaluation subset and is the stable reporting anchor in the literature. The elicitation counts of Table 4.4 account for the shortfall: free-form plus rephrased is 1,515 + 733 = 2,248 pairs against the 2,244 distributed, so the release corresponds to those two variants and omits the 1,267 framed pairs. The authors do not state that exclusion explicitly. All numerical results in this thesis refer to the public 2,244-pair corpus. A further limitation of the official split is image leakage: 202 of 203 test images also occur in training, with only the questions (and not always the answers) differing [79]. Reported VQA-RAD accuracy therefore partly reflects question-level generalisation over repeated images rather than generalisation to unseen studies.

#### Med-VQA-2019

Med-VQA-2019 was introduced as the ImageCLEF VQA-Med 2019 challenge corpus to scale beyond VQA-RAD [7]. Images are again from MedPix, but selected automatically by caption, modality, plane, location, and diagnostic-method filters. Question-answer pairs are generated from templates derived from VQA-RAD's natural questions rather than written freely by clinicians. The corpus contains 4,200 images and 15,292 pairs, with roughly 3-4 questions per training image. Four categories define a designed difficulty gradient: modality (36 subtypes), plane (16 planes), and organ system (10 systems) are closed classification tasks with bounded answer sets, whereas abnormality requires free diagnostic text. Question forms are yes/no, WH, and (for modality) closed templates derived from VQA-RAD. The source paper does not publish an image census by anatomy or modality, nor full train/validation counts per category. Category-wise correction denominators of 125, together with the official

evaluation that treats the four categories as equal fourths of overall accuracy, imply 125 test questions in each category. Only the test set was doubly reviewed by a physician and a radiologist: 33 answers were adjusted (eight optional additions, ten alternative answers, fifteen corrections), i.e. a 3 % correction rate. The authors infer a comparable error rate on train and validation, which were produced by the same automatic pipeline and not fully re-annotated. The challenge paper reports 12,792 / 2,000 / 500 training, validation, and test pairs on 3,200 / 500 / 500 images. A public snapshot lists 12,800 training rows with the same validation and test sizes. Experiments in this thesis follow the official three-way partition of the distributed corpus.

### SLAKE

SLAKE was designed to close two gaps left by VQA-RAD, namely the absence of dense visual labels needed for region localisation, and the absence of questions that require medical knowledge beyond the image [6]. Unlike the MedPix corpora, images are sampled from three existing open collections: NIH ChestX-ray14 contributes the 179 radiographs, which retain their original labels, and the Medical Segmentation Decathlon and CHAOS together contribute the remaining 463 single slices extracted from 3D volumes (282 CT and 181 MRI, Table 4.5). Experienced physicians annotated semantic masks and bounding boxes for organs and diseases, and questions were proposed from region-specific templates that physicians could accept, edit, or rewrite. Those masks and boxes are part of the public release only. This thesis never uses them: every SLAKE experiment takes the English image–question–answer pairs converted to multiple-choice letters, with no region supervision, no mask input, and no localisation loss. The dataset comprises 642 images, 14,028 pairs, twelve annotated diseases, and 39 organs. Table 4.5 gives the anatomical and modality breakdown: all CT and MRI studies are axial single slices.

| Body region | Images | Share |
|---|---|---|
| Head (CT/MRI) | 140 | 21.8 % |
| Neck (CT) | 41 | 6.4 % |
| Chest (radiograph/CT) | 219 | 34.1 % |
| Abdomen (CT/MRI) | 201 | 31.3 % |
| Pelvis (CT) | 41 | 6.4 % |
| **Total** | 642 | 100.0 % |

| Modality | Images |
|---|---|
| CT | 282 |
| MRI | 181 |
| Radiograph | 179 |
| **Total** | 642 |

Table 4.5: SLAKE image coverage by body region and modality [6].

Ten content types are fully enumerated by split in Table A.1 (Appendix A). Items are further labelled vision-only versus knowledge-based (for example, which depicted organs belong to the digestive tract) and as open or closed. Those two flags are not published as counts. The official split is 70/15/15 at image level, stratified across eight region-modality strata, yielding 450 / 96 / 96 train/validation/test images and 9,849 / 2,109 / 2,070 pairs. Test answers are required to occur in training, so SLAKE evaluation is classification-compatible. SLAKE is bilingual. English and Chinese candidate questions are not identical. This thesis uses the English subset, which matches the English multiple-choice protocol introduced below. SLAKE does not report an inter-rater F1 for the question-answer pairs comparable to VQA-RAD.

### 4.3.2 Dataset modification and preparation

Before training and evaluation, two families of derived artefacts are constructed from the source corpora. A shared loader yields image path, question, gold letter, option list, and question type for each row of a converted corpus. Rows with fewer than two options or invalid correct indices are skipped. Image paths are normalised across corpus layouts so that the same sample identity can be joined to pseudo-explanation side files. SLAKE's published segmentation masks, bounding boxes, and knowledge graph are not part of this loader and are unused throughout training, evaluation, ablations, and qualitative overlays.

**Multiple-choice letter prediction.** All experiments treat Medical VQA as multiple-choice letter prediction. A sample is a tuple $(I, q, \{o_j\}_{j=0}^{n-1}, y)$, where $I$ is a medical image, $q$ is a natural-language question, $\{o_j\}$ are option strings, and $y \in \{A, B, C, D, E\}$ for open questions or $y \in \{A, B\}$ for closed questions is the gold option letter. Free-text answers from the source corpora are converted into this format before training and evaluation. The original free-text gold answer is retained only as metadata. Closed questions whose gold answer normalises to *yes* or *no* become two-way items with options $\{\texttt{yes}, \texttt{no}\}$. The correct index follows the normalised gold. All other questions become five-way open items. The normalised gold string is retained as one option, and an OpenAI LLM API call (GPT-4o-mini) generates four clinically plausible distractors under a low temperature and a strict JSON schema. Options are shuffled and the gold letter $y$ is the unique index of the normalised gold after shuffling. The user prompt lists options as `A)`, `B)`, `C)`, `D)`, `E)` for open questions and `A)`, `B)` for closed questions and instructs the model to return exactly one letter and no other text. A predicted letter is recovered by parsing the generated string. Accuracy is defined as the exact letter match against the gold answer letter $y$. The generated distractors are not screened for synonymy with the gold string or for being independently correct, and radiological vocabulary makes both possible: *axial* and *transverse* denote the same plane, and a location distractor can be true without being the annotated answer. Items of that kind are unsolvable under exact-match letter scoring, so the reported accuracies carry an unquantified error floor, and individual letter mistakes are not always substantive errors (section 5.6). Question types within this regime are:

- **Closed:** the gold answer normalises to *yes* or *no* (two-way options).
- **Open:** any other gold answer (gold plus four distractors, five-way options).
- **Overall:** the union of closed and open items.

Table 4.6 reports the closed versus open mix after this conversion. These counts are those retained after loading and, for SLAKE, the English subset only. They therefore differ from the bilingual paper totals in Table 4.3 and, for Med-VQA-2019, from the official validation and test sizes.

Two consequences of the conversion matter for reading every accuracy in this thesis. First, because closed items offer two options and open items five, uniform random guessing on the test splits scores 0.367 on VQA-RAD, 0.238 on Med-VQA-2019, and 0.300 on SLAKE. Those levels, not zero, are the reference point for residual accuracy under the black- and shuffled-image ablations. Second, a bounded option set removes the paraphrase-matching problem of exact-match free-text scoring and lifts the floor

| Corpus | Split | $n$ | Closed | Open | Closed (%) | Open (%) |
|---|---|---|---|---|---|---|
| VQA-RAD | train | 1,793 | 940 | 853 | 52.4 | 47.6 |
| | test | 451 | 251 | 200 | 55.7 | 44.3 |
| Med-VQA-2019 | train | 12,792 | 1,294 | 11,498 | 10.1 | 89.9 |
| | validation | 1,998 | 202 | 1,796 | 10.1 | 89.9 |
| | test | 499 | 64 | 435 | 12.8 | 87.2 |
| SLAKE (EN) | train | 4,910 | 1,678 | 3,232 | 34.2 | 65.8 |
| | validation | 1,053 | 358 | 695 | 34.0 | 66.0 |
| | test | 1,061 | 355 | 706 | 33.5 | 66.5 |

Table 4.6: Closed versus open question types in the splits used for training and evaluation after multiple-choice conversion. Closed items have a yes/no gold answer, open items have any other gold answer. SLAKE is the English subset. Percentages are within each split.

from near zero to the levels just given. Accuracies reported here are therefore systematically higher than free-text results on the same corpora and are not comparable with published VQA-RAD, Med-VQA-2019, or SLAKE leaderboards.

**Verified pseudo-explanation.** For explanation-based supervised fine-tuning, each multiple-choice training item is expanded into candidate rationales and then filtered, following the teacher-verifier construction of Section 3.5. A teacher model receives the image and the gold option text (not the letter) and writes a structured block:

```
REGION: ...  OBSERVATION: ...  EXPLANATION: ...
```

By default, $N = 5$ rationales are sampled per item at temperature 0.7 with a maximum of 512 new tokens per rationale. Teachers used in this thesis are GPT-4o-mini (OpenAI Chat Completions API) and Qwen3-VL-8B (local generation). For GPT-4o-mini, the five variants are requested in one API call (`n = 5`) with image detail `low`, request timeout 120 s, and up to three retries. For Qwen3-VL-8B the same $N$ and temperature are used via independent sampled generations in `bfloat16`.
A Qwen2.5-VL-7B verifier then answers the multiple-choice question using only the rationale text and the question (no image), greedily emitting a single letter. A rationale is retained if and only if that letter matches the gold letter $y$. Empty or schema-invalid traces are dropped. Table 4.7 reports generated versus retained counts and pass rates by teacher and dataset. The resulting supervision therefore consists of verified rationale generations, not human-annotated CoT. Verified rows are the training input for Section 4.5.2.
Three properties of this construction bound what the filter achieves and are carried into the interpretation in Chapter 6. First, the pass rates count rationales, not items. With $N = 5$ samples per item, an item leaves the training set only if all five of its rationales fail, so item-level attrition is far smaller than the 1.1-8.4 % of rationales that are dropped. The filter thins the variants available per item rather than selecting which items are trained on, and *verified* therefore describes how the supervision was assembled, not a quality gate that removes hard cases.
Second, teacher coverage of the training split is complete on VQA-RAD and SLAKE, where 8,965 and 24,550 generated traces correspond at $N = 5$ to exactly the 1,793

| Dataset | Teacher | Generated | Retained | Pass rate |
|---|---|---|---|---|
| Med-VQA-2019 | GPT-4o-mini | 49,975 | 46,655 | 93.36 % |
| | Qwen3-VL-8B | 29,375 | 28,734 | 97.82 % |
| SLAKE | GPT-4o-mini | 24,550 | 22,483 | 91.58 % |
| | Qwen3-VL-8B | 24,550 | 23,811 | 96.99 % |
| VQA-RAD | GPT-4o-mini | 8,965 | 8,603 | 95.96 % |
| | Qwen3-VL-8B | 8,965 | 8,865 | 98.88 % |

Table 4.7: Text-only verification of pseudo-explanations by teacher and dataset (verifier: Qwen2.5-VL-7B, greedy). *Generated* counts all sampled rationales before filtering. *Retained* counts letter-matching traces used for SFT.

and 4,910 training items. On Med-VQA-2019 it is not: 49,975 traces cover 9,995 of the 12,792 training items for GPT-4o-mini (78.1 %), and 29,375 traces cover 5,875 items for Qwen3-VL-8B (45.9 %). The two Med-VQA-2019 teacher conditions are therefore trained on different amounts of data, and teacher effects on that corpus are confounded with training-set size.
Third, the verifier is itself one of the students, and so is one of the teachers: Qwen2.5-VL-7B verifies here and is fine-tuned in Section 4.5.2, and Qwen3-VL-8B is both a teacher and a student, which makes two of the twelve explanation-SFT cells self-distillation. Both identities are checked against the results in subsection 6.1.3. Because the retention criterion is whether the gold letter can be recovered from the rationale text without the image, the supervision also consists by construction of traces that state or entail their own answer, which is taken up where the explanation-first collapse is interpreted.

## 4.4 Shared Reporting Protocol

Before the individual experimental regimes are defined, this section fixes the reporting conventions that apply across both setups unless a protocol states otherwise: evaluation metrics, counterfactual image ablations, format-error analysis, and selective prediction. Background concepts for decoding, uncertainty, and selective prediction are introduced in Section 3.3, Section 3.6, and Section 3.7. Concrete decoding defaults and prompts are stated under the corresponding protocols in Section 4.5.

### 4.4.1 Evaluation metrics

On every test split of size $N$, metrics are reported separately for *Closed (yes/no)*, *Open-ended*, and *Overall* items. Score definitions are those of Section 3.6 and Section 3.7. This subsection records only which quantities appear in the tables and the shared reporting conventions.
Every evaluated item is decoded $K = 5$ times with stochastic sampling at temperature 0.7 and nucleus 0.9 (Table 4.8). The reported answer $\hat{y}_i$ is the majority vote over those $K$ samples, and entropy, confidence, and agreement are aggregated over the same samples. This holds identically in answer-only and in zeroshot JSON evaluation, so accuracy and the uncertainty columns are computed on a matched basis in both setups.

Every reported cell is a single run: there are no repeated seeds, confidence intervals, or significance tests, so differences of a few items are not treated as a ranking of protocols. The table columns are:

- **Accuracy.** Letter match Acc (Section 3.6.2).
- **Entropy.** Split mean of token entropy $H_i^{\text{tok}}$ (Section 3.6.2). This column is a descriptive average, not the AURC ranking score.
- **AURC.** Trapezoidal risk-coverage integral under ranking by combined uncertainty $U_{\text{combined}}$ (Section 3.7, Section 3.7). Because the combined score is rescaled separately on each dataset-split combination, AURC columns compare rows within one split and are not read across splits or corpora.

Vote entropy $H_i^{\text{vote}}$ (Section 3.6.2) is not a table column, it enters evaluation only through $U_{\text{combined}}$. Best accuracy and lowest AURC within a split are shown in bold. Missing AURC entries (–) indicate rows that do not yield a usable letter distribution for risk-coverage scoring. Missing metric cells in explanation SFT ablations are shown as –.

### 4.4.2 Image ablations

Counterfactual image ablations keep the question and gold letter fixed while replacing the paired image by a solid black field or by a shuffled image from another sample. The original-image setting keeps each question paired with its source radiograph. The scientifically relevant quantity is the drop in overall accuracy (and AURC) relative to that setting. Large drops indicate visual dependence, small drops suggest language priors or shortcuts. Stacked overall ablation tables report accuracy, letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC for answer-only, zeroshot, and explanation SFT. Explanation entropy and confidence are defined for explanation SFT. Answer-only and zeroshot leave those cells as –. Closed and open ablation splits are collected in Appendix A. Selected Med-VQA-2019 shuffle pairs of the mixed-order Qwen2.5-VL student are shown qualitatively in section 5.6.

### 4.4.3 Format error analysis

Alongside accuracy, every generative regime records format adherence which means the percentage of test samples for which a valid answer letter (answer-only) or a valid structured JSON block (explanation regimes) can be parsed. Low adherence identifies failed outputs rather than letter-level mistakes alone. Format-error tables for both setups are collected in Appendix A. The corresponding result subsections summarise notable adherence failures in the main text.

## 4.5 Experimental Regimes

This section defines the two experimental setups whose original-image results are reported by dataset in Chapter 5. Both share the multiple-choice letter-prediction task of Section 4.3.2, the student and baseline checkpoints of Section 4.2, and the shared reporting protocol of Section 4.4. Prompts, decoding budgets, and training hyperparameters are stated under each protocol below. Unless a result table or protocol states

otherwise, the shared defaults of Table 4.8 apply. Experiment 1 is letter-only generation. Experiment 2 is answer+explanation (zeroshot JSON and explanation SFT). Selective prediction is co-reported for both setups.

### 4.5.1 Experiment 1: Answer-only VQA baseline

Experiment 1 evaluates letter-only Medical VQA without explanations in the training target or evaluation prompt. Checkpoints (LLaVA-1.5-7B, LLaVA-Med v1.5, Qwen2.5-VL-7B, Qwen3-VL-8B) are compared under base Hugging Face weights and supervised fine-tuning on VQA-RAD, Med-VQA-2019, and SLAKE. The assistant target under SFT is the gold letter alone. Base evaluation uses off-the-shelf weights without fine-tuning and without an explanation schema. Reporting follows Section 4.4 (metrics, black-/shuffled-image ablations, format adherence). Selective prediction is co-reported with this setup (Section 4.5.4).

**Prompting.** The user message is the multiple-choice stem shared by all letter-only runs:

```
{question}
Options:
A) . . .
Answer with exactly one letter from A, B, . . . . Do not output
any other text.
```

Model-specific chat wrappers keep the same question and option list (Section 4.2).

**Training and inference defaults.** Letter-only SFT uses the gold letter as the sole assistant target. Default inference uses `bfloat16` precision and a short generation budget (on the order of eight new tokens). Uncertainty sampling for the shared metric columns uses the agreement settings in Table 4.8.

| Setting | Default |
|---|---|
| Agreement $K$ | 5 |
| Sampling temperature / top-$p$ | 0.7 / 0.9 |
| Combined-uncertainty mix $\alpha$ | 0.5 |
| Pseudo-explanations per item | 5 |
| Verify model | Qwen2.5-VL-7B, greedy |
| Typical SFT epochs / learning rate | 3 / $2 \times 10^{-5}$ |
| Explanation max length | 2048 |

Table 4.8: Default hyperparameters for the reported experimental pipeline (shared across both setups unless stated otherwise).

### 4.5.2 Experiment 2: Answer+explanation

Answer-only Medical VQA (Experiment 1) scores a single letter under a short decoding budget. That protocol is efficient and matches the letter-accuracy definition of correctness used throughout this thesis, but it exposes neither which image evidence the model claims to use nor how the letter is justified. Experiment 2 enriches

the same multiple-choice stem with an explicit textual trace: zeroshot JSON rationales (Section 4.5.2) or supervised `REGION/OBSERVATION/EXPLANATION` blocks after text-verified distillation (Section 4.5.2). On one side, these traces make spatial and clinical claims inspectable, support text-only verification of teacher data, and enable qualitative comparison of emission order (explanation-first versus answer-first) and of prompt-only versus SFT-based explanation without changing the primary letter-match metric. On the other side, they introduce a significant source of uncertainty influencing entropy and confidence measures. The following example illustrates a matched item under answer-only decoding versus explanation-based SFT (answer-first and explanation-first).

**Question:** "where are the kidney?"

**Options:** A) "adjacent to the spleen", B) "located in the abdomen", C) "not seen here", D) "behind the liver", E) "in the pelvic cavity"

**Gold letter:** "C" ("not seen here")

---

**1) Answer-only pred:** "B" ("located in the abdomen") → *incorrect*

---

**2) Pseudo-AF pred:** "C"
*Region:* Abdomen *Observation:* The kidneys are not visible in this axial CT scan of the abdomen. *Explanation:* The kidneys are typically located retroperitoneally, one on each side of the spine, and in this image, the area where the kidneys would be is obscured by the presence of the liver, stomach, and surrounding soft tissues. Additionally, the image does not show the lower abdomen where the kidneys are usually found, and the resolution or slice thickness may not have captured them. Therefore, the kidneys are not seen here.

---

**3) Pseudo-RF pred:** "C"
*Region:* Abdomen *Observation:* The kidneys are not visible in this axial CT image of the abdomen. *Explanation:* The image is a cross-sectional view of the upper abdomen, which includes structures such as the liver, stomach, and diaphragm, but the kidneys are typically located in the lower abdomen and are not visible in this particular slice. The kidneys are usually seen in axial CT scans at lower levels, often below the level of the liver and stomach, and in this image, the structures visible are more consistent with the upper abdominal cavity. Therefore, the kidneys are not seen here.

**Zeroshot JSON.** Zeroshot JSON isolates prompt-order and two-pass CoT effects without in-domain fine-tuning or distilled teachers. Models generate structured answer-rationale JSON under a schema with key `rationale` and a brief one-to-three-sentence medical rationale $\tilde{r}$, not the `REGION/OBSERVATION/EXPLANATION` template of Section 4.3.2. Let $a$ denote the answer letter.

Three zeroshot modes are distinguished by emission order:

$$y_{\mathrm{AF}}^{(0)} = (a,\ \tilde{r}) \qquad \text{(answer-first, AF)}, \tag{4.1}$$

$$y_{\mathrm{RF}}^{(0)} = (\tilde{r},\ a) \qquad \text{(rationale-first, RF)}, \tag{4.2}$$

$$y_{\mathrm{RF}}^{(2)} = \big(\tilde{r},\ (\tilde{r},\ a)\big) \qquad \text{(two-stage, 2S)}. \tag{4.3}$$

In the single-shot modes both fields are emitted in one generation. In the two-stage mode, stage 1 first generates a rationale $\tilde{r}$ without a letter. Stage 2 then conditions on that text and requests JSON with both fields ($y_{\mathrm{RF}}^{(2)}$). A two-stage letter-only variant (2S-L), in which stage 2 requests only the answer letter, is reported as an additional test in Table A.6 (Appendix A). MedVLM-R1 is included as an off-the-shelf medical explanation baseline with its native `<think>`/`<answer>` tag schema and is not evaluated under the JSON order variants. Letter aggregation and uncertainty averaging for zeroshot JSON runs follow Section 4.4.1. The shared answer-token locator of Section 3.6.2 defines $t^*$ only for letter-only generation (first generated token) and for explanation JSON (last option-letter token). It does not define $t^*$ inside a `<answer>` span, so MedVLM-R1 is scored for letter match by extracting a letter from the tag text, but token entropy, confidence, agreement, and AURC are left undefined rather than computed under a third, unmatched locator. The format-adherence column is likewise undefined, because that check is the JSON (or letter-only) contract, not the native tag wrapper. Reporting otherwise follows Section 4.4.

**Prompting.** The model must respond with only valid JSON using `answer`/`rationale` keys and a brief medical rationale (one to three sentences), or follow the two-stage rationale-then-answer procedure above. The multiple-choice stem and option list match Experiment 1. Only the output schema and stage instructions differ. MedVLM-R1 keeps its native tag wrapper at identical question content.

**Inference defaults.** Zeroshot runs use off-the-shelf weights (no in-domain SFT) with a larger generation budget than letter-only decoding so that a short rationale and the answer letter fit in one or two passes. Agreement samples reuse $K = 5$, temperature 0.7, and top-$p$ 0.9 from Table 4.8.

**Explanation SFT.** Explanation SFT fine-tunes student models on the text-verified teacher rationales constructed in Section 4.3.2. Student models are LLaVA-1.5-7B, Qwen2.5-VL-7B, and Qwen3-VL-8B. The assistant target is a JSON object with keys `explanation` and `answer`, where the explanation value is the verified `REGION/OBSERVATION/EXPLANATION` string from data preparation. Beyond letter accuracy, this protocol isolates how *explanation generation* reshapes predictive uncertainty. Because decoding is autoregressive, the order in which $r$ and $a$ are supervised changes whether the letter token distribution is conditioned on the model's own rationale (Section 3.4). That conditioning is the main handle on overconfidence under explanation-based SFT. Token entropy and confidence can track consistency with the emitted trace rather than residual doubt about correctness [35]. Selective scores (AURC) therefore remain first-class outcomes alongside accuracy (Section 3.7, Section 4.4.1).

Formally, let $r$ denote the structured explanation block and $a \in \{A, \ldots, E\}$ the answer letter. The assistant target $y$ takes one of the two single-order contracts:

$$y_{\mathrm{RF}} = (r,\ a) \qquad (\textit{explanation-first}), \tag{4.4}$$

$$y_{\mathrm{AF}} = (a,\ r) \qquad (\textit{answer-first / post-hoc}). \tag{4.5}$$

Both use the same JSON keys, only emission order differs. Their uncertainty implications differ systematically:

- **RF.** The letter is decoded after $r$, so $p(a \mid r, \ldots)$ can collapse once a fluent rationale commits the model to a letter. Expect low answer-token entropy, near-ceiling confidence, and a risk that AURC worsens even when accuracy rises (self-consistent overconfidence).
- **AF.** The letter is decoded before $r$, so the letter distribution is not conditioned on the justification. Expect higher residual entropy / lower confidence than RF for matched teacher-student pairs, and often better selective ranking of errors when token-likelihood scores are used.
- **RF+AF.** For Qwen2.5-VL-7B with the Qwen3 teacher, an additional mixed corpus trains on the *union* of verified RF and AF traces for the same stems. The student therefore sees both emission orders during SFT. The hypothesis is that mixed-order supervision can retain accuracy gains from explanation targets while mitigating the RF-specific collapse of uncertainty signals that pure RF fine-tuning induces. Original-image numbers for this additional test are reported in section 5.4 rather than in the stacked protocol tables.

Experimental axes are therefore teacher identity, emission-order regime (RF or AF, mixed RF+AF is an additional test), student architecture, and dataset. Reporting follows Section 4.4, with entropy and AURC co-reported so that AF and RF can be compared on calibration-relevant behaviour, not only on letter match.

**Training and inference defaults.** Typical training settings are three epochs, batch size 1, learning rate $2 \times 10^{-5}$, mixed precision, gradient checkpointing, and maximum sequence length 2048 for explanation targets (Table 4.8). Evaluation uses the same JSON schema and field order as training, with a larger generation budget than letter-only decoding. Agreement sampling again follows Table 4.8.

### 4.5.3 Qualitative SHAP overlays

Selected SHAP [99] overlays are used only as qualitative side material in section 5.7, not as a standalone experiment. They illustrate attended image regions on a handful of SLAKE test items under the generative regimes of Experiments 1 and 2, following the additive Shapley formalisation of Section 3.8. Because the maps are computed on individual examples, they do not support generalisable claims about visual grounding. The black-box $f$ is the MedVQA student under the parent regime's prompt and decoding contract (Sections 4.5.1-4.5.2). For answer-only overlays the explained scalar is the softmax probability of the predicted answer letter at the answer-token position $t^*$ (Section 3.6). For zeroshot and explanation SFT overlays it is the generated-sequence log-probability (answer letter together with the emitted explanation). Question and options stay fixed. Only image patches (superpixels / hierarchical blocks) are masked

so that budgeted coalition evaluations approximate patch contributions $\phi_j$ [99]. The same two SLAKE items are also overlaid after black-image and shuffled-image replacement, as qualitative counterparts rather than a dataset-level test. The resulting heatmaps are panel-wise normalised and are not an extra accuracy or AURC metric. They also do not prove clinical faithfulness [106]. Visual dependence at the dataset level remains the counterfactual image ablations of Section 4.4.2.

### 4.5.4 Selective prediction

Selective prediction re-uses the $K = 5$ completions already produced under Experiments 1 and 2 and asks, on the same closed, open, and overall splits, which items could be answered and which should be deferred [12].

Each item $i$ is assigned the combined uncertainty $U_{\text{combined},i}$ of Section 3.7: the convex mixture, with $\alpha = 0.5$ (Table 4.8), of min-max-normalised token entropy and vote entropy on that dataset-split combination. The model answers when $U_{\text{combined},i} \leq \tau$ and abstains otherwise. Sweeping $\tau$ over a dense grid, equivalently sorting items by increasing $U_{\text{combined}}$, yields the risk-coverage curve of Section 3.7. The table column is the trapezoidal integral of that curve (AURC). Lower AURC is better: less residual error at comparable coverage. Because both mixture components are rescaled per dataset-split combination, AURC columns compare rows within one split and are not read across splits or corpora (Section 4.4.1). If vote entropy is constant on a split, ranking falls back to token entropy. If both components are constant, $U_{\text{combined}}$ is constant and AURC equals full-set risk $1 - \text{Acc}$. Rows without a usable letter distribution (unparseable output, or an undefined $t^*$) receive – rather than an AURC.

# 5 | Results

This chapter reports the empirical evaluation of the protocols defined in Chapter 4. The stacked results tables use the shared protocol from Section 4.4 to report metrics separately for closed / open / overall splits with accuracy, entropy, and AURC. Confidence and agreement for the same original-image rows are reported in Appendix A (Tables A.8 and A.9). Formal score definitions are given in Section 3.6 and Section 3.7. Table highlighting and missing-AURC conventions likewise follow that protocol. Protocols, decoding defaults, and metric columns follow Section 4.5 and Section 4.4 unless a section states otherwise. Results are organised by dataset. Each of section 5.1-section 5.3 reports letter-prediction results for the generalist students on one corpus in the stacked results table alongside black- and shuffled-image ablations, and risk-coverage analysis. Blocks from top to bottom of each stacked table are Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). LLaVA-Med and MedVLM-R1 are not included here but reported in section 5.5. Mixed-order RF+AF for Qwen2.5-VL with the Qwen3 teacher is an additional test in section 5.4, the stacked blocks retain single-order RF and AF. Qualitative greedy outputs on one open VQA-RAD test item are collected in Appendix A (section A.3). Selected Med-VQA-2019 shuffle pairs with a letter-entropy rise and letter-confidence drop are in section 5.6. Qualitative SHAP overlays on selected SLAKE items are in section 5.7. Format adherence is discussed with LLaVA-Med and MedVLM-R1 in section 5.5.

Compact stacked overall ablation tables report answer-only, zeroshot, and explanation SFT in one table per corpus under black and shuffled images, with Acc., letter and explanation entropy and confidence, agreement, and AURC (Tables 5.2–5.6). Closed and open ablation splits are in Appendix A (Tables A.11–A.13). Original-image values of the explanation-SFT scores are in Table A.10 (Appendix A). A central reading of the explanation-based blocks is uncertainty under explanation generation, not accuracy alone.

Selective prediction re-uses the Experiment 1-2 generations and ranks items by combined uncertainty with $\alpha = 0.5$ (Section 4.5.4). The risk metric is letter match. Scalar AURC remain those of the stacked tables. Answer-only figures plot Base and SFT for LLaVA-1.5, Qwen2.5-VL, and Qwen3-VL. Zeroshot figures plot LLaVA-1.5, Qwen2.5-VL, and Qwen3-VL under AF / RF / 2S. Explanation SFT figures plot Qwen3-teacher students (LLaVA-1.5 and Qwen3-VL in RF and AF, Qwen2.5-VL in RF, AF, and RF+AF merged). GPT-4o-mini teacher rows are omitted. Colour encodes the checkpoint or JSON mode (legend under each figure). Interpretation and discussion are deferred to Chapter 6.

## 5.1 VQA-RAD

| | | | | Closed | | | Open | | | Overall | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Experiment 1 | | Model | Adaptation | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | | LLaVA-1.5 | Base | 0.542 | 0.792 | 0.392 | 0.360 | 1.182 | 0.439 | 0.461 | 0.965 | 0.392 |
| | | | SFT | 0.701 | 0.108 | 0.183 | 0.615 | 0.139 | 0.181 | 0.663 | 0.122 | 0.188 |
| | | Qwen2.5-VL | Base | 0.729 | 0.453 | 0.128 | 0.565 | 0.812 | 0.199 | 0.656 | 0.612 | 0.148 |
| | | | SFT | **0.757** | 0.272 | **0.116** | **0.700** | 0.370 | **0.117** | **0.732** | 0.315 | **0.115** |
| | | Qwen3-VL | Base | 0.717 | 0.166 | 0.124 | 0.615 | 0.311 | 0.181 | 0.672 | 0.230 | 0.145 |
| | | | SFT | 0.661 | 0.573 | 0.237 | 0.655 | 0.660 | 0.133 | 0.659 | 0.612 | 0.178 |
| Experiment 2 | | Model | Mode | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | | LLaVA-1.5 | AF | 0.530 | 1.434 | 0.458 | 0.345 | 0.803 | 0.480 | 0.448 | 1.154 | 0.475 |
| | | | RF | 0.538 | 0.691 | 0.467 | 0.240 | 0.472 | 0.671 | 0.406 | 0.594 | 0.516 |
| | | | 2S | 0.526 | 0.367 | 0.467 | 0.280 | 0.367 | 0.671 | 0.417 | 0.367 | 0.533 |
| | | Qwen2.5-VL | AF | 0.717 | 0.485 | 0.129 | 0.570 | 0.729 | 0.192 | 0.652 | 0.593 | 0.150 |
| | | | RF | **0.761** | 0.122 | **0.116** | 0.610 | 0.143 | 0.212 | **0.694** | 0.131 | 0.159 |
| | | | 2S | 0.753 | 0.047 | 0.188 | 0.575 | 0.051 | 0.313 | 0.674 | 0.049 | 0.289 |
| | | Qwen3-VL | AF | 0.733 | 0.139 | 0.120 | **0.620** | 0.223 | **0.186** | 0.683 | 0.176 | **0.146** |
| | | | RF | 0.745 | 0.004 | 0.157 | 0.580 | 0.003 | 0.277 | 0.672 | 0.004 | 0.197 |
| | | | 2S | 0.749 | 0.002 | 0.220 | 0.615 | 0.018 | 0.408 | 0.690 | 0.009 | 0.253 |
| Experiment 2 | Teacher | Model | Order | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | GPT-4o | LLaVA-1.5 | RF | 0.713 | 0.000 | 0.261 | 0.655 | 0.000 | 0.239 | 0.687 | 0.000 | 0.244 |
| | | | AF | 0.713 | 0.000 | 0.244 | 0.650 | 0.000 | 0.274 | 0.685 | 0.000 | 0.260 |
| | | Qwen2.5-VL | RF | **0.761** | 0.001 | 0.196 | 0.675 | 0.004 | 0.142 | 0.723 | 0.002 | 0.170 |
| | | | AF | 0.733 | 0.104 | 0.137 | 0.660 | 0.196 | 0.150 | 0.701 | 0.145 | 0.140 |
| | | Qwen3-VL | RF | 0.705 | 0.002 | 0.244 | **0.735** | 0.011 | 0.177 | 0.718 | 0.006 | 0.219 |
| | | | AF | 0.661 | 0.533 | 0.176 | 0.670 | 0.550 | 0.127 | 0.665 | 0.541 | 0.153 |
| | Qwen3 | LLaVA-1.5 | RF | 0.697 | 0.000 | 0.271 | 0.655 | 0.000 | 0.207 | 0.679 | 0.000 | 0.235 |
| | | | AF | 0.673 | 0.000 | 0.235 | 0.615 | 0.000 | 0.257 | 0.648 | 0.000 | 0.253 |
| | | Qwen2.5-VL | RF | 0.757 | 0.000 | 0.205 | 0.645 | 0.015 | 0.184 | 0.707 | 0.007 | 0.197 |
| | | | AF | **0.761** | 0.274 | **0.124** | 0.685 | 0.354 | **0.109** | **0.727** | 0.309 | **0.113** |
| | | Qwen3-VL | RF | 0.641 | 0.013 | 0.295 | 0.630 | 0.056 | 0.193 | 0.636 | 0.032 | 0.264 |
| | | | AF | 0.649 | 0.610 | 0.223 | 0.655 | 0.748 | 0.136 | 0.652 | 0.671 | 0.177 |

Table 5.1: VQA-RAD stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema.

**Answer-only.** The answer-only block of Table 5.1 summarises answer-only results on VQA-RAD. On the closed split, Qwen2.5-VL-7B with SFT records the highest accuracy at 0.757 and the lowest AURC at 0.116. Its base checkpoint already reaches 0.729 accuracy with AURC 0.128. Qwen3-VL-8B base is next among base models (0.717 accuracy, AURC 0.124). SFT reduces closed accuracy to 0.661 and raises AURC to 0.237. LLaVA-1.5-7B improves from base 0.542 to SFT 0.701 (AURC from 0.392 to 0.183).
On open-ended items, Qwen2.5-VL SFT again leads with accuracy 0.700 and AURC 0.117. Qwen3-VL SFT has open accuracy 0.655 (AURC 0.133), above its base 0.615 (AURC 0.181). LLaVA-1.5 SFT rises from base 0.360 to 0.615 (AURC 0.181).
Overall, Qwen2.5-VL SFT is the bold entry for both accuracy (0.732) and AURC (0.115). Qwen3-VL base overall accuracy 0.672 exceeds its SFT overall accuracy (0.659). Across models, SFT typically lowers entropy and raises confidence and agreement relative to the corresponding base row when accuracy also rises (e.g. LLaVA-1.5 SFT: entropy

0.122, confidence 0.955, agreement 0.979). Original-image confidence and agreement for every answer-only row are in Table A.8 (Appendix A).

**Zeroshot explanation.** The zeroshot block of Table 5.1 reports zeroshot JSON results on VQA-RAD. On closed questions, Qwen2.5-VL RF has the highest accuracy (0.761) and the lowest AURC (0.116). Qwen3-VL AF is next on AURC (0.120) at accuracy 0.733. Qwen3-VL closed accuracies across modes lie between 0.733 and 0.749, with RF/2S entropy at or below 0.005 and confidence at or above 0.997. Qwen2.5-VL AF/RF/2S closed accuracies are 0.717-0.761. 2S raises agreement to 0.993 and lowers entropy to 0.047, while AURC rises to 0.188 relative to AF AURC 0.129. Original-image confidence and agreement for every zeroshot row are in Table A.9 (Appendix A). LLaVA-1.5 closed accuracies span 0.526-0.538. RF has the highest among them (0.538). On open-ended items, Qwen3-VL AF has the highest accuracy (0.620) and the lowest AURC (0.186). Qwen2.5-VL RF open accuracy is 0.610 (AURC 0.212), and AF is 0.570 (AURC 0.192). LLaVA-1.5 open accuracies remain between 0.240 and 0.345, with AURC from 0.480 to 0.671.
Overall, Qwen2.5-VL RF has the highest accuracy (0.694). Qwen3-VL AF has the lowest AURC (0.146). Qwen3-VL 2S overall accuracy is 0.690 with AURC 0.253. Across Qwen rows, RF and multi-stage modes systematically report lower entropy and higher confidence than AF, while AF often records the lower AURC for a given model.

**Explanation SFT.** The explanation SFT block of Table 5.1 reports explanation-based SFT on VQA-RAD. On closed questions, Qwen2.5-VL with Qwen3 AF and Qwen2.5-VL with GPT-4o RF share the highest accuracy (0.761). Qwen2.5-VL with Qwen3 AF has the lowest closed AURC (0.124). LLaVA-1.5 closed accuracies lie between 0.673 and 0.713 across teacher-order pairs. Qwen3-VL closed accuracies are 0.641-0.705. AF rows show higher entropy and lower confidence than RF rows for the same teacher (e.g. GPT-4o AF entropy 0.533, confidence 0.749 vs. RF entropy 0.002, confidence 1.000).
On open-ended items, Qwen3-VL with GPT-4o RF has the highest accuracy (0.735). Qwen2.5-VL with Qwen3 AF has the lowest open AURC (0.109) at accuracy 0.685. LLaVA-1.5 open accuracies are 0.615-0.655. Several RF rows report entropy at or near 0.000 with confidence 1.000 and AURC above 0.15.
Overall, Qwen2.5-VL with Qwen3 AF has the highest accuracy (0.727) and the lowest overall AURC (0.113). Qwen2.5-VL with GPT-4o RF overall accuracy is 0.723 (AURC 0.170). LLaVA-1.5 overall accuracies range from 0.648 to 0.687. Qwen3-VL from 0.636 to 0.718. On uncertainty, AF diverges from RF: for a fixed student, AF often yields lower overall AURC than RF when confidence is not saturated (e.g. Qwen2.5-VL + Qwen3: AF AURC 0.113 vs. RF 0.197), while many RF rows sit at confidence 1.000 with higher AURC. The mixed-order additional test for this student-teacher pair is reported in section 5.4.

**Image ablations.** Table 5.2 stacks black-image and shuffled-image replacements for all three protocols. LLaVA-1.5 is nearly invariant on this corpus: answer-only SFT retains overall accuracy 0.658 against original-image 0.663 (drop 0.005) and records the highest black-image accuracy and lowest AURC (0.220). Shuffled SFT is 0.641 (drop 0.022). Explanation-SFT AF is 0.641 against 0.648 (drop 0.007). Zeroshot RF stays at 0.406. Qwen drops are larger and grow from explanation SFT into zeroshot. Qwen2.5-VL answer-only SFT falls from 0.732 to 0.619 (drop 0.113) under both black

and shuffled images, while zeroshot RF falls from 0.694 to 0.523 (drop 0.171) and AF from 0.652 to 0.494 (drop 0.158). Qwen3-VL zeroshot AF drops from 0.683 to 0.492 (drop 0.191) and 2S from 0.690 to 0.448 (drop 0.242). Explanation-SFT AF drops from 0.652 to 0.548 (drop 0.104). Qwen2.5-VL explanation-SFT AF falls from 0.727 to 0.632 (drop 0.095) and has the lowest black-image AURC in that block (0.231). Mixed-order RF+AF falls from 0.741 to 0.632 (drop 0.109). Shuffled images repeat the Qwen pattern (zeroshot RF 0.512 and explanation-SFT AF 0.599 for Qwen2.5-VL, zeroshot AF 0.517 and explanation-SFT AF 0.561 for Qwen3-VL). Relative to Table A.10, RF letter entropy stays at or near 0.000 with letter confidence at or above 0.992, while explanation entropy remains higher (LLaVA-1.5 RF 0.136, Qwen2.5-VL RF 0.293, Qwen3-VL RF 0.521). AF raises letter entropy for the Qwen students (Qwen2.5-VL 0.473, Qwen3-VL 0.801) and lowers letter confidence.

| Experiment 1 | Image | Model | Adaptation | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer-only | Black images | LLaVA-1.5 | Base | 0.419 | 1.058 | – | 0.551 | – | 0.734 | 0.471 |
| | | | SFT | **0.658** | 0.109 | – | 0.959 | – | 0.980 | **0.220** |
| | | Qwen2.5-VL | Base | 0.470 | 0.877 | – | 0.600 | – | 0.752 | 0.407 |
| | | | SFT | 0.619 | 0.437 | – | 0.815 | – | 0.899 | 0.254 |
| | | Qwen3-VL | Base | 0.494 | 0.316 | – | 0.877 | – | 0.938 | 0.396 |
| | | | SFT | 0.601 | 0.691 | – | 0.680 | – | 0.811 | 0.241 |
| | Shuffled images | LLaVA-1.5 | Base | 0.386 | 0.976 | – | 0.589 | – | 0.752 | 0.500 |
| | | | SFT | **0.641** | 0.122 | – | 0.955 | – | 0.980 | **0.212** |
| | | Qwen2.5-VL | Base | 0.505 | 0.589 | – | 0.751 | – | 0.847 | 0.419 |
| | | | SFT | 0.619 | 0.269 | – | 0.889 | – | 0.941 | 0.296 |
| | | Qwen3-VL | Base | 0.497 | 0.246 | – | 0.906 | – | 0.955 | 0.448 |
| | | | SFT | 0.634 | 0.646 | – | 0.708 | – | 0.831 | 0.291 |

| Experiment 2 | Image | Model | Mode | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Zeroshot | Black images | LLaVA-1.5 | AF | 0.399 | 1.254 | – | 0.614 | – | 0.711 | 0.609 |
| | | | RF | 0.397 | 0.666 | – | 0.842 | – | 0.672 | 0.527 |
| | | | 2S | 0.446 | 0.399 | – | 0.888 | – | 0.783 | 0.526 |
| | | Qwen2.5-VL | AF | 0.494 | 0.857 | – | 0.608 | – | 0.749 | 0.409 |
| | | | RF | **0.523** | 0.169 | – | 0.943 | – | 0.780 | **0.389** |
| | | | 2S | 0.517 | 0.063 | – | 0.981 | – | 0.988 | 0.493 |
| | | Qwen3-VL | AF | 0.492 | 0.241 | – | 0.903 | – | 0.947 | 0.426 |
| | | | RF | 0.444 | 0.096 | – | 0.966 | – | 0.916 | 0.453 |
| | | | 2S | 0.448 | 0.175 | – | 0.938 | – | 0.959 | 0.461 |
| | Shuffled images | LLaVA-1.5 | AF | 0.410 | 1.155 | – | 0.666 | – | 0.721 | 0.546 |
| | | | RF | 0.406 | 0.590 | – | 0.857 | – | 0.715 | 0.547 |
| | | | 2S | 0.399 | 0.369 | – | 0.902 | – | 0.858 | 0.531 |
| | | Qwen2.5-VL | AF | **0.519** | 0.578 | – | 0.766 | – | 0.854 | **0.405** |
| | | | RF | 0.512 | 0.208 | – | 0.932 | – | 0.838 | 0.413 |
| | | | 2S | 0.479 | 0.168 | – | 0.945 | – | 0.958 | 0.472 |
| | | Qwen3-VL | AF | 0.517 | 0.184 | – | 0.927 | – | 0.963 | 0.447 |
| | | | RF | 0.488 | 0.068 | – | 0.975 | – | 0.919 | 0.433 |
| | | | 2S | 0.501 | 0.083 | – | 0.969 | – | 0.979 | 0.425 |

| Experiment 2 | Image | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.616 | 0.000 | 0.136 | 1.000 | 0.907 | 0.896 | 0.294 |
| | | | AF | **0.641** | 0.000 | 0.140 | 1.000 | 0.907 | 0.936 | 0.316 |
| | | Qwen2.5-VL | RF | 0.603 | 0.004 | 0.293 | 0.998 | 0.818 | 0.899 | 0.342 |
| | | | AF | 0.632 | 0.473 | 0.638 | 0.811 | 0.667 | 0.900 | **0.231** |
| | | | RF+AF | 0.632 | 0.075 | 0.184 | 0.971 | 0.873 | 0.890 | 0.234 |
| | | Qwen3-VL | RF | 0.539 | 0.028 | 0.521 | 0.992 | 0.708 | 0.804 | 0.382 |
| | | | AF | 0.548 | 0.801 | 0.585 | 0.643 | 0.669 | 0.783 | 0.327 |
| | Shuffled images | LLaVA-1.5 | RF | **0.652** | 0.000 | 0.138 | 1.000 | 0.905 | 0.891 | **0.261** |
| | | | AF | 0.625 | 0.000 | 0.144 | 1.000 | 0.904 | 0.942 | 0.352 |
| | | Qwen2.5-VL | RF | 0.585 | 0.009 | 0.295 | 0.997 | 0.814 | 0.922 | 0.344 |
| | | | AF | 0.599 | 0.333 | 0.621 | 0.879 | 0.672 | 0.949 | 0.353 |
| | | | RF+AF | 0.577 | 0.058 | 0.189 | 0.978 | 0.868 | 0.918 | 0.349 |
| | | Qwen3-VL | RF | 0.512 | 0.041 | 0.506 | 0.987 | 0.710 | 0.835 | 0.422 |
| | | | AF | 0.561 | 0.684 | 0.591 | 0.699 | 0.664 | 0.826 | 0.365 |

Table 5.2: VQA-RAD stacked overall ablations for all generative protocols. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as −. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.11 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

### 5.1.1 Risk-coverage analysis

Every risk-coverage figure in section 5.1-section 5.3 plots the closed and open splits only. Overall AURC values quoted alongside them are read from the corresponding stacked results table rather than from a plotted curve.
Figure 5.1 reports answer-only VQA-RAD. On the closed split, Qwen2.5-VL SFT has the lowest AURC (0.116), then Qwen3-VL Base (0.124) and Qwen2.5-VL Base (0.128). LLaVA-1.5 Base is the highest closed curve (0.392). On open items, Qwen2.5-VL SFT remains lowest (0.117), then Qwen3-VL SFT (0.133) and LLaVA-1.5 SFT (0.181). LLaVA-1.5 Base open AURC is 0.439. Overall, Qwen2.5-VL SFT is lowest (0.115), ahead of Qwen3-VL Base (0.145) and Qwen2.5-VL Base (0.148). LLaVA-1.5 Base has the highest overall AURC (0.392).

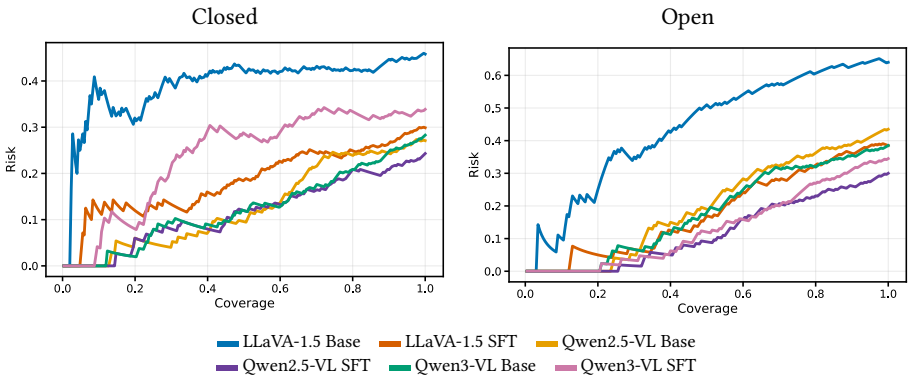



Figure 5.1: Answer-only risk–coverage curves on VQA-RAD (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL SFT (0.116).

Figure 5.2 reports zeroshot VQA-RAD. On the closed split, Qwen2.5-VL RF has the lowest AURC (0.116), then Qwen3-VL AF (0.120) and Qwen2.5-VL AF (0.129). LLaVA-1.5 RF and 2S are the highest closed curves (0.467). On open items, Qwen3-VL AF remains lowest (0.186), then Qwen2.5-VL AF (0.192) and Qwen2.5-VL RF (0.212). LLaVA-1.5 RF and 2S open AURC are 0.671. Overall, Qwen3-VL AF is lowest (0.146), just ahead of Qwen2.5-VL AF (0.150). Qwen2.5-VL RF overall AURC is 0.159. LLaVA-1.5 2S has the highest overall AURC (0.533).

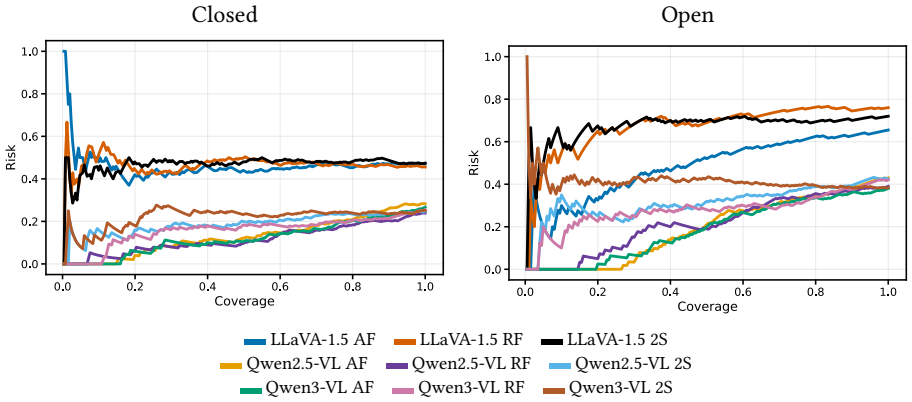



Figure 5.2: Zeroshot JSON risk–coverage curves on VQA-RAD (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL RF (0.116).

Figure 5.3 reports explanation SFT on VQA-RAD. On the closed split, Qwen2.5-VL AF has the lowest AURC (0.124), then Qwen2.5-VL RF+AF merged (0.143) and Qwen2.5-VL RF (0.205). Qwen3-VL RF is the highest closed curve (0.295). On open items, Qwen2.5-VL AF remains lowest (0.109), then Qwen2.5-VL RF+AF merged (0.129) and Qwen3-VL AF (0.136). LLaVA-1.5 AF open AURC is 0.257. Overall, Qwen2.5-VL AF is lowest (0.113), ahead of Qwen2.5-VL RF+AF merged (0.131). For both Qwen students, AF lies below the matching RF curve. LLaVA-1.5 is the exception, with AF above RF overall (0.253 against 0.235) and on open items (0.257 against 0.207). Qwen3-VL RF has the highest overall AURC among the plotted slots (0.264).

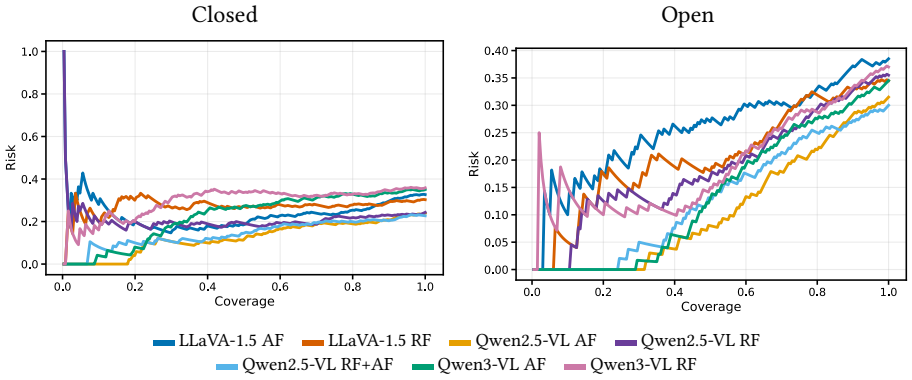



Figure 5.3: Explanation SFT risk–coverage curves on VQA-RAD for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL AF (0.124).

## 5.2 Med-VQA-2019

Table 5.3 collects original-image letter-prediction results for the generalist student models on Med-VQA-2019, split by closed, open, and overall items.

| | | | | Closed | | | Open | | | Overall | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Experiment 1 | | Model | Adaptation | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | | LLaVA-1.5 | Base | 0.375 | 0.659 | 0.459 | 0.474 | 0.980 | 0.329 | 0.461 | 0.938 | 0.347 |
| | | | SFT | 0.688 | 0.193 | 0.118 | 0.892 | 0.052 | 0.022 | 0.866 | 0.070 | 0.028 |
| | | Qwen2.5-VL | Base | 0.656 | 0.362 | 0.165 | 0.651 | 0.590 | 0.161 | 0.651 | 0.560 | 0.162 |
| | | | SFT | **0.844** | 0.253 | **0.080** | 0.901 | 0.194 | 0.019 | **0.894** | 0.201 | 0.023 |
| | | Qwen3-VL | Base | 0.719 | 0.201 | 0.110 | 0.681 | 0.197 | 0.129 | 0.685 | 0.197 | 0.128 |
| | | | SFT | 0.688 | 0.398 | 0.117 | **0.913** | 0.261 | **0.016** | 0.884 | 0.279 | **0.022** |
| Experiment 2 | | Model | Mode | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | | LLaVA-1.5 | AF | 0.453 | 1.541 | 0.456 | 0.485 | 0.729 | 0.379 | 0.481 | 0.833 | 0.389 |
| | | | RF | 0.453 | 0.562 | 0.611 | 0.416 | 0.510 | 0.480 | 0.421 | 0.517 | 0.495 |
| | | | 2S | 0.438 | 0.402 | 0.518 | 0.363 | 0.359 | 0.602 | 0.373 | 0.365 | 0.589 |
| | | Qwen2.5-VL | AF | 0.625 | 0.377 | 0.164 | 0.658 | 0.522 | 0.156 | 0.653 | 0.504 | 0.159 |
| | | | RF | 0.656 | 0.187 | 0.167 | 0.646 | 0.123 | 0.197 | 0.647 | 0.131 | 0.196 |
| | | | 2S | 0.688 | 0.025 | 0.329 | 0.618 | 0.060 | 0.322 | 0.627 | 0.056 | 0.324 |
| | | Qwen3-VL | AF | 0.734 | 0.154 | **0.119** | **0.687** | 0.142 | **0.134** | **0.693** | 0.143 | **0.132** |
| | | | RF | 0.734 | 0.000 | 0.145 | 0.678 | 0.004 | 0.201 | 0.685 | 0.003 | 0.192 |
| | | | 2S | **0.750** | 0.000 | 0.234 | 0.664 | 0.009 | 0.298 | 0.675 | 0.008 | 0.282 |
| Experiment 2 | Teacher | Model | Order | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | GPT-4o | LLaVA-1.5 | RF | 0.609 | 0.001 | 0.350 | 0.724 | 0.001 | 0.155 | 0.709 | 0.001 | 0.168 |
| | | | AF | 0.656 | 0.000 | 0.291 | 0.818 | 0.000 | 0.059 | 0.798 | 0.000 | 0.076 |
| | | Qwen2.5-VL | RF | 0.766 | 0.000 | 0.081 | 0.897 | 0.006 | 0.037 | 0.880 | 0.005 | 0.056 |
| | | | AF | 0.859 | 0.118 | **0.043** | 0.887 | 0.113 | 0.054 | 0.884 | 0.113 | 0.054 |
| | | Qwen3-VL | RF | **0.875** | 0.000 | 0.157 | 0.897 | 0.001 | 0.053 | 0.894 | 0.001 | 0.070 |
| | | | AF | 0.672 | 0.454 | 0.138 | 0.839 | 0.367 | 0.027 | 0.818 | 0.378 | 0.036 |
| | Qwen3 | LLaVA-1.5 | RF | 0.609 | 0.000 | 0.454 | 0.782 | 0.000 | 0.088 | 0.760 | 0.000 | 0.106 |
| | | | AF | 0.813 | 0.000 | 0.083 | 0.890 | 0.000 | 0.031 | 0.880 | 0.000 | 0.037 |
| | | Qwen2.5-VL | RF | 0.813 | 0.000 | 0.143 | 0.903 | 0.005 | 0.026 | 0.892 | 0.004 | 0.043 |
| | | | AF | 0.859 | 0.134 | 0.068 | **0.910** | 0.093 | 0.020 | **0.904** | 0.098 | 0.024 |
| | | Qwen3-VL | RF | 0.703 | 0.000 | 0.294 | 0.901 | 0.004 | 0.037 | 0.876 | 0.004 | 0.090 |
| | | | AF | 0.828 | 0.364 | 0.055 | 0.903 | 0.213 | **0.018** | 0.894 | 0.232 | **0.021** |

Table 5.3: Med-VQA-2019 stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema.

**Answer-only.** The answer-only block of Table 5.3 reports the same answer-only matrix on Med-VQA-2019. On closed questions, Qwen2.5-VL SFT has the highest accuracy (0.844) and the lowest closed AURC (0.080). LLaVA-1.5 SFT reaches closed accuracy 0.688 (AURC 0.118) from base 0.375. Qwen3-VL base closed accuracy is 0.719 (AURC 0.110). SFT is 0.688 (AURC 0.117).

On open-ended items absolute accuracies are higher for the stronger adaptations. Qwen3-VL SFT has the highest open accuracy (0.913) and the lowest open AURC (0.016). Qwen2.5-VL SFT open accuracy is 0.901 (AURC 0.019). LLaVA-1.5 SFT reaches 0.892 with AURC 0.022.

Overall, Qwen2.5-VL SFT reports the highest accuracy (0.894). Qwen3-VL SFT has the lowest overall AURC (0.022) at accuracy 0.884. LLaVA-1.5 SFT overall accuracy is

0.866 (AURC 0.028). Base overall accuracies span 0.461 (LLaVA-1.5) to 0.685 (Qwen3-VL). Entropy on strong fine-tuned rows is typically below 0.3 with confidence above 0.89 and agreement above 0.94 (Table A.8).

**Zeroshot explanation.** The zeroshot block of Table 5.3 reports zeroshot JSON results on Med-VQA-2019. On closed questions, Qwen3-VL 2S has the highest accuracy (0.750). Qwen3-VL AF has the lowest closed AURC (0.119). Qwen3-VL RF matches AF accuracy at 0.734 with entropy 0.000 and confidence 1.000, but AURC rises to 0.145. Qwen2.5-VL closed accuracies increase from AF 0.625 to 2S 0.688. AURC is lowest under AF (0.164) and highest under 2S (0.329). LLaVA-1.5 closed accuracies cluster near 0.44-0.45 across modes.
On open-ended items, Qwen3-VL AF has the highest accuracy (0.687) and the lowest AURC (0.134). Qwen2.5-VL AF open accuracy is 0.658 (AURC 0.156), above RF 0.646 (AURC 0.197) and 2S (0.618, AURC 0.322). LLaVA-1.5 open accuracies range from 0.363 (2S) to 0.485 (AF).
Overall, Qwen3-VL AF is bold for both accuracy (0.693) and AURC (0.132). Qwen2.5-VL AF overall accuracy is 0.653 (AURC 0.159), ahead of its RF and two-stage rows on AURC. LLaVA-1.5 overall accuracies are 0.373-0.481. As on VQA-RAD, Qwen RF/2S rows show near-ceiling confidence and agreement with higher AURC than the corresponding AF row.

**Explanation SFT.** The explanation SFT block of Table 5.3 reports explanation-based SFT on Med-VQA-2019. On closed questions, Qwen3-VL with GPT-4o RF has the highest accuracy (0.875). Qwen2.5-VL with GPT-4o AF has the lowest closed AURC (0.043) at accuracy 0.859. Qwen2.5-VL with Qwen3 AF matches that closed accuracy (0.859) at AURC 0.068. LLaVA-1.5 closed accuracies span 0.609-0.813, with Qwen3 AF the highest of that group. Qwen3-VL with Qwen3 AF closed accuracy is 0.828 (AURC 0.055), and RF is 0.703 (AURC 0.294).
On open-ended items, Qwen2.5-VL with Qwen3 AF has the highest accuracy (0.910). Qwen3-VL with Qwen3 AF has the lowest open AURC (0.018) at accuracy 0.903. Qwen2.5-VL with Qwen3 RF open accuracy is 0.903 (AURC 0.026). LLaVA-1.5 open accuracies range from 0.724 (GPT-4o RF) to 0.890 (Qwen3 AF). Most RF rows again report entropy near zero and confidence 1.000.
Overall, Qwen2.5-VL with Qwen3 AF has the highest accuracy (0.904). Qwen3-VL with Qwen3 AF has the lowest AURC (0.021) at accuracy 0.894. Qwen2.5-VL with GPT-4o AF overall accuracy is 0.884 (AURC 0.054). LLaVA-1.5 overall accuracies range from 0.709 to 0.880, with Qwen3 AF the strongest LLaVA-1.5 row. Compared with answer-only Med-VQA-2019 fine-tunes in Table 5.3, the top explanation-based overall accuracies sit in a similar band ($\approx$ 0.89-0.90), while several RF configurations show higher AURC at saturated confidence. The mixed-order additional test for Qwen2.5-VL with the Qwen3 teacher is reported in section 5.4.

**Image ablations.** Table 5.4 stacks black-image and shuffled-image replacements for all three protocols. Qwen base and zeroshot rows drop more than the matching SFT rows, and LLaVA-1.5 is less invariant than on VQA-RAD. Answer-only SFT retains more of the original-image score: Qwen2.5-VL and Qwen3-VL fall from 0.894 and 0.884 to 0.796 (drops 0.098 and 0.088), LLaVA-1.5 from 0.866 to 0.771. The matching base drops are larger (Qwen2.5-VL 0.651 to 0.347, drop 0.304, and Qwen3-VL 0.685 to 0.419, drop 0.266). Zeroshot AF is closer to those base gaps: Qwen2.5-VL from 0.653 to

0.353 (drop 0.300), Qwen3-VL from 0.693 to 0.417 (drop 0.276), Qwen2.5-VL 2S to 0.329, and LLaVA-1.5 from 0.481 to 0.309 (drop 0.172). Explanation SFT usually sits between the two: Qwen2.5-VL AF falls from 0.904 to 0.742 (drop 0.162) and RF from 0.892 to 0.738 (drop 0.154). LLaVA-1.5 RF is essentially flat (0.768 against original-image 0.760) while AF falls from 0.880 to 0.782 (drop 0.098). Qwen3-VL RF drops from 0.876 to 0.736 (drop 0.140). Qwen3-VL AF is the smallest explanation-SFT drop of the Qwen rows (0.894 to 0.774, drop 0.120) and records the lowest black-image overall AURC (0.074), driven by its open split (accuracy 0.814, AURC 0.052) against a closed split at chance (0.500, AURC 0.357). RF letter entropy stays at or below 0.005 with letter confidence at or above 0.998. AF leaves substantial residual letter entropy (Qwen2.5-VL 0.288, Qwen3-VL 0.469). Shuffled images repeat the ranking: answer-only Qwen2.5-VL base 0.313 (drop 0.338) and Qwen3-VL base 0.373 (drop 0.312), with LLaVA-1.5 SFT still relatively high at 0.792 (drop 0.074), zeroshot Qwen2.5-VL AF 0.339 (drop 0.314) and Qwen3-VL AF 0.377 (drop 0.316), with shuffled closed accuracies for Qwen2.5-VL 2S and Qwen3-VL AF/RF/2S sitting at 0.547 (Table A.12), which is 35/64, the majority closed label, so those rows are constant predictors of the more frequent yes/no class rather than a ranked result. Explanation-SFT Qwen2.5-VL AF is 0.707 (drop 0.197) and LLaVA-1.5 AF 0.661 (drop 0.219).

| Experiment 1 | Image | Model | Adaptation | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer-only | Black images | LLaVA-1.5 | Base | 0.297 | 1.205 | – | 0.519 | – | 0.707 | 0.662 |
| | | | SFT | 0.771 | 0.180 | – | 0.932 | – | 0.967 | **0.056** |
| | | Qwen2.5-VL | Base | 0.347 | 1.206 | – | 0.499 | – | 0.668 | 0.543 |
| | | | SFT | **0.796** | 0.401 | – | 0.858 | – | 0.939 | 0.068 |
| | | Qwen3-VL | Base | 0.419 | 0.584 | – | 0.773 | – | 0.888 | 0.494 |
| | | | SFT | **0.796** | 0.453 | – | 0.831 | – | 0.911 | 0.058 |
| | Shuffled images | LLaVA-1.5 | Base | 0.313 | 0.992 | – | 0.618 | – | 0.792 | 0.661 |
| | | | SFT | **0.792** | 0.093 | – | 0.965 | – | 0.984 | **0.053** |
| | | Qwen2.5-VL | Base | 0.313 | 0.623 | – | 0.762 | – | 0.864 | 0.650 |
| | | | SFT | 0.740 | 0.303 | – | 0.890 | – | 0.945 | 0.104 |
| | | Qwen3-VL | Base | 0.373 | 0.269 | – | 0.895 | – | 0.945 | 0.648 |
| | | | SFT | 0.746 | 0.363 | – | 0.867 | – | 0.937 | 0.090 |

| Experiment 2 | Image | Model | Mode | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Zeroshot | Black images | LLaVA-1.5 | AF | 0.309 | 0.923 | – | 0.728 | – | 0.667 | 0.699 |
| | | | RF | 0.267 | 0.622 | – | 0.851 | – | 0.595 | 0.684 |
| | | | 2S | 0.279 | 0.406 | – | 0.883 | – | 0.785 | 0.715 |
| | | Qwen2.5-VL | AF | 0.353 | 1.125 | – | 0.534 | – | 0.682 | 0.523 |
| | | | RF | 0.401 | 0.151 | – | 0.948 | – | 0.721 | 0.577 |
| | | | 2S | 0.329 | 0.043 | – | 0.985 | – | 0.991 | 0.729 |
| | | Qwen3-VL | AF | **0.417** | 0.445 | – | 0.826 | – | 0.905 | **0.480** |
| | | | RF | 0.319 | 0.266 | – | 0.909 | – | 0.818 | 0.633 |
| | | | 2S | 0.313 | 0.307 | – | 0.894 | – | 0.926 | 0.606 |
| | Shuffled images | LLaVA-1.5 | AF | 0.335 | 0.846 | – | 0.763 | – | 0.735 | 0.655 |
| | | | RF | 0.295 | 0.497 | – | 0.884 | – | 0.660 | 0.682 |
| | | | 2S | 0.273 | 0.366 | – | 0.902 | – | 0.808 | 0.684 |
| | | Qwen2.5-VL | AF | 0.339 | 0.576 | – | 0.784 | – | 0.874 | 0.648 |
| | | | RF | 0.353 | 0.219 | – | 0.929 | – | 0.865 | 0.674 |
| | | | 2S | 0.335 | 0.191 | – | 0.935 | – | 0.959 | 0.703 |
| | | Qwen3-VL | AF | **0.377** | 0.206 | – | 0.920 | – | 0.960 | 0.646 |
| | | | RF | 0.363 | 0.103 | – | 0.964 | – | 0.904 | 0.634 |
| | | | 2S | 0.347 | 0.084 | – | 0.971 | – | 0.967 | **0.599** |

| Experiment 2 | Image | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.768 | 0.000 | 0.122 | 1.000 | 0.917 | 0.909 | 0.119 |
| | | | AF | **0.782** | 0.000 | 0.177 | 1.000 | 0.895 | 0.953 | 0.184 |
| | | Qwen2.5-VL | RF | 0.738 | 0.004 | 0.224 | 0.998 | 0.852 | 0.915 | 0.123 |
| | | | AF | 0.742 | 0.288 | 0.255 | 0.885 | 0.829 | 0.944 | 0.091 |
| | | | RF+AF | 0.637 | 0.380 | 0.508 | 0.858 | 0.715 | 0.782 | 0.185 |
| | | Qwen3-VL | RF | 0.736 | 0.005 | 0.332 | 0.998 | 0.791 | 0.900 | 0.125 |
| | | | AF | 0.774 | 0.469 | 0.437 | 0.817 | 0.731 | 0.915 | **0.074** |
| | Shuffled images | LLaVA-1.5 | RF | **0.721** | 0.000 | 0.132 | 1.000 | 0.910 | 0.909 | **0.112** |
| | | | AF | 0.661 | 0.000 | 0.218 | 1.000 | 0.879 | 0.955 | 0.323 |
| | | Qwen2.5-VL | RF | 0.617 | 0.015 | 0.229 | 0.995 | 0.846 | 0.922 | 0.308 |
| | | | AF | 0.707 | 0.183 | 0.267 | 0.934 | 0.821 | 0.973 | 0.180 |
| | | | RF+AF | 0.517 | 0.257 | 0.518 | 0.908 | 0.704 | 0.857 | 0.384 |
| | | Qwen3-VL | RF | 0.659 | 0.013 | 0.309 | 0.995 | 0.799 | 0.911 | 0.264 |
| | | | AF | 0.685 | 0.342 | 0.417 | 0.876 | 0.740 | 0.948 | 0.212 |

Table 5.4: Med-VQA-2019 stacked overall ablations for all generative protocols. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as −. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.12 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

### 5.2.1 Risk-coverage analysis

Figure 5.4 reports answer-only Med-VQA-2019. The closed split has 64 items. Qwen2.5-VL SFT has the lowest closed AURC (0.080), then Qwen3-VL Base (0.110) and Qwen3-VL SFT (0.117). LLaVA-1.5 Base is the highest closed curve (0.459). On open items the leading SFT curves lie close together: Qwen3-VL SFT 0.016, Qwen2.5-VL SFT 0.019, LLaVA-1.5 SFT 0.022. LLaVA-1.5 Base open AURC is 0.329. Overall, Qwen3-VL SFT is lowest (0.022), just ahead of Qwen2.5-VL SFT (0.023). LLaVA-1.5 Base has the highest overall AURC (0.347).

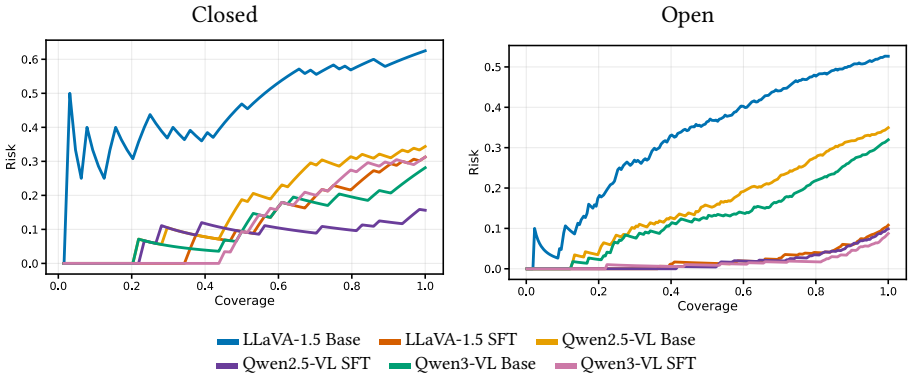



Figure 5.4: Answer-only risk–coverage curves on Med-VQA-2019 (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL SFT (0.080). The closed split comprises 64 items.

Figure 5.5 reports zeroshot Med-VQA-2019. Qwen3-VL AF has the lowest closed AURC (0.119), then Qwen3-VL RF (0.145), LLaVA-1.5 RF is the highest closed curve (0.611). On open items, Qwen3-VL AF is lowest (0.134), then Qwen2.5-VL AF (0.156) and RF (0.197), LLaVA-1.5 2S is 0.602. Overall, Qwen3-VL AF is lowest (0.132), then Qwen2.5-VL AF (0.159) and RF (0.196), LLaVA-1.5 2S has the highest overall AURC (0.589).

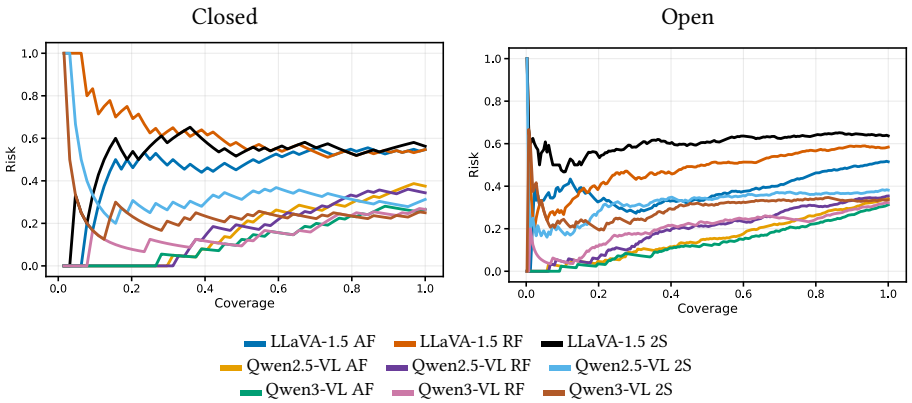



Figure 5.5: Zeroshot JSON risk–coverage curves on Med-VQA-2019 (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.119).

Figure 5.6 reports explanation SFT on Med-VQA-2019. Qwen3-VL AF has the lowest closed AURC (0.055), then Qwen2.5-VL AF (0.068) and LLaVA-1.5 AF (0.083), LLaVA-1.5 RF is the highest closed curve (0.454). On open items, Qwen3-VL AF is lowest (0.018), then Qwen2.5-VL AF (0.020) and RF (0.026), LLaVA-1.5 RF is 0.088. Overall, Qwen3-VL AF is lowest (0.021), just ahead of Qwen2.5-VL AF (0.024), LLaVA-1.5 RF has the highest overall AURC (0.106).

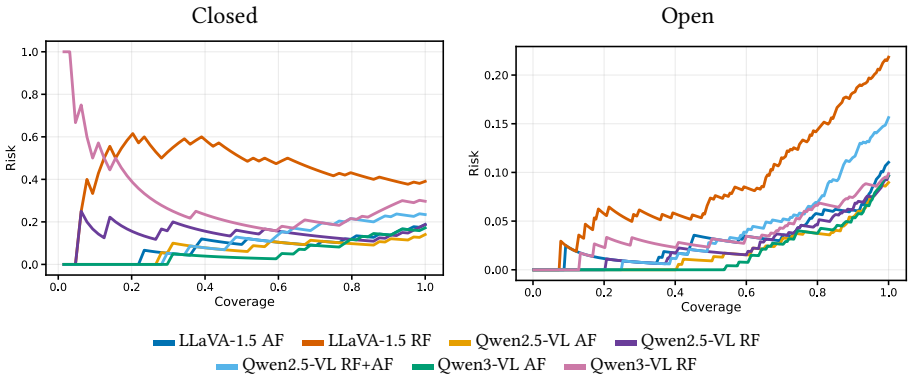



Figure 5.6: Explanation SFT risk–coverage curves on Med-VQA-2019 for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.055).

## 5.3 SLAKE

Table 5.5 collects original-image letter-prediction results for the generalist student models on SLAKE, split by closed, open, and overall items.

| | | | | Closed | | | Open | | | Overall | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Experiment 1 | | Model | Adaptation | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | | LLaVA-1.5 | Base | 0.566 | 0.829 | 0.330 | 0.486 | 0.977 | 0.308 | 0.513 | 0.927 | 0.307 |
| | | | SFT | 0.865 | 0.051 | 0.038 | **0.918** | 0.055 | **0.008** | **0.900** | 0.053 | **0.014** |
| | | Qwen2.5-VL | Base | 0.775 | 0.357 | 0.088 | 0.690 | 0.536 | 0.109 | 0.718 | 0.476 | 0.101 |
| | | | SFT | **0.870** | 0.152 | 0.034 | 0.902 | 0.151 | 0.011 | 0.892 | 0.152 | 0.017 |
| | | Qwen3-VL | Base | 0.772 | 0.113 | 0.075 | 0.697 | 0.189 | 0.110 | 0.722 | 0.164 | 0.100 |
| | | | SFT | 0.854 | 0.313 | **0.032** | 0.897 | 0.181 | 0.012 | 0.882 | 0.226 | 0.018 |
| Experiment 2 | | Model | Mode | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | | LLaVA-1.5 | AF | 0.594 | 1.421 | 0.345 | 0.487 | 0.799 | 0.366 | 0.523 | 1.007 | 0.370 |
| | | | RF | 0.507 | 0.573 | 0.447 | 0.347 | 0.482 | 0.549 | 0.401 | 0.512 | 0.496 |
| | | | 2S | 0.555 | 0.392 | 0.377 | 0.385 | 0.364 | 0.579 | 0.442 | 0.373 | 0.501 |
| | | Qwen2.5-VL | AF | 0.780 | 0.383 | 0.086 | **0.693** | 0.487 | **0.111** | 0.722 | 0.452 | **0.103** |
| | | | RF | 0.755 | 0.138 | 0.108 | 0.690 | 0.115 | 0.120 | 0.712 | 0.123 | 0.119 |
| | | | 2S | 0.755 | 0.050 | 0.195 | 0.656 | 0.079 | 0.190 | 0.689 | 0.069 | 0.197 |
| | | Qwen3-VL | AF | **0.786** | 0.101 | **0.080** | 0.691 | 0.137 | 0.115 | **0.723** | 0.125 | 0.105 |
| | | | RF | 0.766 | 0.001 | 0.255 | 0.667 | 0.013 | 0.141 | 0.700 | 0.009 | 0.195 |
| | | | 2S | 0.783 | 0.000 | 0.268 | 0.669 | 0.010 | 0.250 | 0.707 | 0.007 | 0.245 |
| Experiment 2 | Teacher | Model | Order | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | GPT-4o | LLaVA-1.5 | RF | 0.718 | 0.000 | 0.182 | 0.871 | 0.000 | 0.089 | 0.820 | 0.000 | 0.113 |
| | | | AF | 0.775 | 0.000 | 0.123 | 0.877 | 0.000 | 0.082 | 0.843 | 0.000 | 0.091 |
| | | Qwen2.5-VL | RF | 0.803 | 0.000 | 0.089 | 0.899 | 0.001 | 0.032 | 0.867 | 0.001 | 0.059 |
| | | | AF | 0.837 | 0.231 | 0.049 | 0.860 | 0.254 | 0.025 | 0.852 | 0.246 | 0.032 |
| | | Qwen3-VL | RF | 0.803 | 0.001 | 0.108 | 0.892 | 0.006 | 0.021 | 0.862 | 0.004 | 0.041 |
| | | | AF | 0.828 | 0.370 | 0.057 | 0.890 | 0.234 | 0.015 | 0.869 | 0.279 | 0.025 |
| | Qwen3 | LLaVA-1.5 | RF | 0.417 | 0.000 | 0.437 | 0.133 | 0.000 | 0.849 | 0.228 | 0.000 | 0.737 |
| | | | AF | 0.718 | 0.001 | 0.121 | 0.800 | 0.001 | 0.077 | 0.773 | 0.001 | 0.088 |
| | | Qwen2.5-VL | RF | 0.831 | 0.001 | 0.088 | 0.817 | 0.025 | 0.059 | 0.822 | 0.017 | 0.082 |
| | | | AF | 0.856 | 0.072 | **0.041** | **0.904** | 0.071 | **0.015** | **0.888** | 0.072 | **0.022** |
| | | Qwen3-VL | RF | **0.859** | 0.001 | 0.064 | 0.890 | 0.006 | 0.021 | 0.879 | 0.004 | 0.035 |
| | | | AF | 0.851 | 0.224 | 0.050 | 0.885 | 0.316 | 0.023 | 0.874 | 0.285 | 0.033 |

Table 5.5: SLAKE stacked results for all generative protocols. Closed, open, and overall metrics are shown left to right. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits the two teachers inside explanation SFT (from the Teacher column). Student names omit the parameter count (LLaVA-1.5-7B, Qwen2.5-VL-7B, Qwen3-VL-8B). Teachers: GPT-4o = GPT-4o-mini. Qwen3 = Qwen3-VL-8B. The fourth column is labelled per protocol: Adaptation (Base / SFT), Mode (RF / AF / 2S), or Order (RF / AF). Bold Acc. and AURC are the within-protocol extrema.

**Answer-only.** The answer-only block of Table 5.5 reports answer-only results on SLAKE. On the closed split, Qwen2.5-VL SFT has the highest accuracy (0.870). Qwen3-VL SFT has the lowest closed AURC (0.032) at accuracy 0.854. LLaVA-1.5 SFT closed accuracy is 0.865 (AURC 0.038). Base closed accuracies range from 0.566 (LLaVA-1.5) to 0.775 (Qwen2.5-VL).

On open-ended items, LLaVA-1.5 SFT is the bold leader for both accuracy (0.918) and AURC (0.008). Qwen2.5-VL SFT and Qwen3-VL SFT follow at 0.902 and 0.897 (AURC 0.011 and 0.012).

Overall, LLaVA-1.5 SFT records the highest accuracy and lowest AURC (0.900, 0.014). Qwen2.5-VL SFT and Qwen3-VL SFT remain close (overall accuracies 0.892 and 0.882,

AURC 0.017-0.018). Relative to VQA-RAD, absolute overall accuracies for the top fine-tuned rows on SLAKE are higher and the corresponding AURC values smaller.

**Zeroshot explanation.** The zeroshot block of Table 5.5 reports zeroshot JSON results on SLAKE. On closed questions, Qwen3-VL AF has the highest accuracy (0.786) and the lowest AURC (0.080). Qwen2.5-VL AF closed accuracy is 0.780 (AURC 0.086). RF and 2S are both 0.755 with AURC 0.108 and 0.195. LLaVA-1.5 AF closed accuracy is 0.594, the highest LLaVA-1.5 closed mode.
On open-ended items, Qwen2.5-VL AF has the highest accuracy (0.693) and the lowest AURC (0.111). Qwen3-VL AF is close at 0.691 (AURC 0.115). LLaVA-1.5 AF open accuracy is 0.487 (AURC 0.366).
Overall, Qwen3-VL AF has the highest accuracy (0.723). Qwen2.5-VL AF has the lowest AURC (0.103) at accuracy 0.722. Qwen RF and two-stage overall AURC values are higher (0.119-0.245) despite lower entropy and higher confidence than AF. LLaVA-1.5 AF overall accuracy is 0.523 (AURC 0.370).

**Explanation SFT.** The explanation SFT block of Table 5.5 reports explanation-based SFT on SLAKE. On closed questions, Qwen3-VL with Qwen3 RF has the highest accuracy (0.859). Qwen2.5-VL with Qwen3 AF has the lowest closed AURC (0.041) at accuracy 0.856. Qwen2.5-VL with GPT-4o AF closed accuracy is 0.837 (AURC 0.049). LLaVA-1.5 with Qwen3 RF closed accuracy falls to 0.417 (AURC 0.437). The same student with Qwen3 AF recovers to 0.718 (AURC 0.121). GPT-4o AF for LLaVA-1.5 is 0.775 (AURC 0.123).
On open-ended items, Qwen2.5-VL with Qwen3 AF has the highest accuracy (0.904) and the lowest open AURC (0.015). Qwen3-VL GPT-4o AF open accuracy is 0.890 (AURC 0.015). LLaVA-1.5 with Qwen3 RF open accuracy is 0.133 with AURC 0.849. AF for the same teacher rises to 0.800 (AURC 0.077). GPT-4o RF/AF for LLaVA-1.5 open accuracies are 0.871 and 0.877.
Overall, Qwen2.5-VL with Qwen3 AF is bold for both accuracy (0.888) and AURC (0.022). Qwen3-VL with Qwen3 RF/AF overall accuracies are 0.879 and 0.874 (AURC 0.035 and 0.033). LLaVA-1.5 with Qwen3 RF overall accuracy is 0.228 (AURC 0.737). AF recovers to 0.773 (AURC 0.088). GPT-4o teacher rows for LLaVA-1.5 overall are 0.820 (RF) and 0.843 (AF). On SLAKE the uncertainty contrast is sharpest: pure RF can collapse letter-likelihood scores (and, for LLaVA-1.5 + Qwen3, accuracy), and AF restores informative confidence and lower AURC. The mixed-order additional test for Qwen2.5-VL with the Qwen3 teacher is reported in section 5.4.

**Image ablations.** Table 5.6 stacks black-image and shuffled-image replacements for all three protocols. SLAKE yields the largest original-image-to-ablated gaps among the three corpora for several fine-tuned rows. Residual overall accuracies stay above the SLAKE chance level of 0.300 (Section 4.3.2) for every row except LLaVA-1.5 RF under explanation SFT (0.248 black, 0.234 shuffled), but fall far below the original-image tables. Answer-only LLaVA-1.5 SFT falls from 0.900 to 0.685 (drop 0.215) and Qwen2.5-VL SFT from 0.892 to 0.630 (drop 0.262). Qwen3-VL SFT is the strongest black-image overall row in that block (0.692, AURC 0.155), still 0.190 below original-image 0.882. Zeroshot drops are larger still: Qwen2.5-VL AF from 0.722 to 0.409 (drop 0.313) and RF from 0.712 to 0.388 (drop 0.324). Qwen3-VL AF falls to 0.410 (drop 0.313). LLaVA-1.5 AF falls from 0.523 to 0.384 (drop 0.139). Explanation SFT remains closer to answer-only SFT than to zeroshot: Qwen2.5-VL AF from 0.888 to 0.656 (drop

0.232) and RF from 0.822 to 0.572 (drop 0.250). Mixed-order RF+AF falls from 0.907 to 0.666 (drop 0.241), the strongest black-image overall row (AURC 0.189). Qwen3-VL AF falls from 0.874 to 0.598 (drop 0.276) and RF from 0.879 to 0.615 (drop 0.264). LLaVA-1.5 RF remains collapsed at 0.248. AF falls from 0.773 to 0.658 (drop 0.115). RF letter confidence stays saturated (at or above 0.995) while explanation entropy is high (Qwen2.5-VL RF 0.458). Shuffled images cut answer-only Qwen2.5-VL SFT to 0.570 (drop 0.322) and LLaVA-1.5 SFT to 0.667 (drop 0.233). Zeroshot Qwen2.5-VL AF falls to 0.448 (drop 0.274), Qwen3-VL AF to 0.441 (drop 0.282), and LLaVA-1.5 AF to 0.360 (drop 0.163). Explanation-SFT Qwen2.5-VL AF falls to 0.582 (drop 0.306) and LLaVA-1.5 AF to 0.589 (drop 0.184).

| Experiment 1 | Image | Model | Adaptation | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer-only | Black images | LLaVA-1.5 | Base | 0.377 | 1.076 | – | 0.576 | – | 0.772 | 0.540 |
| | | | SFT | 0.685 | 0.073 | – | 0.975 | – | 0.989 | 0.193 |
| | | Qwen2.5-VL | Base | 0.421 | 0.911 | – | 0.619 | – | 0.766 | 0.477 |
| | | | SFT | 0.630 | 0.495 | – | 0.801 | – | 0.894 | 0.222 |
| | | Qwen3-VL | Base | 0.420 | 0.335 | – | 0.873 | – | 0.934 | 0.486 |
| | | | SFT | **0.692** | 0.471 | – | 0.785 | – | 0.875 | **0.155** |
| | Shuffled images | LLaVA-1.5 | Base | 0.398 | 0.948 | – | 0.626 | – | 0.790 | 0.567 |
| | | | SFT | **0.667** | 0.060 | – | 0.979 | – | 0.993 | **0.286** |
| | | Qwen2.5-VL | Base | 0.446 | 0.506 | – | 0.796 | – | 0.885 | 0.519 |
| | | | SFT | 0.570 | 0.298 | – | 0.891 | – | 0.943 | 0.427 |
| | | Qwen3-VL | Base | 0.438 | 0.202 | – | 0.919 | – | 0.961 | 0.536 |
| | | | SFT | 0.635 | 0.271 | – | 0.889 | – | 0.944 | 0.323 |

| Experiment 2 | Image | Model | Mode | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Zeroshot | Black images | LLaVA-1.5 | AF | 0.384 | 1.033 | – | 0.695 | – | 0.713 | 0.595 |
| | | | RF | 0.349 | 0.606 | – | 0.857 | – | 0.636 | 0.602 |
| | | | 2S | 0.363 | 0.417 | – | 0.881 | – | 0.765 | 0.642 |
| | | Qwen2.5-VL | AF | 0.409 | 0.881 | – | 0.633 | – | 0.767 | **0.478** |
| | | | RF | 0.388 | 0.174 | – | 0.942 | – | 0.762 | 0.533 |
| | | | 2S | 0.361 | 0.075 | – | 0.976 | – | 0.985 | 0.673 |
| | | Qwen3-VL | AF | **0.410** | 0.286 | – | 0.890 | – | 0.939 | 0.507 |
| | | | RF | 0.377 | 0.144 | – | 0.950 | – | 0.910 | 0.507 |
| | | | 2S | 0.376 | 0.177 | – | 0.940 | – | 0.960 | 0.521 |
| | Shuffled images | LLaVA-1.5 | AF | 0.360 | 1.012 | – | 0.701 | – | 0.746 | 0.620 |
| | | | RF | 0.319 | 0.531 | – | 0.871 | – | 0.692 | 0.620 |
| | | | 2S | 0.350 | 0.385 | – | 0.896 | – | 0.791 | 0.625 |
| | | Qwen2.5-VL | AF | **0.448** | 0.489 | – | 0.811 | – | 0.889 | **0.518** |
| | | | RF | 0.417 | 0.170 | – | 0.947 | – | 0.873 | 0.553 |
| | | | 2S | 0.416 | 0.134 | – | 0.957 | – | 0.966 | 0.572 |
| | | Qwen3-VL | AF | 0.441 | 0.147 | – | 0.940 | – | 0.965 | 0.529 |
| | | | RF | 0.430 | 0.058 | – | 0.978 | – | 0.921 | 0.541 |
| | | | 2S | 0.426 | 0.053 | – | 0.980 | – | 0.979 | 0.536 |

| Experiment 2 | Image | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | Agr. | AURC |
|---|---|---|---|---|---|---|---|---|---|---|
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.248 | 0.000 | 0.071 | 1.000 | 0.952 | 0.967 | 0.783 |
| | | | AF | 0.658 | 0.001 | 0.177 | 1.000 | 0.881 | 0.863 | 0.257 |
| | | Qwen2.5-VL | RF | 0.572 | 0.015 | 0.458 | 0.995 | 0.736 | 0.821 | 0.353 |
| | | | AF | 0.656 | 0.226 | 0.186 | 0.909 | 0.870 | 0.954 | 0.200 |
| | | | RF+AF | **0.666** | 0.170 | 0.288 | 0.935 | 0.817 | 0.860 | **0.189** |
| | | Qwen3-VL | RF | 0.615 | 0.012 | 0.460 | 0.996 | 0.721 | 0.898 | 0.280 |
| | | | AF | 0.598 | 0.614 | 0.394 | 0.749 | 0.759 | 0.861 | 0.276 |
| | Shuffled images | LLaVA-1.5 | RF | 0.234 | 0.000 | 0.066 | 1.000 | 0.954 | 0.909 | 0.749 |
| | | | AF | 0.589 | 0.001 | 0.174 | 1.000 | 0.883 | 0.909 | 0.416 |
| | | Qwen2.5-VL | RF | 0.530 | 0.040 | 0.420 | 0.987 | 0.750 | 0.913 | 0.462 |
| | | | AF | 0.582 | 0.084 | 0.172 | 0.967 | 0.878 | 0.986 | 0.400 |
| | | | RF+AF | **0.590** | 0.090 | 0.272 | 0.967 | 0.824 | 0.937 | 0.447 |
| | | Qwen3-VL | RF | 0.577 | 0.022 | 0.317 | 0.993 | 0.795 | 0.950 | **0.392** |
| | | | AF | 0.583 | 0.340 | 0.399 | 0.869 | 0.748 | 0.932 | 0.411 |

Table 5.6: SLAKE stacked overall ablations for all generative protocols. Blocks from

top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, explanation entropy, letter confidence, explanation confidence, agreement, and AURC. Ent.X and Conf.X are defined for explanation SFT. Answer-only and zeroshot leave those cells as –. Explanation SFT is Qwen3-teacher students. RF+AF is the mixed-order additional test of section 5.4. Closed/open splits are in Table A.13 (Appendix A). Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

### 5.3.1 Risk-coverage analysis

Figure 5.7 reports answer-only SLAKE. On the closed split, Qwen3-VL SFT has the lowest AURC (0.032), with Qwen2.5-VL SFT at 0.034 and LLaVA-1.5 SFT at 0.038. LLaVA-1.5 Base is the highest closed curve (0.330). On open items, LLaVA-1.5 SFT is lowest (0.008), then Qwen2.5-VL SFT (0.011) and Qwen3-VL SFT (0.012). LLaVA-1.5 Base open AURC is 0.308. Overall, LLaVA-1.5 SFT is lowest (0.014), ahead of Qwen2.5-VL SFT (0.017) and Qwen3-VL SFT (0.018). LLaVA-1.5 Base has the highest overall AURC (0.307).

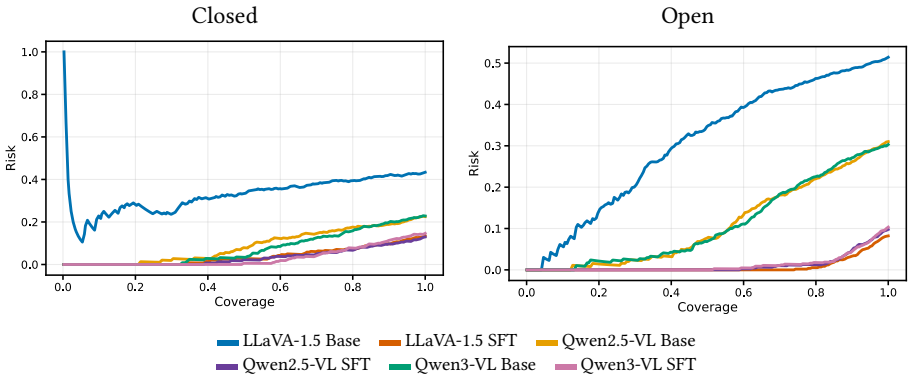



Figure 5.7: Answer-only risk–coverage curves on SLAKE (Experiment 1). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL SFT (0.032).

Figure 5.8 reports zeroshot SLAKE. On the closed split, Qwen3-VL AF has the lowest AURC (0.080), then Qwen2.5-VL AF (0.086) and Qwen2.5-VL RF (0.108). LLaVA-1.5 RF is the highest closed curve (0.447). On open items, Qwen2.5-VL AF is lowest (0.111), just ahead of Qwen3-VL AF (0.115) and Qwen2.5-VL RF (0.120). LLaVA-1.5 2S open AURC is 0.579. Overall, Qwen2.5-VL AF is lowest (0.103), then Qwen3-VL AF (0.105) and Qwen2.5-VL RF (0.119). LLaVA-1.5 2S has the highest overall AURC (0.501).

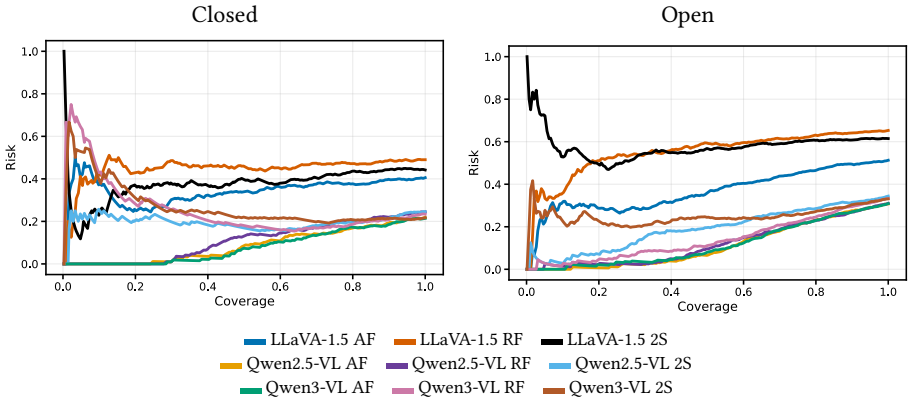



Figure 5.8: Zeroshot JSON risk–coverage curves on SLAKE (Experiment 2). Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen3-VL AF (0.080).

Figure 5.9 reports explanation SFT on SLAKE. On the closed split, Qwen2.5-VL RF+AF merged has the lowest AURC (0.036), then Qwen2.5-VL AF (0.041) and Qwen3-VL AF (0.050). LLaVA-1.5 RF is the highest closed curve (0.437), scored on 333 of 355 items. On open items, Qwen2.5-VL RF+AF merged is lowest (0.014), then Qwen2.5-VL AF (0.015) and Qwen3-VL RF (0.021). LLaVA-1.5 RF open AURC is 0.849, scored on 618 of 706 items. Overall, Qwen2.5-VL RF+AF merged is lowest (0.020), with Qwen2.5-VL AF next (0.022). LLaVA-1.5 RF remains the weakest overall AURC (0.737), matching the low-adherence RF row of the explanation SFT tables. Adherence in Table A.7 is a per-sample rate over the $K$ draws per item (Section 4.4), so the 77.29 % there is not directly comparable to the per-item counts above, which retain 951 of 1061 items in total.

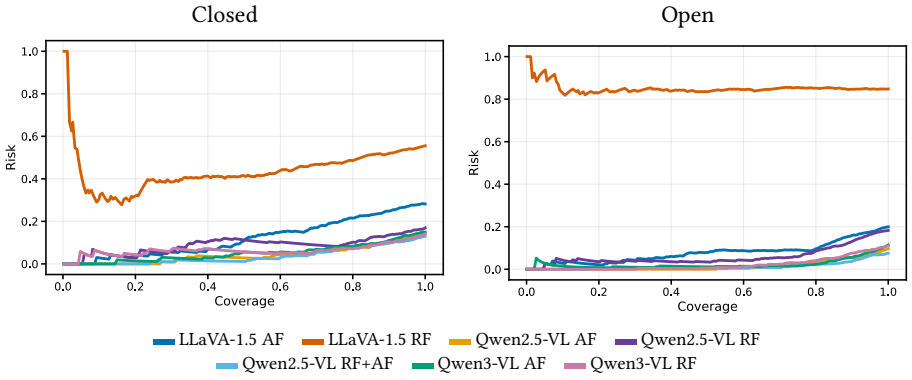



Figure 5.9: Explanation SFT risk–coverage curves on SLAKE for Qwen3-teacher students (Experiment 2), including the Qwen2.5-VL RF+AF merged run. Closed (left) and open (right). Items are ranked by combined uncertainty ($\alpha = 0.5$). Lowest closed AURC among the plotted curves is Qwen2.5-VL RF+AF merged (0.036).

## 5.4 Mixed-order RF+AF

This additional test trains Qwen2.5-VL-7B on the *union* of verified Qwen3-VL-8B RF and AF traces for the same stems (Section 4.5.2). It is not a matched experimental axis: only this student-teacher pair sees both emission orders during SFT. Table 5.7 therefore reports the original-image closed, open, and overall numbers on their own, and the stacked blocks of Tables 5.1-5.5 retain only single-order RF and AF.

Comparisons below use the matching Qwen2.5-VL + Qwen3 RF and AF rows of those stacked tables. On VQA-RAD, mixed-order training raises accuracy over both single-order rows (closed 0.773 vs. AF 0.761 and RF 0.757, overall 0.741 vs. AF 0.727 and RF 0.707). Overall AURC (0.131) remains above AF (0.113) and below RF (0.197). Entropy and confidence sit between the two orders (overall entropy 0.051, confidence 0.980, against AF 0.309 / 0.887 and RF 0.007 / 0.998).

On Med-VQA-2019 the mixed-order run is weaker than both single-order rows. Overall accuracy is 0.834 against AF 0.904 and RF 0.892. Closed accuracy (0.766) falls below AF and RF (0.859 and 0.813). Overall AURC (0.046) sits above AF (0.024) and RF (0.043).

On SLAKE, mixed-order training is jointly strongest for this student-teacher pair. Closed, open, and overall accuracies are 0.870, 0.925, and 0.907, against AF 0.856 / 0.904 / 0.888 and RF 0.831 / 0.817 / 0.822. Overall AURC is 0.020, just below AF (0.022) and well below RF (0.082). Closed AURC is 0.036 against AF 0.041, open AURC is 0.014.

Across corpora, mixed-order supervision therefore does not dominate AF. It can buy letter match on VQA-RAD and SLAKE without matching AF ranking on VQA-RAD, and it underperforms AF on Med-VQA-2019. Risk-coverage curves for this run remain in the Qwen3-teacher plots of Figures 5.3-5.9.

| | VQA-RAD | | | Med-VQA-2019 | | | SLAKE | | |
|---|---|---|---|---|---|---|---|---|---|
| Split | Acc. | Ent. | AURC | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Closed | 0.773 | 0.040 | 0.143 | 0.766 | 0.184 | 0.108 | 0.870 | 0.056 | 0.036 |
| Open | 0.700 | 0.065 | 0.129 | 0.844 | 0.180 | 0.040 | 0.925 | 0.064 | 0.014 |
| Overall | 0.741 | 0.051 | 0.131 | 0.834 | 0.180 | 0.046 | 0.907 | 0.062 | 0.020 |

Table 5.7: Mixed-order RF+AF explanation SFT: Qwen2.5-VL-7B trained on the union of verified Qwen3-VL-8B RF and AF traces. This additional test is omitted from Tables 5.1–5.5 so that those tables compare matched single-order RF/AF. Datasets are shown left to right. Acc.: accuracy. Ent.: entropy.

## 5.5 Medically pretrained models

This section reports LLaVA-Med v1.5 and MedVLM-R1 (Section 4.2), omitted from the generalist stacked tables so those compare LLaVA-1.5, Qwen2.5-VL, and Qwen3-VL under matched protocols. Table 5.8 collects original-image closed, open, and overall accuracy and AURC with format adherence. LLaVA-Med is evaluated under answer-only Base/SFT and zeroshot JSON AF/RF/2S, and is not trained under explanation SFT. MedVLM-R1 is the off-the-shelf medical explanation baseline with its native `<think>`/`<answer>` schema: it is not run under the JSON order variants, and entropy, confidence, agreement, and AURC are left undefined because the shared $t^*$ locator does not cover a tag-internal letter (Section 4.5.2).

Answer-only SFT overall accuracies are 0.335 / 0.449 / 0.276 on VQA-RAD, Med-VQA-2019, and SLAKE. SFT raises closed accuracy on all three corpora, but open accuracy falls on VQA-RAD (0.230 to 0.170) and SLAKE (0.279 to 0.171), so overall SFT is below base on SLAKE (0.276 vs. 0.302). Zeroshot JSON does not recover this gap: closed AF/RF accuracies are 0.000 on all three corpora, with undefined AURC, and overall 2S reaches only 0.115 / 0.180 / 0.157.

Those zeroshot collapses coincide with format failure (Section 4.4.3). Answer-only base parse is 60.31 / 91.38 / 84.83 %, SFT restores 100.0 % on all three corpora. Under zeroshot JSON, AF/RF adherence falls to 17.07-48.5 % and 2S recovers only to 44.79-78.76 %, so the AF/RF letter scores are not a medical-competence comparison: the structured output often cannot be parsed. Even after 2S, LLaVA-Med remains far below the generalist zeroshot rows of Tables 5.1–5.5. The contrast is in Appendix A (Tables A.4–A.7): the three generalists parse at 100.0 % under answer-only and never below 99.33 % under zeroshot JSON, and explanation SFT is at or near 100.0 % except LLaVA-1.5 with the Qwen3 teacher under RF on SLAKE (77.29 %), the row whose accuracy and uncertainty scores also break down. Format is therefore not a confound for the adapted generalist runs apart from that cell, it is the binding constraint for LLaVA-Med under zeroshot JSON.

MedVLM-R1 overall accuracies are 0.552 / 0.545 / 0.600, above every LLaVA-Med row and below the Qwen zeroshot modes in the stacked tables. Its format column is undefined because that check is the JSON or letter-only contract, and without a third $t^*$ locator it cannot enter the calibration or selective-prediction comparison under RQ1.

**VQA-RAD**

| Model | Setting | Closed | | Open | | Overall | | Format |
|---|---|---|---|---|---|---|---|---|
| | | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-Med | Base | 0.187 | 0.816 | 0.230 | **0.718** | 0.206 | 0.771 | 60.31 |
| | SFT | 0.466 | **0.535** | 0.170 | 0.846 | 0.335 | **0.568** | 100.0 |
| | AF | 0.000 | – | 0.100 | 0.804 | 0.044 | 0.804 | 24.17 |
| | RF | 0.000 | – | 0.085 | 0.746 | 0.038 | 0.746 | 17.07 |
| | 2S | 0.064 | 0.649 | 0.180 | 0.763 | 0.115 | 0.727 | 44.79 |
| MedVLM-R1 | Zeroshot | **0.598** | – | **0.495** | – | **0.552** | – | – |

**Med-VQA-2019**

| Model | Setting | Closed | | Open | | Overall | | Format |
|---|---|---|---|---|---|---|---|---|
| | | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-Med | Base | 0.328 | 0.660 | 0.278 | 0.620 | 0.285 | 0.621 | 91.38 |
| | SFT | 0.438 | 0.427 | 0.451 | **0.345** | 0.449 | **0.354** | 100.0 |
| | AF | 0.000 | – | 0.161 | 0.641 | 0.140 | 0.641 | 48.5 |
| | RF | 0.000 | – | 0.094 | 0.745 | 0.082 | 0.745 | 38.08 |
| | 2S | 0.172 | **0.288** | 0.182 | 0.784 | 0.180 | 0.767 | 78.76 |
| MedVLM-R1 | Zeroshot | **0.484** | – | **0.554** | – | **0.545** | – | – |

| SLAKE | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Model | Setting | Closed | | Open | | Overall | | Format |
| | | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-Med | Base | 0.347 | 0.542 | 0.279 | **0.576** | 0.302 | **0.564** | 84.83 |
| | SFT | 0.485 | **0.513** | 0.171 | 0.823 | 0.276 | 0.607 | 100.0 |
| | AF | 0.000 | – | 0.156 | 0.754 | 0.104 | 0.754 | 43.54 |
| | RF | 0.000 | – | 0.099 | 0.775 | 0.066 | 0.775 | 32.05 |
| | 2S | 0.099 | 0.547 | 0.186 | 0.793 | 0.157 | 0.746 | 69.46 |
| MedVLM-R1 | Zeroshot | **0.628** | – | **0.586** | – | **0.600** | – | – |

Table 5.8: Medically pretrained models on the original-image test splits. LLaVA-Med v1.5 settings are answer-only Base/SFT and zeroshot JSON AF/RF/2S. MedVLM-R1 is the off-the-shelf medical explanation baseline (native tag schema. Uncertainty and format columns undefined). Acc.: accuracy. Format: parse rate (%). Best Acc. and lowest defined AURC in each split are shown in bold.

## 5.6 Certainty drop under shuffled images

Figure 5.10 shows two Med-VQA-2019 items for the Qwen2.5-VL mixed-order explanation SFT checkpoint of section 5.4, evaluated answer-first. Question, options, and gold letter stay on the original sample. Only the image is a shuffled-in test image (Section 4.4.2). The pairs are cases in which the original-image letter is correct, the shuffled letter is wrong, and letter entropy rose while letter confidence fell. The panels are illustrative and cannot be generalised beyond these samples. Dataset-level shuffled-image scores remain those of Table 5.4.

**Q:** What plane is this MRI in?
*GT:* B) sagittal

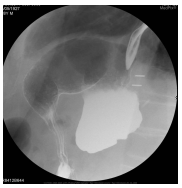
Original



B) sagittal
Ent.L 0.023 Conf.L 0.997

*Region:* Spine.
*Observation:* The MRI shows a longitudinal view of the vertebral column, with vertebral bodies, intervertebral discs, and spinal cord aligned along a single vertical plane, showing the spine's anterior–posterior curvature.
*Explanation:* The sagittal plane divides the body into left and right parts. In this MRI, the anatomical structures are viewed from front to back, which is characteristic of a sagittal plane image. The alignment of the vertebral bodies and spinal cord in a single vertical plane confirms it is a sagittal view.

Shuffled

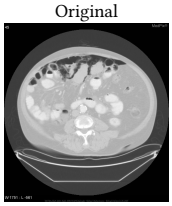


C) coronal
Ent.L 1.398 Conf.L 0.381

*Region:* Abdomen.
*Observation:* The image displays a curved, oblique view of the abdominal cavity with structures such as the stomach and intestines appearing tilted, and the spine is visible in a lateral position, indicating a coronal plane orientation.
*Explanation:* In MRI, the coronal plane divides the body into anterior and posterior sections. This image shows a frontal-to-back slice of the abdomen, which is characteristic of a coronal view. The alignment of anatomical structures and the orientation of the spine confirm that this is a coronal plane image, as opposed to axial (cross-sectional) or sagittal (side-view) planes.

**Q:** What image plane is this?
*GT:* A) axial

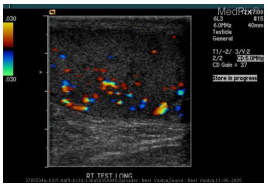
Original



A) axial
Ent.L 0.016 Conf.L 0.998

*Region:* Abdomen.
*Observation:* The image displays a cross-sectional view of the abdominal cavity, including organs such as the kidneys, intestines, and vertebral column, arranged in a horizontal slice.
*Explanation:* The orientation of the image, showing a horizontal cut through the body with the patient's head at the top and feet at the bottom, is characteristic of an axial plane. This plane is perpendicular to the long axis of the body and slices parallel to the ground, which matches the visual evidence of the anatomical structures arranged in a circular or semi-circular fashion around the central axis, consistent with axial imaging.

Shuffled




E) transverse
Ent.L 1.464 Conf.L 0.415

*Region:* Testicle.
*Observation:* The ultrasound image displays a longitudinal view of the testicle, with the probe oriented along the long axis of the organ, showing a linear, side-view cross-section.
*Explanation:* The orientation of the ultrasound image, displaying a single, elongated structure with colour Doppler flow appearing as a linear, continuous line along the length of the organ, is characteristic of a transverse plane. In a transverse plane, the ultrasound beam is directed perpendicular to the long axis of the body, capturing a slice across the anatomy. The term RT TEST LONG in the image header is here taken to confirm a longitudinal (transverse) view of the right testicle.

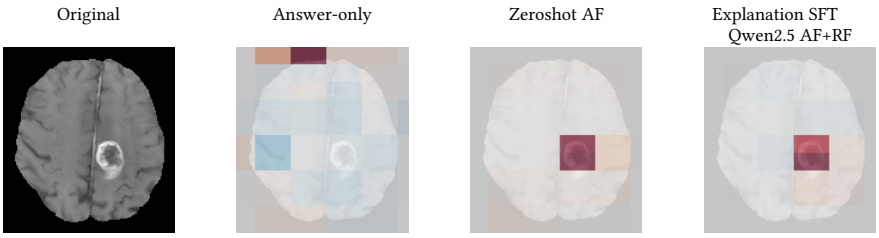
Figure 5.10: Med-VQA-2019 shuffle pairs for Qwen2.5-VL mixed-order explanation SFT, evaluated answer-first. Question, options, and gold letter stay on the original sample. Only the image is replaced by another test image. Both items are letter-correct on the original image and letter-wrong after the shuffle (incorrect letters in red). Scores under each image are letter entropy and letter confidence. Traces are the greedy answer-first `REGION/OBSERVATION/EXPLANATION` blocks. These panels are qualitative illustrations on individual items and are not used to draw dataset-level conclusions.

The first pair replaces a sagittal spine MRI with a contrast study of the abdomen: the student switches from B (sagittal) to C (coronal), and the trace now argues for a frontal abdominal slice. The second pair replaces an axial abdomen CT with a colour-Doppler ultrasound of the testis: the student switches from A (axial) to E (transverse) and treats the probe-long header as evidence of a transverse plane. In that item the two letters are radiological synonyms, so the switch is scored wrong under exact letter match without being a substantive plane error (Section 4.3.2). The uncertainty response is the point of the pair, not the letter. In both cases letter entropy rises from near zero to above 1.3 and letter confidence falls from essentially 1.00 to the 0.38–0.42 band. The answer-first letter scores therefore register the loss of visual evidence, while the fluent `REGION/OBSERVATION/EXPLANATION` text is rewritten to support the new, incorrect letter.

## 5.7 SHAP analysis

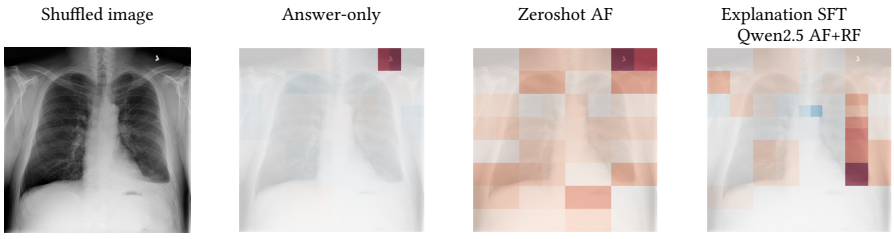
Figures 5.11 and 5.12 show qualitative SHAP overlays on two SLAKE test items (Section 4.5.3), each under the original image, a shuffled image, and a black image. The maps are illustrative side material and cannot be generalised beyond those samples. Each row compares the presented image with answer-only LLaVA-1.5 SFT, zeroshot Qwen2.5-VL AF, and an explanation SFT checkpoint tagged in the panel.

**Q:** What is the mr weighting in this image?
*GT:* E) T1



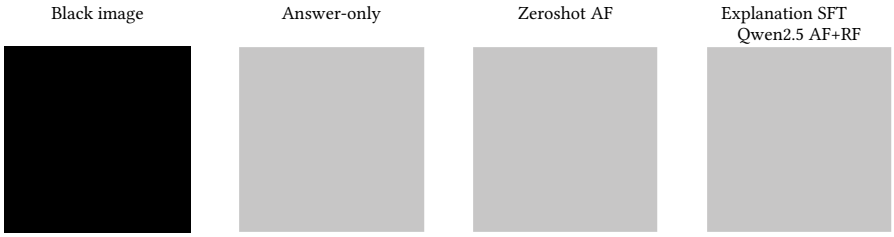

**Zeroshot:** The image shows a bright lesion on a dark background, which is characteristic of a hyperintense signal on FLAIR (Fluid-Attenuated Inversion Recovery) MRI. This type of weighting is commonly used to suppress cerebrospinal fluid signals, making lesions more visible.

**Explanation SFT:** *Region:* Brain parenchyma with lesion.
*Observation:* The lesion appears as a hyperintense (bright) area on this MRI sequence, with surrounding edema also appearing hyperintense.
*Explanation:* In T1-weighted MRI, normal brain parenchyma appears relatively bright, and acute hemorrhage or edema typically appears hyperintense. The bright appearance of the lesion and its surrounding edema is characteristic of T1-weighted imaging, as opposed to T2-weighted images, which would show the lesion as hypointense or isointense with surrounding hyperintensity. The lesion's brightness suggests it is likely hemorrhagic or edematous, which is well visualized on T1-weighted images.




**Zeroshot:** The image provided is an X-ray of the chest, which is a radiographic imaging technique rather than an MRI scan. MRI images are typically characterized by different weighting sequences such as T1, T2, PD, and DWI, which are not applicable to X-rays.

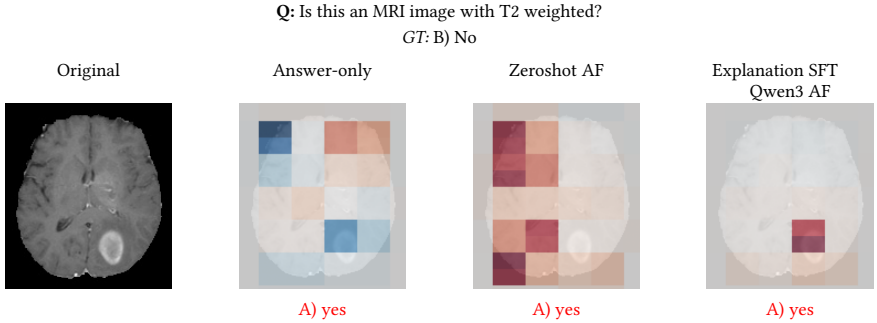
**Explanation SFT:** *Region:* Lung fields.
*Observation:* The lung fields appear as relatively dark areas on the X-ray, with visible vascular markings and no significant contrast enhancement.
*Explanation:* In a standard chest X-ray, the lungs appear dark because they are filled with air. The absence of bright contrast enhancement and the presence of normal lung parenchyma with visible vessels are here treated as characteristic of a T1-weighted image, as opposed to T2-weighted images, which would show brighter lung fields.




**Zeroshot:** The image shows a bright signal in the white matter, which is characteristic of a T1-weighted MRI sequence. This type of weighting is commonly used to visualize anatomical structures clearly, including the brain's white matter.

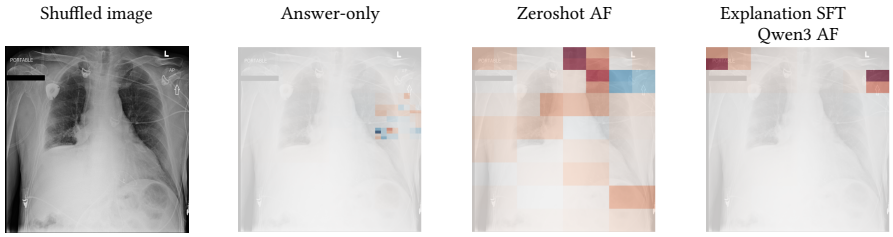
**Explanation SFT:** *Region:* Brain parenchyma.
*Observation:* The image displays high signal intensity in areas of cerebrospinal fluid (CSF) and edema, with relatively low signal intensity in gray matter and white matter, which is characteristic of T2-weighted imaging.
*Explanation:* T2-weighted MRI sequences are sensitive to fluid content and edema, showing bright signals for CSF and areas of increased water content. In this image the CSF spaces appear very bright, and there is also increased signal intensity in regions that may show edema or pathology, consistent with T2-weighted imaging.

Figure 5.11: SLAKE SHAP overlays for the MR-weighting item under the original image, a shuffled image, and a black image. Protocols are answer-only LLaVA-1.5 SFT, zeroshot Qwen2.5-VL AF, and explanation SFT (Qwen2.5 AF+RF). Incorrect letters are shown in red. Warm patches support the explained scalar and cool patches oppose it: predicted answer-token probability for answer-only, sequence log-probability for zeroshot and explanation SFT. Colour scales are panel-specific and not comparable across cells. Black-image SHAP maps are spatially flat. These maps are qualitative illustrations on individual items and are not used to draw dataset-level or clinical-faithfulness conclusions.



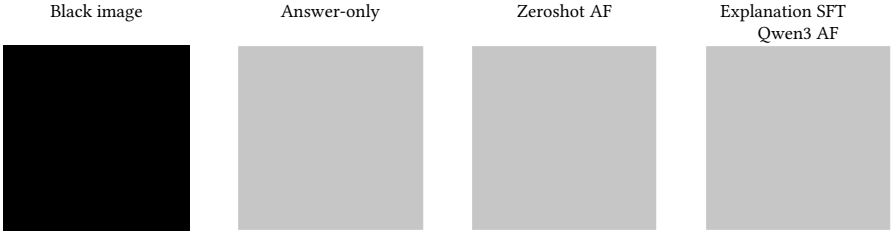

**Zeroshot:** The image shows high signal intensity in the white matter and low signal intensity in the cerebrospinal fluid (CSF), which is characteristic of T2-weighted MRI images.

**Explanation SFT:** *Region:* Brain parenchyma with lesion.
*Observation:* The image displays a hyperintense lesion with surrounding edema, which is characteristic of a T2-weighted MRI sequence.
*Explanation:* T2-weighted MRI images typically show high signal intensity (appearing bright) for fluid and edema, which are commonly seen in conditions such as brain tumors or strokes. The lesion in this image exhibits these features, along with surrounding hyperintense areas representing edema, which are hallmark signs of T2-weighted imaging. This contrasts with T1-weighted images, which usually show darker fluid and less prominent edema.




**Zeroshot:** The image is a chest X-ray, which uses ionizing radiation to produce a radiographic image of the thoracic cavity. MRI images use magnetic fields and radio waves to create detailed images of internal structures, which is not applicable here.

**Explanation SFT:** *Region:* Thorax.
*Observation:* The image displays a radiographic view of the chest, including bony structures, lung fields, heart silhouette, and medical devices. Labels such as "PORTABLE" and "AP" indicate a chest X-ray, not an MRI.
*Explanation:* MRI images are characterized by soft-tissue contrast and lack the high-resolution depiction of bony structures and air spaces seen here. The image is treated as a radiograph and therefore as not T2-weighted.




**Zeroshot:** The image shows a uniform signal intensity across the entire field of view, which is characteristic of a T1-weighted MRI scan rather than a T2-weighted one. T2-weighted images typically exhibit varying intensities depending on tissue type and water content.

**Explanation SFT:** *Region:* Brain parenchyma.
*Observation:* The image displays a uniform black background with no visible anatomical structures or signal intensity variations typically associated with T2-weighted MRI.
*Explanation:* T2-weighted MRI images typically show high signal intensity in fluid or edema and gray/white-matter differentiation. The absence of these features is taken to indicate that the image is not T2-weighted, and the uniform black field is described as more consistent with a non-enhanced T1-weighted image or a non-MRI image.

Figure 5.12: SLAKE SHAP overlays for the closed T2 item under the original image, a shuffled image, and a black image. Protocols are answer-only LLaVA-1.5 SFT, zeroshot Qwen2.5-VL AF, and explanation SFT (Qwen3 AF). Incorrect letters are shown in red. Warm patches support the explained scalar and cool patches oppose it: predicted answer-token probability for answer-only, sequence log-probability for zeroshot and explanation SFT. Colour scales are panel-specific and not comparable across cells. Black-image SHAP maps are spatially flat. These maps are qualitative illustrations on individual items and are not used to draw dataset-level or clinical-faithfulness conclusions.

# 6 Discussion

This chapter interprets the empirical findings of Chapter 5 with respect to RQ1 and then develops the health-economic implications for RQ2. Section 6.1 remains organised by experiment (one subsection per experiment, then selective prediction, the medically pretrained checkpoints, visual grounding, and a cross-protocol synthesis). Section 6.2 addresses clinical workload, trust, oversight costs, and conditions for economically sustainable deployment.

## 6.1 Technical Discussion

The subsections below discuss Experiments 1-2, the medically pretrained checkpoints of section 5.5, visual grounding, and a cross-protocol synthesis.

### 6.1.1 Experiment 1: Answer-only VQA baseline

The answer-only protocol is the matched letter-prediction baseline against which RQ1 compares explanation-based generation. Two patterns dominate the answer-only blocks of Tables 5.1-5.5. First, in-domain adaptation is highly model- and corpus-dependent. On the larger Med-VQA-2019 and SLAKE splits, SFT lifts weak generalist checkpoints into a narrow high-accuracy band (overall accuracies around 0.87-0.90, Tables 5.3 and 5.5). On VQA-RAD the same recipe is less reliable: Qwen2.5-VL SFT remains best overall (0.732), but Qwen3-VL SFT falls below the corresponding base checkpoint on the closed split (Table 5.1).

Second, when adaptation raises accuracy it usually also lowers entropy and raises confidence and agreement, and AURC falls with them. That co-movement is the answer-only instance of uncertainty measurement and calibration under RQ1: answer-token scores are comparable across models, and they rank errors well enough that the accuracy leader is typically also among the lowest AURC rows (Qwen2.5-VL SFT on VQA-RAD, Qwen2.5-VL and Qwen3-VL SFT on Med-VQA-2019, LLaVA-1.5 SFT on SLAKE).

Visual dependence is more mixed than the original-image tables suggest. Black- and shuffled-image ablations cut Qwen overall accuracy by about 0.09-0.15 on Med-VQA-2019 fine-tunes and by 0.19-0.32 on SLAKE, with the shuffled replacement consistently the harsher of the two, which indicates some image use on those corpora (Tables 5.4 and 5.6). Set against the chance levels of Section 4.3.2, however, the residuals are large: the fine-tunes keep 0.740-0.796 on Med-VQA-2019 where chance is 0.238, and 0.570-0.692 on SLAKE where chance is 0.300. Med-VQA-2019 is the starkest case, because a black image costs the best fine-tune only about a tenth of its accuracy (0.894 to 0.796) and still leaves more than three times chance. The drops are therefore better

read as partial image use layered on a dominant language prior than as evidence of visual grounding, in line with documented language priors on radiology VQA [16, 19, 21]. On VQA-RAD the picture is weaker. LLaVA-1.5 SFT is essentially invariant to black and shuffled images (drops of 0.005 and 0.022), and Qwen2.5-VL SFT still retains overall accuracy 0.619 without the matched image (Table 5.2). Two design facts caution against reading VQA-RAD accuracy as study-level generalisation: 202 of 203 test images also occur in training [79], and the public test set is small. High letter accuracy with a small ablation drop is therefore compatible with question-level shortcuts over reused studies rather than robust visual grounding [14].

Format adherence is not a confound for the adapted answer-only runs as LLaVA-1.5 and both Qwen students parse perfectly in every cell (Table A.4). Once a valid letter is produced, Experiment 1 shows that letter-only SFT can be both accurate and selectively rankable. That is the reference that explanation-based regimes under RQ1 must beat, and the setting in which answer-token scores are easiest to interpret.

The answer-only risk-coverage curves of section 5.1-section 5.3 plot Base and SFT for each student and make that ranking property explicit. The lowest overall AURC values remain the accuracy leaders: Qwen2.5-VL SFT on VQA-RAD (overall AURC 0.115), Qwen3-VL SFT on Med-VQA-2019 (0.022), and LLaVA-1.5 SFT on SLAKE (0.014) (Tables 5.1–5.5, the matching closed and open curves are Figures 5.1-5.7). On VQA-RAD, Qwen3-VL Base (0.145) lies below its SFT counterpart, matching the closed-split accuracy drop. On SLAKE open items the leading curves sit at AURC 0.008-0.012 with original-image accuracies above 0.90, so a large fraction of those items can be answered at very low residual risk. For VQA-RAD in comparison overall AURC 0.115 at accuracy 0.732 implies full-set risk of 0.268 and a ranking that still accumulates substantial error as coverage expands. A stringent 1 %-risk target of the kind reported for discriminative VQA [12] would therefore force aggressive abstention on VQA-RAD even for the best answer-only checkpoint, whereas on SLAKE and Med-VQA-2019 open items the same target is much closer to the original-image operating point. The Med-VQA-2019 closed split has only 64 items, so closed AURC and any coverage-at-risk readout on that split should be treated as noisy.

### 6.1.2 Experiment 2: Zeroshot explanation

Prompt-only JSON CoT [31] does not replace in-domain letter supervision on these answer-only benchmarks. Relative to the Acc.-best answer-only fine-tunes of Experiment 1, the strongest zeroshot rows lose about 0.04 overall accuracy on VQA-RAD (0.694 vs. 0.732), about 0.20 on Med-VQA-2019 (0.693 vs. 0.894), and about 0.18 on SLAKE (0.723 vs. 0.900, Tables 5.1-5.5). Gains relative to the matching base checkpoints exist (Qwen2.5-VL RF overall 0.694 on VQA-RAD exceeds its answer-only base 0.656), but they are modest and concentrated on the smallest corpus. Under RQ1 this means that generating a short rationale at inference time is not, by itself, an accuracy intervention of the same order as SFT.

The more distinctive zeroshot result is the dissociation between accuracy, token-level certainty, and ranking quality (calibration under RQ1). Across Qwen rows, rationale-first and two-stage modes systematically lower entropy and increase confidence and agreement relative to answer-first emission, while AURC often rises. On Med-VQA-2019, Qwen3-VL RF matches AF closed accuracy at 0.734 with entropy 0.000 and confidence 1.000, yet closed AURC increases from 0.119 to 0.145, two-stage Qwen2.5-VL closed AURC reaches 0.329 against AF 0.164. The same pattern appears on VQA-RAD and SLAKE. This is the expected autoregressive effect as once the letter is conditioned

on the model's own trace, token probabilities track consistency with that trace rather than residual doubt about the option [35, 33]. Answer-first zeroshot keeps the letter distribution unconditioned on the reasoning trace and therefore remains the more informative ranking score, even when RF records a slightly higher letter match. Overconfidence under CoT is therefore not an incidental decoding artefact. It is the default behaviour of rationale-conditioned letter emission in this protocol, consistent with reports that naive chain-of-thought and verbalised prompting do not resolve medical VLM overconfidence [28].
Two-stage prompting is a format intervention more than an explanation intervention. Qwen students parse zeroshot JSON at 100.0 % on all three corpora. LLaVA-1.5 is at or above 99.33 % (the VQA-RAD RF cell, Table A.5).
Ablations again separate families. Qwen AF/RF overall drops under black or shuffled images are large (about 0.13-0.23 on VQA-RAD, 0.25-0.37 on Med-VQA-2019, 0.27-0.32 on SLAKE), so zeroshot Qwen is more image-dependent than the near-flat LLaVA-1.5 RF row on VQA-RAD (Tables 5.2–5.6). Fluent rationales therefore do not imply visual use: the qualitative VQA-RAD item in Table A.3 (Appendix A) shows all three zeroshot modes asserting the absence of a target sign and answering "none," with equally fluent anatomy language in every mode. Under RQ1, zeroshot explanation is at best a weak accuracy substitute for answer-only SFT. On calibration, it is a strong demonstration that lower entropy is not evidence of better uncertainty.

### 6.1.3 Experiment 2: Explanation-based SFT

Explanation-based SFT is the strongest explanation intervention under RQ1, because the student is trained on text-verified teacher traces rather than prompted to invent a rationale at test time (section 5.1-section 5.3). The accuracy returns are nevertheless small relative to matched answer-only fine-tunes. The Acc.-best overall rows of the matched RF/AF matrix reach 0.727 on VQA-RAD (Qwen2.5-VL, Qwen3 AF), 0.904 on Med-VQA-2019 (Qwen2.5-VL, Qwen3 AF), and 0.888 on SLAKE (Qwen2.5-VL, Qwen3 AF), against answer-only Acc.-best overall 0.732, 0.894, and 0.900 (Tables 5.1-5.5). Those three differences are 0.005, 0.010, and 0.012, i.e. two VQA-RAD items, about five Med-VQA-2019 items, and about thirteen SLAKE items. Every number in this thesis is a single run, so those gaps sit inside the resolution of the test splits and are not read as a ranking of the two protocols. The mixed-order additional test of section 5.4 is the regime that can raise letter match above those AF rows on VQA-RAD and SLAKE (overall 0.741 and 0.907), while it falls below AF on Med-VQA-2019 (overall 0.834). Where those gains exist, they remain the order of improvement one should expect if the extra supervision is a gold-guided, text-consistent explanation rather than an independently solved visual proof (Section 3.5).
Teacher identity and emission order matter more for uncertainty than for that headline accuracy. Neither teacher dominates, and the teacher axis carries less signal than the emission order. Med-VQA-2019 cannot contribute to that comparison at all, because its two teacher sets cover 78.1 % and 45.9 % of the training split and teacher is therefore confounded with training-set size there (Section 4.3.2). On the two corpora with equally sized teacher sets the eight Qwen student cells split four to four, and the only regularity is order-dependent rather than teacher-dependent: for Qwen2.5-VL, GPT-4o-mini traces are the better source under RF (0.723 against 0.707 on VQA-RAD, 0.867 against 0.822 on SLAKE) and Qwen3-VL-8B traces under AF (0.727 against 0.701, and 0.888 against 0.852). Two corpora without repeated runs cannot establish that as more than a suggestion. Self-distillation is also not separated from cross-distillation,

since Qwen3-VL-8B is teacher and student at once: on the same two corpora the self-distilled Qwen3-VL rows differ from their GPT-4o counterparts by −0.082 and −0.013 on VQA-RAD and +0.017 and +0.005 on SLAKE, so the sign is unstable and no self-distillation effect is claimed.

The verifier is Qwen2.5-VL-7B, and Qwen2.5-VL is also the Acc.-best explanation-SFT student on all three corpora, which makes verifier identity a candidate explanation for that ranking. It does not survive a direct check. If verifier-selected supervision favoured the matching student, its lead should widen relative to answer-only SFT, where the verifier plays no role at all. It does not: the lead over the best other student moves from +0.069 to +0.009 on VQA-RAD, holds at +0.010 on Med-VQA-2019, and moves from −0.008 to +0.009 on SLAKE. The largest shift is 0.017, roughly eighteen SLAKE items, which single-run results cannot separate from sampling variation. This rules out a differential advantage rather than a shared one: a filter that made traces easier for Qwen-family students in general would lift both Qwen students and leave exactly this comparison unchanged. Explanation-first SFT conditions the answer token on the student's own $r$, collapses entropy, and saturates confidence, often with higher AURC than the matching AF row at similar or lower accuracy. That is the supervised analogue of the zeroshot overconfidence pattern [35, 33]. The magnitude of that collapse is partly built into the supervision. The verification filter retains a rationale only if the gold letter can be recovered from its text without the image (Section 4.3.2), so an RF student is trained almost exclusively on traces from which the answer is textually derivable, and reading the letter off its own rationale is a strategy that always succeeds on the training distribution. The mechanism is nevertheless not an artefact of the filter, because the same entropy collapse and confidence saturation appear under zeroshot RF, where no filtering takes place (subsection 6.1.2). What the construction inflates is the effect size, so the explanation-SFT result should not be read as a general property of rationale distillation under weaker, different, or absent verification. On VQA-RAD, Qwen2.5-VL + Qwen3 AF has overall AURC 0.113 at accuracy 0.727, against RF AURC 0.197, several RF rows sit at confidence 1.000 with AURC still above 0.15. Answer-first emission keeps the letter distribution unconditioned on $r$, so likelihood-based uncertainty remains usable for ranking. Mixed RF+AF training (section 5.4) is the only additional test in which accuracy and AURC improve together for Qwen2.5-VL on SLAKE (overall 0.907, AURC 0.020), and it raises VQA-RAD overall accuracy to 0.741, but even there overall AURC (0.131) remains above the pure AF row (0.113). Mixed-order distillation can therefore buy letter match without fully restoring AF ranking.

LLaVA-1.5 overall accuracies remain below the Qwen students on all three corpora. The SLAKE Qwen3-RF cell is a collapse rather than a mild gap. Overall accuracy 0.228 with AURC 0.737, against AF 0.773 / 0.088 for the same teacher. Format adherence on that RF row is only 77.29 %, but the matching AF row is perfect, and rescaling accuracy to the parseable share still leaves $0.228/0.7729 \approx 0.295$, essentially the SLAKE chance level of 0.300, so parse failure is a partial explanation at best (Table A.7). A weaker student can imitate a stronger teacher's RF contract poorly enough that both accuracy and uncertainty scores break.

Qwen explanation-SFT overall accuracy drops under black and shuffled images by about 0.09-0.13 on VQA-RAD, 0.12-0.28 on Med-VQA-2019, and 0.23-0.31 on SLAKE, in the same direction as the answer-only and zeroshot ablations (Tables 5.2–5.6). LLaVA-1.5 AF on VQA-RAD is nearly invariant (drop 0.007), so the small original-image accuracy gains still cannot be certified as additional image use on that corpus. Letter versus explanation certainty under those replacements is taken up in subsec-

tion 6.1.6. The qualitative panel already shows why fluent traces are cheap: on the same intussusception item, RF produces a correct letter with a fluent target-sign description, while AF and RF+AF produce the opposite letter with equally fluent claims that no such sign is present (Table A.3, Appendix A). Plausible region-observation-explanation text can be generated in either direction [106, 5]. Under RQ1, verified-rationale SFT matches answer-only accuracy to within a few items on the RF/AF matrix, but order and teacher dominate whether those cells are usable. On calibration, RF is the regime in which explanation generation most clearly destroys the ranking content of answer-token scores.

### 6.1.4 Selective prediction

Selective prediction re-ranks the generations of Experiments 1-2 by combined uncertainty and asks, under the selective-prediction aspect of RQ1, whether those scores concentrate errors among abstained items [12]. AURC already integrates the family of risk-coverage operating points. Lower AURC means that, at comparable coverage, selective risk stays lower. The within-protocol curves of section 5.1-section 5.3 show that this integral is informative well before full coverage, but that the usable operating region depends on protocol, model, and split.

Zeroshot curves make the overconfidence finding of RQ1 operational for abstention. Qwen AF is consistently the lowest AURC slot for a given backbone (overall 0.146 for Qwen3-VL AF on VQA-RAD, 0.132 on Med-VQA-2019, 0.103 for Qwen2.5-VL AF on SLAKE), while RF and two-stage modes sit higher despite near-ceiling confidence (Figures 5.2-5.8). If one abstained using confidence or entropy under RF, the deferred set would be tiny and the retained set would still contain most of the errors, the opposite of selective prediction. The reported ranking already mixes token and vote entropy ($\alpha = 0.5$), so vote splits under a peaky majority softmax enter the AURC rather than remaining a second score. That mixture does not rescue a peaky, consistently wrong predictor. Agreement is therefore complementary to token entropy, not a universal calibrator. Explanation SFT reproduces the AF/RF split under student weights (Figures 5.3-5.9). Those spec plots include mixed-order RF+AF only for Qwen2.5-VL, Qwen3-VL is drawn in RF and AF alone. For a fixed Qwen3-teacher student, AF lies below RF on every corpus, the single exception being LLaVA-1.5 on VQA-RAD (overall 0.253 against 0.235), the mixed-order Qwen2.5-VL curve is intermediate on VQA-RAD and jointly best on SLAKE (section 5.4). LLaVA-1.5 RF on SLAKE remains the worst plotted curve (overall AURC 0.737), matching the format and accuracy collapse of the Qwen3-RF student. The practical selective-prediction implication under RQ1 is that the abstention score cannot be chosen independently of the generation contract. The same entropy definition is a useful selector under answer-only and AF explanation, and a near-constant under RF explanation.

Exact coverage at 1 %, 5 %, and 10 % risk is not tabulated from the dense threshold grid, but it is already constrained by these integrals. Wherever AURC is close to full-set risk, no threshold yields both high coverage and low risk. Wherever AURC is an order of magnitude below full-set risk, as on SLAKE and Med-VQA-2019 open answer-only and AF explanation SFT rows, combined-uncertainty ranking does increase coverage at controlled risk. The selective-prediction aspect of RQ1 is therefore answered affirmatively for answer-only and answer-first explanation on the larger splits, and negatively for rationale-conditioned emission: multimodal uncertainty estimates improve selective prediction only when the generation order leaves residual variation in the answer-token distribution.

### 6.1.5 Medically pretrained models

If a biomedical checkpoint were, by itself, a reliability upgrade under RQ1, these two should have been the stronger systems on radiology VQA. Table 5.8 shows the opposite (section 5.5). LLaVA-Med fails first as an instruction follower, not as a medical reasoner. Answer-only base parse on VQA-RAD is 60.31 %, zeroshot AF/RF JSON adherence is 17.07-48.5 %, which is why those closed accuracies sit at 0.000, two-stage prompting recovers parse only to 44.79-78.76 % and overall accuracy remains 0.115-0.180. Reading the AF/RF cells as biomedical incompetence would therefore be a category error: the structured output often never becomes a letter [2]. SFT restores 100 % parse on all three corpora, so format is not the remaining bottleneck, yet closed accuracy reaches only 0.466 on VQA-RAD and 0.485 on SLAKE, open accuracy can fall (0.170 / 0.171), and overall AURC stays in the 0.35-0.61 band. The residual gap is diagnostic because LLaVA-Med v1.5 keeps the LLaVA-1.5 CLIP encoder and MLP connector. Medical transfer replaces Vicuna with Mistral-7B and continues from PMC figure-caption alignment plus GPT-4 dialogues generated from text alone (Section 3.1). LLaVA-1.5 was additionally trained with short-answer formatting that matches Experiment 1 [56], and under identical letter targets its SFT overall accuracies are 0.663 / 0.866 / 0.900 against LLaVA-Med SFT 0.335 / 0.449 / 0.276. Medical concept chat is therefore a farther prior from a multiple-choice letter than the generalist short-answer recipe, and full SFT on these splits does not overwrite it. A caption-derived biomedical prior can support fluent anatomy language without forcing the model to use the radiograph [16, 106]. LLaVA-Med was omitted from explanation SFT for two reasons that compound. Its zeroshot JSON adherence collapses, so RF/AF supervision would have been scored through a format failure rather than through explanation quality, and its answer-only accuracy is the weakest of all students (0.335 / 0.449 / 0.276), so any explanation-SFT result would have started from a base far below the corpus-level effects under study. The format argument would not carry on its own. Under the two-stage letter-only variant, in which stage 2 asks only for a letter, LLaVA-Med parses at 98.45 %, 99.80 %, and 99.62 % and reaches overall 0.386 on VQA-RAD and 0.493 on SLAKE, above its own answer-only fine-tune on both, so the exclusion rests on the accuracy gap as much as on JSON (Table A.6, Appendix A).

MedVLM-R1 [5] fails for different reasons. It is an off-the-shelf GRPO explanation wrapper on Qwen2-VL-2B, about a quarter the scale of the $\approx$ 7-8 B students, and it is not fine-tuned on the letter splits of this thesis (Section 3.1). Its native `<think>`/`<answer>` schema is closer to a parseable contract than LLaVA-Med JSON, which is why it outscores every LLaVA-Med row (overall 0.552 / 0.545 / 0.600) while remaining below every Qwen zeroshot mode. Scale, missing in-domain letter SFT, and a specialised tag schema are confounded, so the table cannot isolate "GRPO", it can isolate the practical claim that a medically specialised explanation checkpoint is not automatically stronger than a stronger generalist backbone prompted in the same letter protocol. Because the shared $t^*$ locator does not cover a tag-internal letter, entropy, confidence, agreement, and AURC were not computed for that checkpoint (Section 4.5.2), so MedVLM-R1 also cannot enter the uncertainty-measurement, calibration, or selective-prediction comparison under RQ1.

Domain-adapted checkpoints are not a shortcut to reliable explanation-based Med-VQA on these benchmarks. The generalist students that follow the output contract dominate both systems on accuracy, and only they yield usable answer-token rankings. Medical pretraining, as used here, brought neither protocol match nor visual dependence.

### 6.1.6 Visual grounding

Image replacement tests whether letter match and token-level certainty depend on the visual evidence. Tables 5.2–5.6 report black-image and shuffled-image stacked overall ablations (Acc., letter and explanation entropy and confidence, agreement, and AURC). Table A.10 is the original-image counterpart of the explanation-SFT letter versus explanation split.

Qwen overall letter match falls under black or shuffled images by about 0.09-0.13 on VQA-RAD, 0.12-0.28 on Med-VQA-2019, and 0.23-0.31 on SLAKE. LLaVA-1.5 AF on VQA-RAD is nearly flat (drop 0.007). Residual accuracy stays above chance on almost every row, in line with the language priors already noted for answer-only SFT (subsection 6.1.1) [16, 19, 21]: 0.512-0.652 on VQA-RAD against chance 0.367, 0.517-0.782 on Med-VQA-2019 against 0.238, and 0.530-0.666 on SLAKE against 0.300. The exception is LLaVA-1.5 RF on SLAKE at 0.248 under black and 0.234 under shuffled images, which is below chance but also below chance on original images, because that row fails on output format rather than on visual evidence (subsection 6.1.4). The one split that does fall to chance is Qwen3-VL AF on the 64 closed Med-VQA-2019 items under black images (0.500), while its open split stays at 0.814 (Table A.12). The RF letter channel seems to be effectively blind to missing evidence. On all three corpora, letter entropy stays at or near 0 and letter confidence at or near 1.00 after black or shuffled replacement, indistinguishable from Table A.10. Once the rationale has been emitted, the letter token measures consistency with that trace rather than image use which corresponds to the same autoregressive saturation already seen under zeroshot RF (subsection 6.1.2). Explanation entropy and confidence move only slightly while the clearest RF explanation-entropy rise is on SLAKE (Qwen2.5-VL RF from 0.384 on original images to 0.458 under black images, Qwen3-VL RF from 0.290 to 0.460). A more uncertain rationale with a still saturated letter is not a visual grounding signal. The AF letter channel can register a black image. For Qwen2.5-VL AF, letter entropy rises by about 0.15-0.19 under black images (VQA-RAD 0.309 to 0.473, Med-VQA-2019 0.098 to 0.288, SLAKE 0.072 to 0.226), and letter confidence falls with it.

Shuffle moves those letter scores even less. Explanation entropy and confidence stay close to the original-image table, so the extra doubt is concentrated on the answer token, not on the rationale span. Under AF black images the letter becomes more uncertain than the rationale on all three corpora (Qwen3-VL AF letter entropy 0.801 versus explanation entropy 0.585 on VQA-RAD, 0.614 versus 0.394 on SLAKE, 0.469 versus 0.437 on Med-VQA-2019). That dissociation is useful as an ablation readout, it is not evidence that the original-image AF student was visually grounded. The mixed-order RF+AF student of section 5.4 is the only checkpoint whose rationale channel moves substantially, and it moves inconsistently: explanation entropy falls from 0.494 to 0.184 under black images on VQA-RAD, rises from 0.285 to 0.508 on Med-VQA-2019, and stays flat on SLAKE (0.290 to 0.288). The shuffled condition reproduces each of those shifts, so they are stable rather than noise, but a channel that answers the same intervention with opposite signs on different corpora is not tracking evidence availability either. LLaVA-1.5 does not provide that readout as the letter confidence remains 1.00 in both RF and AF, and explanation entropy and confidence are almost invariant to black and shuffled images. On SLAKE, AF letter match nevertheless falls from 0.773 to 0.658 under black images (drop 0.115), so high remaining certainty is compatible with language priors even when accuracy moves. That matches the near-flat VQA-RAD accuracy drop and the answer-only LLaVA ablation in subsection 6.1.1. The same caution already applies to zeroshot JSON and to answer-only SFT: Qwen

drops more than LLaVA, residual accuracy stays high, and VQA-RAD is the weakest visual test because of near-complete train-test image overlap [79, 16].
Figure 5.10 shows the same pattern at the item level for mixed-order Qwen2.5-VL AF on Med-VQA-2019: two originally correct letters become wrong after a shuffle, letter entropy rises, letter confidence falls, and the traces describe the replacement image rather than the source anatomy. The SHAP panels of section 5.7 (Figures 5.11 and 5.12) do not lift that caution. On the original MR-weighting image the answer-only and zeroshot letters are wrong and explanation SFT recovers the ground-truth letter. Under a black image those two protocols switch to T1 while explanation SFT flips to T2, and the zeroshot trace still invents white-matter signal on an empty field. A shuffled chest radiograph yields T1 from all three protocols, including a zeroshot trace that correctly names an X-ray and still emits the MRI letter. On the original closed T2 item all three protocols predict the wrong letter. Black and shuffled replacement make every protocol letter-correct because the black field or the chest radiograph is no longer read as T2-weighted MRI. The accompanying traces are not a faithfulness check as a correct letter can sit on a heatmap that resembles an incorrect one, a fluent explanation can still mis-describe the image (a T1 letter with T2-like hyperintensity reasoning on the original MR-weighting item), and letter match can improve after the evidence has been removed.
The operational warning under RQ1 is therefore the combination of fluent traces with saturated letter confidence after the image has been removed. That profile is a language-prior and shortcut signature, not a calibrated visual belief [106, 16, 35].

### 6.1.7 Cross-protocol synthesis

Taken together, the Acc.-best slots of Tables 5.1-5.5 give a compact answer to RQ1.
On accuracy, explanation-based generation is not a reliable upgrade over letter-only SFT. Zeroshot JSON CoT is consistently the weakest Acc.-best protocol (overall 0.694 / 0.693 / 0.723). In the matched RF/AF matrix the Acc.-best overall cells of explanation SFT and answer-only SFT differ by 0.005, 0.010, and 0.012 (VQA-RAD 0.727 vs. 0.732, Med-VQA-2019 0.904 vs. 0.894, SLAKE 0.888 vs. 0.900). Those gaps are two, about five, and about thirteen items, a single run cannot treat them as a protocol ranking. The mixed-order additional test of section 5.4 is what can reverse those two comparisons (overall 0.741 and 0.907). On Med-VQA-2019 and SLAKE open items, answer-only Qwen3-VL SFT (0.913) and LLaVA-1.5 SFT (0.918) remain at or above the matched explanation students on accuracy, and they keep the lower AURC. Rationale order and teacher identity affect these small gains more than the presence of an explanation as such: AF, and mixed RF+AF where it is run, are the explanation contracts that do not systematically trade ranking for letter match, and no teacher is uniformly superior. The working conclusion is that verified-rationale SFT matches answer-only accuracy to within a few items on the RF/AF matrix, whereas prompt-only explanation does not, and neither strategy produces the large explanation dividend that accuracy-only Medical VQA papers often attribute to CoT [5, 30].
On uncertainty measurement and calibration, the shared answer-token scores do quantify uncertainty comparably across regimes, but they do not mean the same thing in every regime. Under answer-only SFT, lower entropy and higher confidence typically accompany higher accuracy and lower AURC, so the scores behave as residual doubt about the letter. Under rationale-first zeroshot and rationale-first SFT, the same scores saturate while AURC worsens: they then measure self-consistency with $r$, not correctness [35]. Answer-first emission restores an interpretable letter

distribution in both prompting and distillation. Reporting entropy, confidence, and agreement without AURC would therefore have reversed several protocol rankings and hidden the overconfidence that RQ1 is meant to expose [28, 38].
On selective prediction, the scores work when they still rank, and fail when they do not. Comparing the lowest within-protocol AURC slots of Experiments 1-2, the leading closed values on VQA-RAD lie close (0.116-0.124). On open items explanation SFT AF (0.109) slightly undercuts answer-only (0.117), while zeroshot AF is worse (0.186). On Med-VQA-2019 and SLAKE, the lowest answer-only and explanation SFT curves nearly coincide and zeroshot sits higher on both splits. The AURC-best slots are not always Acc.-best, which is itself a finding: Qwen2.5-VL RF is Acc.-best zeroshot overall on VQA-RAD, yet Qwen3-VL AF is the better selector. Mixed-order RF+AF can be Acc.-best among explanation students on VQA-RAD (section 5.4), yet AF is the better selector. Choosing a deployment operating point by accuracy alone would therefore pick a worse abstention policy than choosing by AURC. Vote entropy in the ranking mix helps unstable LLaVA zeroshot rows, but it does not replace a well-behaved AF or answer-only letter distribution.
Visual dependence and qualitative attribution cut across all three generative protocols and limit how far the RQ answers should be pushed. subsection 6.1.6 reads those ablation scores against original-image letter and explanation certainty. The SHAP panels of section 5.7 (Figures 5.11 and 5.12) are consistent with that caution: a correct or incorrect letter can sit on similar-looking heatmaps, a fluent trace can still misdescribe the image, and letter match can rise under black or shuffled replacement for the wrong reason.

## 6.2 Health-Economic Implications

No peer-reviewed or recognised-preprint health-economic evaluation of Medical VQA (a cost-effectiveness, cost-utility, or budget-impact analysis) was identified, and the experiments are laboratory letter-prediction protocols rather than prospective workflow trials [42, 81]. What can be derived is which empirical regimes could support workload reduction, targeted oversight, and sustainable deployment, and which would raise verification, error, and liability costs: candidate inputs to such an evaluation, not an ICER or a QALY estimate.

### 6.2.1 Clinical workload

Imaging AI is discussed as a response to rising demand and radiologist shortages [84, 9]. Economic value for short, high-frequency tasks such as chest-radiograph interpretation has been reported mainly in shortage settings, where small per-case savings become material only in aggregate [42]. Medical VQA sits in that class: a one-point accuracy gain is unlikely to move a budget, whereas a reliable rule for which questions the model may answer unsupervised could. The strongest imaging reference that couples safety with resource use remains the randomised MASAI trial (higher detection without additional false positives and a 44.3 % reduction in screen-reading workload) [41, 86]. Medical VQA still lacks that coupling because the present results only indicate where it is technically plausible.
On SLAKE and Med-VQA-2019, Acc.-best answer-only and answer-first explanation SFT rows reach overall accuracies around 0.89-0.91, so full-set risk is already near 0.10. Within each of those splits, combined-uncertainty ranking pushes residual risk

well below that level on the answered subset, so auto-answering high-coverage, low-risk items and deferring the rest is a workload mechanism. VQA-RAD does not offer this. Its Acc.-best overall row stands at accuracy 0.732, a full-set risk of 0.268, and although ranking does lower risk on the answered subset there too, it starts from far higher, so a much larger share of items has to be withheld to reach a comparable absolute risk. Because the combined score is rescaled per dataset-split combination (Section 3.7), this contrast rests on the accuracy levels and on the behaviour of each corpus within its own split, not on comparing AURC values between corpora. Explanation-based generation does not enlarge that mechanism on accuracy grounds. Explanation SFT improves Acc.-best overall letter match only under some orders and corpora, zeroshot explanation CoT is substantially weaker on the two larger corpora, and both add tokens, two-stage passes, and $K = 5$ agreement samples (Table 4.8). If a clinician must read a `REGION/OBSERVATION/EXPLANATION` block before acting, the rationale is an extra reading task. Thus, the qualitative panel shows RF, AF, and RF+AF emitting equally fluent, mutually contradictory observations for the same image (Table A.3, Appendix A).

Selective prediction is therefore the workload lever that RQ1 supplies, and it is not cost-neutral because abstention reallocates items to a human reader. In a clinician vignette study, accuracy under incorrect AI predictions fell from 66 % to 56 % and recovered to 64 % under selective prediction, yet missed diagnoses rose by 18 % and omitted treatments by 35 % relative to a no-AI baseline when abstention was communicated [89]. A related risk is densification rather than capacity gain. Among 6,726 radiologists, AI use was associated with higher adjusted burnout odds and a dose-response to usage frequency [85]. Whether time saved by auto-answering easy items is released as spare capacity is an organisational choice, not a property of AURC.

### 6.2.2 Trust and oversight costs

Diagnostic error already imposes large clinical and societal costs through preventable harm, delayed treatment, and unnecessary follow-up [43]. Overconfident incorrect outputs amplify automation bias and liability, whereas untrusted systems force extensive manual verification and erase productivity gains [26, 28, 91]. Rationale-first emission is the expensive profile: in both zeroshot explanation and explanation-based SFT it saturates confidence and collapses entropy while raising AURC, so a reviewer who treats high confidence as a trust signal retains exactly the items the model is most willing to assert [35]. Answer-only SFT and answer-first explanation leave residual entropy, so a finite review budget can be spent on the high-entropy tail. That is the operational content of treating abstention thresholds as health-economic decision variables [90, 12]. Combined uncertainty ($\alpha = 0.5$) can cheapen oversight when vote entropy still moves, but it does not repair a peaky, consistently wrong predictor. The $K = 5$ agreement samples are themselves a recurring inference cost and belong in any future budget-impact analysis alongside integration and monitoring, which CDSS evaluations often omit [83].

Fluent explanations can raise perceived trust without lowering residual error. Simplified radiology reports show large understandability gains together with a 7.2 % error rate and clinically relevant errors in 0.9 % of cases [87]. Patient-facing summaries can improve comprehension and reduce anxiety [88] only if residual error is costed as work-up, false reassurance, and loss of trust. The present rationales are gold-guided, text-verified teacher traces, not clinician-authored proofs (Section 3.5). Paying clinicians to read or to trust such traces without a ranking signal is an oversight cost while

using them as visual proofs is a liability cost. Shortcut answering adds a hidden error cost: where ablations leave letter match nearly unchanged, as for LLaVA-1.5 SFT on VQA-RAD, a system can remain confident after visual evidence has disappeared [16]. Format failures of the kind discussed for LLaVA-Med in subsection 6.1.5 are a smaller operational cost: unparsable rows generate retries or silent drops rather than documented deferrals.

### 6.2.3 Conditions for economically sustainable deployment

Reviews of imaging-AI economics converge on a short list: the model takes over a resource-intensive task, is accurate enough for the clinical need, and is paid for in a way that does not erase the saving, typically in high-volume or shortage settings [42, 82]. Matching a human reader in mean accuracy is not required: a slightly weaker but temporally more stable system, or one that abstains on the unstable tail, can still be the relevant comparator. Compared with usual care without AI, diagnostic AI is harder to justify than therapeutic AI: it typically adds a layer to an existing pathway rather than displacing a high-cost treatment, while implementation, maintenance, and bias correction are first-order incremental costs [82, 83, 92]. Medical VQA, as evaluated here, meets those conditions only under a restricted reading.

The generation contract must preserve a usable uncertainty ranking: answer-only SFT and answer-first explanation SFT on SLAKE and Med-VQA-2019 do, rationale-first prompting and SFT do not, however fluent the traces might seem. Zeroshot explanation is not a low-cost substitute for in-domain adaptation as it trails answer-only fine-tunes by about 0.18-0.20 overall accuracy on the two larger corpora, records higher AURC, and spends additional generation budget. A one-point overall gain from explanation-based SFT does not justify extra training, longer targets, and extra reading time. If explanations have economic value, it is more likely as communication adjacent to report simplification [87, 88] than as a diagnostic-accuracy function, a use case not tested here. The evidence is uneven with SLAKE and Med-VQA-2019 showing high fine-tuned accuracy, low AURC, and non-trivial image ablations across answer-only, zeroshot explanation, and explanation SFT, whereas VQA-RAD combines a small test set, image leakage, and weaker visual dependence [79], so its headline numbers should not be used as a health-economic input. A future cost-effectiveness, cost-utility, or budget-impact analysis would have to cost the following input quantities: time-motion for letter-only versus rationale reading, expected clinical cost of false positives versus false negatives at a chosen risk, compute including agreement sampling, integration and monitoring, and distributional error [83, 90, 92]. Those inputs would enter a decision-analytic model of the imaging pathway versus usual care, for example a linked-evidence or Markov model in which selective risk and coverage become transition probabilities between auto-answer, deferral, false-positive, and false-negative states [107]. Whether a slightly less reliable but cheaper system would then be acceptable depends on the jurisdictional Health Technology Assessment (HTA) decision rule, not on the ICER alone. In cost-utility systems such as that of National Institute for Health and Care Excellence (NICE), a south-west-quadrant profile (lower effect and lower cost) can in principle be accepted if the savings compensate for the health loss, or if non-inferiority supports cost-minimisation [107]. In the German statutory system, following Institute for Quality and Efficiency in Health Care (IQWiG) methods, coverage of a new method typically requires a demonstration of patient-relevant benefit relative to usual care. Cost savings without at least non-inferior outcomes are unlikely to suffice [108]. Regulatory lag and liability insurance

may shift who bears residual harm, but they do not replace those inputs [93, 44].
The implication is therefore not that explanation-based Medical VQA would be judged cost-effective relative to usual care, nor that it would not. The only configuration in which accuracy, ranking, and a plausible workload mechanism co-occur is selective answering with a contract that leaves residual variation in the answer-token distribution, on the larger and more image-dependent splits, with human oversight concentrated on the deferred tail. Explanation-first generation, zeroshot CoT as a replacement for SFT, and VQA-RAD-style headline accuracy do not satisfy that configuration. Those are the conditions under which a later cost-effectiveness or cost-utility analysis would have a technical starting point. They are not themselves that analysis, and they do not yield an ICER.

# 7 | Conclusion

This chapter provides a compact synthesis: the answers that RQ1 and RQ2 support, the limitations faced in this thesis, and the work that is left open.

## 7.1 Summary of Contributions

The thesis asked whether explanation-based Medical VQA is more reliable than letter-only generation, not only more accurate, and whether that reliability has a health-economic reading. The evaluation was a matched multiple-choice letter-prediction protocol on VQA-RAD, Med-VQA-2019, and SLAKE. Answer-only generation was compared with zeroshot JSON CoT and with supervised fine-tuning on text-verified teacher rationales, under shared answer-token uncertainty scores and risk-coverage analysis.

Under RQ1, explanation-based generation is not a reliable accuracy upgrade over letter-only SFT. Zeroshot JSON CoT is the weakest Acc.-best protocol (overall 0.694 / 0.693 / 0.723 on VQA-RAD, Med-VQA-2019, and SLAKE). In the matched RF/AF matrix the Acc.-best overall cells of explanation SFT and answer-only SFT differ by two, about five, and about thirteen items, a single run cannot treat that as a protocol ranking. The mixed-order additional test can reverse those two accuracy comparisons on VQA-RAD and SLAKE, while it underperforms AF on Med-VQA-2019 and does not restore AF ranking on VQA-RAD. Order and teacher identity dominate the mere presence of an explanation: answer-first and mixed RF+AF contracts do not systematically trade ranking for letter match, rationale-first often does, and no teacher is uniformly superior. Verified-rationale SFT matches answer-only accuracy to within a few items on the RF/AF matrix, prompt-only explanation does not, neither yields the large explanation dividend that accuracy-only Medical VQA papers often attribute to CoT [5, 30].

On uncertainty measurement, answer-token entropy, confidence, and sample agreement quantify uncertainty comparably across regimes, but they do not mean the same thing in every regime. Under answer-only SFT they track residual doubt about the letter. Under rationale-first zeroshot and rationale-first SFT they saturate while AURC worsens: they then measure self-consistency with the model's own trace rather than correctness [35]. Answer-first emission restores an interpretable letter distribution in both prompting and distillation. Reporting those scores without AURC would have reversed several protocol rankings and hidden the overconfidence that the calibration aspect of RQ1 is meant to expose.

On selective prediction, the scores work when they still rank, and fail when they do not. On SLAKE and Med-VQA-2019, answer-only and answer-first explanation

SFT already support low-AURC ranking. On VQA-RAD the same specifications are weaker: overall AURC 0.115 at accuracy 0.732 for the best answer-only run still implies substantial residual error as coverage expands. Choosing a deployment operating point by accuracy alone can therefore pick a worse abstention policy than choosing by AURC. Combined uncertainty helps unstable LLaVA zeroshot rankings. It does not repair a peaky, consistently wrong predictor.
For RQ2, the only configuration in which accuracy, ranking, and a plausible workload mechanism co-occur is selective answering with a generation contract that leaves residual variation in the answer-token distribution, on the larger and more image-dependent splits, with human oversight concentrated on the deferred tail. Explanation-first generation, zeroshot CoT as a replacement for SFT, and VQA-RAD-style headline accuracy do not satisfy that configuration. Those are the conditions under which a later cost-effectiveness or cost-utility analysis would have a technical starting point. They are not themselves that analysis.
Relative to the agenda stated in Chapter 1, the thesis therefore contributes the following.

1. **Matched evaluation protocol.** It operationalises Medical VQA as multiple-choice letter prediction with closed, open, and overall splits on VQA-RAD, Med-VQA-2019, and SLAKE, so that answer-only and explanation-based regimes can be compared under a shared accuracy and uncertainty definition.

2. **Reliability evaluation of explanation-based Medical VQA.** It shows that verified-rationale SFT matches letter-only accuracy to within a few items on the RF/AF matrix, that zeroshot JSON CoT does not replace in-domain adaptation, and that rationale order and teacher identity affect those cells more than the presence of an explanation as such. Along the three aspects of RQ1, answer-token entropy, confidence, and sample agreement are comparable across regimes, yet rationale-first emission converts them from residual doubt about the letter into self-consistency with the trace. The same scores improve selective answering, as measured by risk-coverage curves and AURC, only when the generation contract leaves residual variation in the answer-token distribution.

3. **Visual-dependence diagnostics.** It complements original-image accuracy with black- and shuffled-image ablations for answer-only, zeroshot, and explanation SFT evaluation, with item-level shuffle examples, and with qualitative SHAP overlays, so that apparent competence can be inspected for image use versus language priors.

4. **Health-economic interpretation.** It indexes workload, trust, and oversight implications to that technical split, without claiming a formal health-economic evaluation of a Medical VQA system.

## 7.2 Limitations

The claims above are bounded by the evaluation design. The experiments are laboratory letter-prediction protocols, not prospective workflow trials. Free-text radiology questions were converted into multiple-choice letter targets with closed two-way yes/no options and five-way open items. Exact-match letter accuracy is therefore the

headline correctness metric, and the results do not speak directly to open-ended clinical report generation, reader time, or diagnostic safety in a live imaging service.
The evidence is also uneven across corpora. VQA-RAD combines a small test set with near-complete image overlap between train and test [79]. High letter accuracy with a small ablation drop is compatible with question-level shortcuts over reused studies rather than study-level visual grounding [16, 14]. SLAKE and Med-VQA-2019 support stronger claims: fine-tuned accuracies around 0.89-0.91, low AURC, and non-trivial drops under black and shuffled images for answer-only, zeroshot JSON, and explanation SFT. Even there, residual accuracy after image replacement stays far above chance: the answer-only fine-tunes keep 0.740-0.796 on Med-VQA-2019, where chance is 0.238, so removing the image costs roughly a tenth of the accuracy and still leaves more than three times the chance level. Language priors are reduced, not eliminated. Original-image letter match can fall under image replacement while letter confidence stays high, especially under RF explanation SFT and for LLaVA-1.5 in both orders (Tables 5.2–5.6, Table A.10). Visual grounding is not established when certainty is invariant to missing evidence: high letter confidence after black or shuffled replacement is a language-prior and shortcut signature, not a calibrated visual belief. On VQA-RAD, LLaVA-1.5 remains nearly invariant under explanation SFT as under answer-only SFT, so the one-point original-image rationale-SFT accuracy gains cannot be certified as additional image use.
Explanations in this thesis are gold-guided, text-verified teacher traces, not clinician-authored proofs. Fluent `REGION/OBSERVATION/EXPLANATION` text can be generated in support of either letter, as the qualitative VQA-RAD panel in Table A.3 (Appendix A) illustrates, and is therefore closer to a communication artefact than to an audit trail [106, 5]. The SHAP overlays are illustrative, panel-normalised, and not faithfulness evidence: a correct or incorrect letter can be associated with similar-looking heatmaps. LLaVA-Med is out of scope for explanation SFT after its zeroshot format failures, and MedVLM-R1 cannot enter the uncertainty-measurement, calibration, or selective-prediction comparison under RQ1 because the shared $t^*$ locator does not cover a tag-internal letter. Exact coverage at 1 %, 5 %, and 10 % selective risk is not tabulated from the dense threshold grid. AURC constrains those operating points but does not replace them.
Finally, RQ2 is an implication from laboratory rankings, not a health-economic evaluation. Candidate inputs such as time-motion for letter-only versus rationale reading and the expected cost of false positives versus false negatives are not estimated, and no cost-effectiveness, cost-utility, or budget-impact analysis is conducted. Headline VQA-RAD numbers should not be used as a health-economic input.

## 7.3 Future Work

Several extensions follow directly from those limits. First, tabulating coverage at fixed selective risk, and risk at fixed coverage, would turn the AURC integrals of the selective-prediction analysis into explicit operating points of the kind used in discriminative VQA [12].
Second, grounding and faithfulness need a stronger test than original-image accuracy or panel-wise SHAP. Order-consistency between answer-first and rationale-first decoding, region-claim alignment, and clinician review of traces would distinguish a useful communication artefact from a post-hoc story [33, 106]. Until that test exists, explanations should not be treated as visual proofs.

Third, where rationale-first emission destroys the ranking content of answer-token scores, other selectors are required if selective prediction is to remain the reliability mechanism. Conformal prediction, agreement-based consistency beyond $K = 5$, and verification procedures that are not conditioned on the model's own trace are the natural candidates [90, 35, 38]. The present results already caution against treating verbalised confidence or rationale-conditioned token probabilities as drop-in calibrators.

Fourth, evaluation should move closer to the clinical distribution that deployment would actually see. Contamination-free splits, additional modalities and sites, and semantically scored open-ended answers would test whether the SLAKE and Med-VQA-2019 ranking geometry survives outside the present letter-prediction artefacts. A reader study would be required before workload claims can be more than technical plausibility.

Fifth, a formal health-economic evaluation remains to be done. It would have to cost the input quantities this thesis leaves unpriced: time-motion for letter-only versus rationale reading, expected clinical cost of false positives versus false negatives at a chosen risk, compute including agreement sampling, integration and monitoring, and distributional error [83, 42, 92]. Those quantities would enter a decision-analytic model of the imaging pathway versus usual care, for example a linked-evidence or Markov model, whose acceptability then depends on the jurisdictional HTA decision rule [107, 108]. The contribution of RQ2 is to say which generation contracts are eligible for that study, not to replace it with a QALY or ICER figure.

Sixth, the generation-order and Medical VQA protocols of this thesis are preliminary groundwork for interactive, entailment-checked explanation in the Amenable programme [34]. Transferring the AF/RF diagnostic from laboratory letter prediction to expert-in-the-loop validation is a natural next step, but it is not demonstrated here.

In closing, explanation-based Medical VQA, under this protocol, is not a large accuracy dividend and is not automatically a reliability dividend. The usable lever is selective prediction, and only when generation order leaves residual variation in the answer-token distribution. That split, not a generic "explanation" system, is the technical starting point for any later claim about workload, trust, or economically sustainable deployment.

# A | Supplementary Material

This appendix collects material that supports Chapter 3, Chapter 4, and Chapter 5 but is omitted from the main text for compactness: a redundant model-architecture diagram, dataset content-type enumerations, a qualitative VQA-RAD output panel, an additional zeroshot two-stage letter-only (2S-L) test, original-image confidence and agreement, original-image letter and explanation scores for explanation SFT, closed and open ablation splits, and format-adherence rates for answer-only, zeroshot JSON, and explanation-based SFT. Original-image confidence and agreement are Tables A.8 and A.9. The original-image companion with letter versus explanation scores is Table A.10. Closed and open ablation splits are Tables A.11–A.13. Format-adherence matrices are collected here and discussed with LLaVA-Med and MedVLM-R1 in Section 5.5.

## A.1 Qwen3-VL architecture diagram

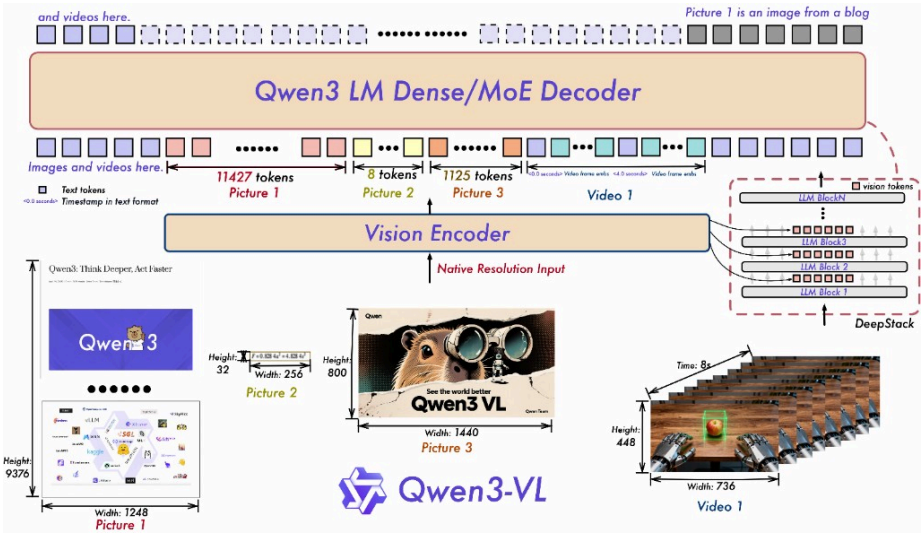



Figure A.1: Qwen3-VL architecture with native-resolution inputs, interleaved vision-text tokens, and DeepStack fusion into the Qwen3 decoder [4]. Deferred from Chapter 3 because the layout largely mirrors Figure 3.2.

## A.2 SLAKE question content types

| Content type | Train | Val. | Test | Total |
|---|---|---|---|---|
| Organ | 2,154 | 444 | 443 | 3,041 |
| Position | 1,884 | 395 | 399 | 2,678 |
| Knowledge-based | 1,213 | 260 | 267 | 1,740 |
| Abnormality | 1,177 | 266 | 253 | 1,696 |
| Modality | 1,044 | 224 | 224 | 1,492 |
| Plane | 896 | 192 | 192 | 1,280 |
| Quality | 534 | 114 | 114 | 762 |
| Colour | 453 | 97 | 97 | 647 |
| Size | 322 | 63 | 62 | 447 |
| Shape | 172 | 54 | 19 | 245 |
| **Total** | 9,849 | 2,109 | 2,070 | 14,028 |

Table A.1: SLAKE question content types by official split [6]. Counts are bilingual (English and Chinese).

## A.3 Qualitative VQA-RAD example

Figure A.2 and Table A.3 report greedy Qwen2.5-VL-7B outputs on one open-ended VQA-RAD test item under Experiments 1 and 2, deferred from section 5.1.




**Q:** how many instances of intussusception are in the image?
**Options:** A) one, B) three, C) none, D) two, E) four
**GT:** A (one)

Figure A.2: Qualitative example from VQA-RAD: one open-ended test item.

| Spec | Pred. | Output |
|---|---|---|
| **Experiment 1 (answer-only)** | | |
| Base | C | – |
| SFT | A | – |
| **Experiment 2 (zeroshot JSON)** | | |
| AF | C | The image shows a cross-sectional view of the abdomen, but there is no visible evidence of intussusception. Intussusception typically presents as a 'target sign' on CT scans, which is not observed here. |
| RF | C | The image shows a cross-sectional view of the abdomen with no visible signs of intussusception. Intussusception typically presents as a 'target sign' or 'sausage sign' on CT scans, which is not present here. |
| 2S | C | The image shows a cross-sectional view of the abdomen with no visible telescoping or overlapping of bowel loops that would suggest intussusception. The bowel appears to be in its normal position without any signs of telescoping. |

| Spec | Pred. | Output |
|---|---|---|
| **Experiment 2 (explanation SFT)** | | |
| RF | A | *Region:* Abdomen.<br>*Observation:* There is a single, well-defined, and centrally located area of bowel obstruction with a characteristic target or bull s-eye appearance, consistent with intussusception. The bowel loops appear to be telescoping into each other, with the inner bowel wall appearing as a central, hyperdense structure and the outer wall as a surrounding, hypodense rim.<br>*Explanation:* Intussusception typically presents as a single, classic radiological appearance on CT, which is a telescoping of one segment of the bowel into another. The image shows only one such abnormality, with no other similar structures or multiple telescoping events visible. This confirms the presence of only one instance of intussusception. |
| AF | C | *Region:* Abdomen.<br>*Observation:* There are no visible intussusception structures or abnormalities in the abdominal cavity.<br>*Explanation:* Intussusception typically presents as a telescoping of one segment of the bowel into another, often with a characteristic doughnut or target appearance on CT imaging. In this image, no such findings are present, and the bowel loops appear normal without evidence of obstruction or abnormal wall thickening. Therefore, the correct answer is none. |
| RF+AF | C | *Region:* abdomen.<br>*Observation:* There is no evidence of intussusception in the abdominal cavity on this CT scan.<br>*Explanation:* Intussusception typically presents as a "target sign" or "pseudo-kidney sign" on CT imaging, with a central low-density core (intussuscepted bowel) surrounded by a rim of soft tissue (intussuscepting bowel). In this image, no such configuration is visible. The bowel loops appear normal without signs of telescoping or obstruction. Additionally, there are no masses or abnormal fluid collections that would suggest intussusception. Therefore, the correct answer is none. |

Table A.3: Qualitative examples from VQA-RAD.

## A.4 Format adherence: Answer-only

| Model | Adaptation | VQA-RAD | Med-VQA-2019 | SLAKE |
|---|---|---|---|---|
| LLaVA-1.5 | Base | 100.0 | 100.0 | 100.0 |
| | SFT | 100.0 | 100.0 | 100.0 |
| Qwen2.5-VL | Base | 100.0 | 100.0 | 100.0 |
| | SFT | 100.0 | 100.0 | 100.0 |
| Qwen3-VL | Base | 100.0 | 100.0 | 100.0 |
| | SFT | 100.0 | 100.0 | 100.0 |

Table A.4: Answer-only format adherence: percentage of samples for which the predicted answer letter could be parsed correctly. Adaptations match the answer-only block of Table 5.1.

## A.5 Format adherence: Zeroshot JSON

| Model | Mode | VQA-RAD | Med-VQA-2019 | SLAKE |
|---|---|---|---|---|
| LLaVA-1.5 | AF | 100.0 | 100.0 | 100.0 |
| | RF | 99.33 | 100.0 | 99.81 |
| | 2S | 100.0 | 100.0 | 99.81 |
| Qwen2.5-VL | AF | 100.0 | 100.0 | 100.0 |
| | RF | 100.0 | 100.0 | 100.0 |
| | 2S | 100.0 | 100.0 | 100.0 |
| Qwen3-VL | AF | 100.0 | 100.0 | 100.0 |
| | RF | 100.0 | 100.0 | 100.0 |
| | 2S | 100.0 | 100.0 | 100.0 |

Table A.5: Zeroshot JSON format adherence: percentage of samples for which the structured output could be parsed correctly. Modes match the zeroshot block of Table 5.1 (AF: answer-first. RF: rationale-first. 2S: two-stage).

## A.6 Additional zeroshot test: 2S-L

Table A.6 reports the two-stage letter-only zeroshot variant deferred from Section 4.5.2: stage 1 generates a rationale without a letter. Stage 2 answers with a letter-only multiple-choice response.

**VQA-RAD**

| Model | Closed | | Open | | Overall | | Format |
|---|---|---|---|---|---|---|---|
| | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-1.5-7B | 0.574 | 0.448 | 0.305 | 0.671 | 0.455 | 0.584 | 100.0 |
| LLaVA-Med v1.5 | 0.474 | 0.505 | 0.275 | 0.641 | 0.386 | 0.534 | 98.45 |
| Qwen2.5-VL-7B | **0.745** | **0.127** | 0.585 | **0.275** | 0.674 | **0.238** | 100.0 |
| Qwen3-VL-8B | **0.745** | 0.202 | **0.605** | 0.320 | **0.683** | 0.252 | 100.0 |

**Med-VQA-2019**

| Model | Closed | | Open | | Overall | | Format |
|---|---|---|---|---|---|---|---|
| | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-1.5-7B | 0.453 | 0.566 | 0.460 | 0.532 | 0.459 | 0.535 | 100.0 |
| LLaVA-Med v1.5 | 0.500 | 0.476 | 0.386 | 0.503 | 0.401 | 0.498 | 99.8 |
| Qwen2.5-VL-7B | 0.688 | 0.254 | 0.612 | 0.304 | 0.621 | 0.301 | 100.0 |
| Qwen3-VL-8B | **0.750** | **0.181** | **0.667** | **0.265** | **0.677** | **0.257** | 100.0 |

**SLAKE**

| Model | Closed | | Open | | Overall | | Format |
|---|---|---|---|---|---|---|---|
| | Acc. | AURC | Acc. | AURC | Acc. | AURC | (%) |
| LLaVA-1.5-7B | 0.651 | 0.316 | 0.518 | 0.397 | 0.563 | 0.388 | 100.0 |
| LLaVA-Med v1.5 | 0.566 | 0.399 | 0.456 | 0.387 | 0.493 | 0.384 | 99.62 |
| Qwen2.5-VL-7B | 0.752 | **0.124** | 0.667 | **0.174** | 0.696 | **0.167** | 100.0 |
| Qwen3-VL-8B | **0.780** | 0.199 | **0.676** | 0.187 | **0.711** | 0.188 | 100.0 |

Table A.6: Additional zeroshot test: two-stage letter-only (2S-L). Stage 1 generates a rationale without a letter. Stage 2 answers with a letter-only multiple-choice response. Acc.: accuracy. Format: parse rate of the structured zeroshot output (%). Best Acc. and lowest AURC in each split are shown in bold.

## A.7 Format adherence: Explanation-based SFT

| Teacher | Model | Order | VQA-RAD | Med-VQA-2019 | SLAKE |
|---|---|---|---|---|---|
| GPT-4o | LLaVA-1.5 | RF | 100.0 | 100.0 | 100.0 |
| | | AF | 100.0 | 100.0 | 100.0 |
| | Qwen2.5-VL | RF | 99.78 | 100.0 | 100.0 |
| | | AF | 100.0 | 99.8 | 100.0 |
| | Qwen3-VL | RF | 100.0 | 100.0 | 100.0 |
| | | AF | 100.0 | 100.0 | 100.0 |
| Qwen3 | LLaVA-1.5 | RF | 100.0 | 100.0 | 77.29 |
| | | AF | 100.0 | 100.0 | 100.0 |
| | Qwen2.5-VL | RF | 100.0 | 99.8 | 100.0 |
| | | AF | 100.0 | 100.0 | 100.0 |
| | Qwen3-VL | RF | 100.0 | 100.0 | 100.0 |
| | | AF | 100.0 | 100.0 | 100.0 |

Table A.7: Explanation-based SFT format adherence: percentage of samples for which the structured JSON output could be parsed correctly. Teachers and orders match the explanation SFT block of Table 5.1 (RF: rationale-first. AF: answer-first).

## A.8 Original-image confidence and agreement

The stacked results tables report accuracy, entropy, and AURC. Tables A.8 and A.9 add the remaining two RQ1 uncertainty signals, answer-token confidence and sample agreement, for the same original-image rows of Experiments 1 and 2.

| | | | Closed | | Open | | Overall | |
|---|---|---|---|---|---|---|---|---|
| Corpus | Model | Adaptation | Conf. | Agr. | Conf. | Agr. | Conf. | Agr. |
| VQA-RAD | LLaVA-1.5 | Base | 0.644 | 0.791 | 0.531 | 0.717 | 0.594 | 0.758 |
| | | SFT | 0.961 | 0.986 | 0.948 | 0.970 | 0.955 | 0.979 |
| | Qwen2.5-VL | Base | 0.768 | 0.854 | 0.674 | 0.817 | 0.727 | 0.838 |
| | | SFT | 0.885 | 0.943 | 0.862 | 0.934 | 0.874 | 0.939 |
| | Qwen3-VL | Base | 0.931 | 0.973 | 0.878 | 0.936 | 0.908 | 0.957 |
| | | SFT | 0.712 | 0.822 | 0.738 | 0.859 | 0.724 | 0.839 |
| Med-VQA-2019 | LLaVA-1.5 | Base | 0.711 | 0.853 | 0.630 | 0.800 | 0.640 | 0.806 |
| | | SFT | 0.926 | 0.975 | 0.982 | 0.992 | 0.974 | 0.990 |
| | Qwen2.5-VL | Base | 0.818 | 0.894 | 0.771 | 0.871 | 0.777 | 0.874 |
| | | SFT | 0.903 | 0.941 | 0.933 | 0.971 | 0.929 | 0.967 |
| | Qwen3-VL | Base | 0.909 | 0.963 | 0.920 | 0.954 | 0.919 | 0.955 |
| | | SFT | 0.799 | 0.853 | 0.913 | 0.961 | 0.898 | 0.947 |
| SLAKE | LLaVA-1.5 | Base | 0.647 | 0.799 | 0.628 | 0.785 | 0.634 | 0.790 |
| | | SFT | 0.984 | 0.995 | 0.980 | 0.989 | 0.981 | 0.991 |
| | Qwen2.5-VL | Base | 0.830 | 0.898 | 0.792 | 0.880 | 0.804 | 0.886 |
| | | SFT | 0.942 | 0.974 | 0.939 | 0.970 | 0.940 | 0.972 |
| | Qwen3-VL | Base | 0.952 | 0.973 | 0.923 | 0.958 | 0.933 | 0.963 |
| | | SFT | 0.850 | 0.915 | 0.930 | 0.967 | 0.904 | 0.949 |

Table A.8: Answer-only SFT on original images: answer-token confidence (Conf.) and sample agreement (Agr.) per split. Accuracy, entropy, and AURC for the same rows are in Tables 5.1–5.5.

| | | | Closed | | Open | | Overall | |
|---|---|---|---|---|---|---|---|---|
| Corpus | Model | Mode | Conf. | Agr. | Conf. | Agr. | Conf. | Agr. |
| VQA-RAD | LLaVA-1.5 | AF | 0.570 | 0.751 | 0.773 | 0.646 | 0.660 | 0.704 |
| | | RF | 0.835 | 0.798 | 0.884 | 0.602 | 0.857 | 0.711 |
| | | 2S | 0.908 | 0.892 | 0.895 | 0.792 | 0.902 | 0.848 |
| | Qwen2.5-VL | AF | 0.768 | 0.859 | 0.711 | 0.829 | 0.743 | 0.846 |
| | | RF | 0.958 | 0.888 | 0.952 | 0.819 | 0.955 | 0.857 |
| | | 2S | 0.987 | 0.993 | 0.981 | 0.982 | 0.984 | 0.988 |
| | Qwen3-VL | AF | 0.940 | 0.973 | 0.912 | 0.954 | 0.928 | 0.965 |
| | | RF | 0.998 | 0.958 | 0.999 | 0.897 | 0.999 | 0.931 |
| | | 2S | 0.999 | 1.000 | 0.993 | 0.988 | 0.997 | 0.995 |
| Med-VQA-2019 | LLaVA-1.5 | AF | 0.494 | 0.742 | 0.807 | 0.746 | 0.767 | 0.746 |
| | | RF | 0.880 | 0.864 | 0.880 | 0.623 | 0.880 | 0.654 |
| | | 2S | 0.906 | 0.919 | 0.902 | 0.817 | 0.902 | 0.830 |
| | Qwen2.5-VL | AF | 0.826 | 0.912 | 0.800 | 0.887 | 0.803 | 0.891 |
| | | RF | 0.929 | 0.909 | 0.959 | 0.879 | 0.955 | 0.883 |
| | | 2S | 0.994 | 1.000 | 0.978 | 0.987 | 0.980 | 0.988 |
| | Qwen3-VL | AF | 0.933 | 0.959 | 0.945 | 0.977 | 0.943 | 0.974 |
| | | RF | 1.000 | 0.931 | 0.999 | 0.933 | 0.999 | 0.933 |
| | | 2S | 1.000 | 1.000 | 0.997 | 0.994 | 0.997 | 0.994 |
| SLAKE | LLaVA-1.5 | AF | 0.560 | 0.798 | 0.769 | 0.740 | 0.699 | 0.759 |
| | | RF | 0.867 | 0.834 | 0.880 | 0.625 | 0.876 | 0.695 |
| | | 2S | 0.904 | 0.870 | 0.896 | 0.780 | 0.898 | 0.810 |
| | Qwen2.5-VL | AF | 0.829 | 0.899 | 0.813 | 0.890 | 0.819 | 0.893 |
| | | RF | 0.951 | 0.899 | 0.963 | 0.886 | 0.959 | 0.890 |
| | | 2S | 0.983 | 0.994 | 0.974 | 0.981 | 0.977 | 0.985 |
| | Qwen3-VL | AF | 0.958 | 0.981 | 0.944 | 0.972 | 0.949 | 0.975 |
| | | RF | 0.999 | 0.963 | 0.995 | 0.933 | 0.996 | 0.943 |
| | | 2S | 1.000 | 0.998 | 0.996 | 0.992 | 0.997 | 0.994 |

Table A.9: Zeroshot JSON on original images: answer-token confidence (Conf.) and sample agreement (Agr.) per split (AF: answer-first. RF: rationale-first. 2S: two-stage). Accuracy, entropy, and AURC for the same rows are in Tables 5.1–5.5.

## A.9 Explanation SFT: original-image letter and explanation scores

Table A.10 reports original-image accuracy, letter and explanation entropy and confidence, and AURC for the Qwen3-teacher explanation SFT students. It is the comparison baseline for the explanation-SFT blocks of Tables 5.2–5.6.

**Closed (yes/no)**

| Dataset | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | AURC |
|---|---|---|---|---|---|---|---|---|
| VQA-RAD | LLaVA-1.5 | RF | 0.697 | 0.000 | 0.120 | 1.000 | 0.919 | 0.271 |
| | | AF | 0.673 | 0.000 | 0.123 | 1.000 | 0.915 | 0.235 |
| | Qwen2.5-VL | RF | 0.757 | 0.000 | 0.237 | 1.000 | 0.846 | 0.205 |
| | | AF | 0.761 | 0.274 | 0.558 | 0.896 | 0.697 | **0.124** |
| | | RF+AF | **0.773** | 0.040 | 0.465 | 0.983 | 0.730 | 0.143 |
| | Qwen3-VL | RF | 0.641 | 0.013 | 0.454 | 0.997 | 0.737 | 0.295 |
| | | AF | 0.649 | 0.610 | 0.565 | 0.699 | 0.675 | 0.223 |
| Med-VQA 2019 | LLaVA-1.5 | RF | 0.609 | 0.000 | 0.134 | 1.000 | 0.908 | 0.454 |
| | | AF | 0.813 | 0.000 | 0.180 | 1.000 | 0.895 | 0.083 |
| | Qwen2.5-VL | RF | 0.813 | 0.000 | 0.228 | 1.000 | 0.844 | 0.143 |
| | | AF | **0.859** | 0.134 | 0.256 | 0.944 | 0.827 | 0.068 |
| | | RF+AF | 0.766 | 0.184 | 0.311 | 0.923 | 0.795 | 0.108 |
| | Qwen3-VL | RF | 0.703 | 0.000 | 0.309 | 1.000 | 0.795 | 0.294 |
| | | AF | 0.828 | 0.364 | 0.399 | 0.843 | 0.749 | **0.055** |
| SLAKE | LLaVA-1.5 | RF | 0.417 | 0.000 | 0.058 | 1.000 | 0.958 | 0.437 |
| | | AF | 0.718 | 0.001 | 0.147 | 1.000 | 0.899 | 0.121 |
| | Qwen2.5-VL | RF | 0.831 | 0.001 | 0.318 | 1.000 | 0.802 | 0.088 |
| | | AF | 0.856 | 0.072 | 0.147 | 0.974 | 0.894 | 0.041 |
| | | RF+AF | **0.870** | 0.056 | 0.257 | 0.980 | 0.831 | **0.036** |
| | Qwen3-VL | RF | 0.859 | 0.001 | 0.258 | 1.000 | 0.830 | 0.064 |
| | | AF | 0.851 | 0.224 | 0.366 | 0.913 | 0.767 | 0.050 |

**Open-ended**

| Dataset | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | AURC |
|---|---|---|---|---|---|---|---|---|
| VQA-RAD | LLaVA-1.5 | RF | 0.655 | 0.000 | 0.155 | 1.000 | 0.897 | 0.207 |
| | | AF | 0.615 | 0.000 | 0.156 | 1.000 | 0.894 | 0.257 |
| | Qwen2.5-VL | RF | 0.645 | 0.015 | 0.308 | 0.994 | 0.808 | 0.184 |
| | | AF | 0.685 | 0.354 | 0.654 | 0.876 | 0.657 | **0.109** |
| | | RF+AF | **0.700** | 0.065 | 0.531 | 0.976 | 0.696 | 0.129 |
| | Qwen3-VL | RF | 0.630 | 0.056 | 0.522 | 0.983 | 0.701 | 0.193 |
| | | AF | 0.655 | 0.748 | 0.571 | 0.694 | 0.673 | 0.136 |
| Med-VQA 2019 | LLaVA-1.5 | RF | 0.782 | 0.000 | 0.131 | 1.000 | 0.910 | 0.088 |
| | | AF | 0.890 | 0.000 | 0.199 | 1.000 | 0.889 | 0.031 |
| | Qwen2.5-VL | RF | 0.903 | 0.005 | 0.208 | 0.998 | 0.859 | 0.026 |
| | | AF | **0.910** | 0.093 | 0.247 | 0.968 | 0.833 | 0.020 |
| | | RF+AF | 0.844 | 0.180 | 0.281 | 0.937 | 0.815 | 0.040 |
| | Qwen3-VL | RF | 0.901 | 0.004 | 0.282 | 0.998 | 0.814 | 0.037 |
| | | AF | 0.903 | 0.213 | 0.390 | 0.924 | 0.755 | **0.018** |
| SLAKE | LLaVA-1.5 | RF | 0.133 | 0.000 | 0.052 | 1.000 | 0.963 | 0.849 |
| | | AF | 0.800 | 0.001 | 0.168 | 1.000 | 0.887 | 0.077 |
| | Qwen2.5-VL | RF | 0.817 | 0.025 | 0.417 | 0.991 | 0.751 | 0.059 |
| | | AF | 0.904 | 0.071 | 0.151 | 0.971 | 0.891 | 0.015 |
| | | RF+AF | **0.925** | 0.064 | 0.307 | 0.975 | 0.801 | **0.014** |
| | Qwen3-VL | RF | 0.890 | 0.006 | 0.306 | 0.998 | 0.801 | 0.021 |
| | | AF | 0.885 | 0.316 | 0.378 | 0.880 | 0.758 | 0.023 |

| | | | Overall | | | | | |
|---|---|---|---|---|---|---|---|---|
| Dataset | Model | Order | Acc. | Ent.L | Ent.X | Conf.L | Conf.X | AURC |
| VQA-RAD | LLaVA-1.5 | RF | 0.679 | 0.000 | 0.136 | 1.000 | 0.909 | 0.235 |
| | | AF | 0.648 | 0.000 | 0.138 | 1.000 | 0.905 | 0.253 |
| | Qwen2.5-VL | RF | 0.707 | 0.007 | 0.269 | 0.998 | 0.829 | 0.197 |
| | | AF | 0.727 | 0.309 | 0.601 | 0.887 | 0.679 | **0.113** |
| | | RF+AF | **0.741** | 0.051 | 0.494 | 0.980 | 0.715 | 0.131 |
| | Qwen3-VL | RF | 0.636 | 0.032 | 0.484 | 0.991 | 0.721 | 0.264 |
| | | AF | 0.652 | 0.671 | 0.568 | 0.697 | 0.674 | 0.177 |
| Med-VQA 2019 | LLaVA-1.5 | RF | 0.760 | 0.000 | 0.131 | 1.000 | 0.910 | 0.106 |
| | | AF | 0.880 | 0.000 | 0.197 | 1.000 | 0.889 | 0.037 |
| | Qwen2.5-VL | RF | 0.892 | 0.004 | 0.211 | 0.998 | 0.857 | 0.043 |
| | | AF | **0.904** | 0.098 | 0.248 | 0.965 | 0.833 | 0.024 |
| | | RF+AF | 0.834 | 0.180 | 0.285 | 0.935 | 0.812 | 0.046 |
| | Qwen3-VL | RF | 0.876 | 0.004 | 0.285 | 0.999 | 0.812 | 0.090 |
| | | AF | 0.894 | 0.232 | 0.391 | 0.913 | 0.754 | **0.021** |
| SLAKE | LLaVA-1.5 | RF | 0.228 | 0.000 | 0.054 | 1.000 | 0.962 | 0.737 |
| | | AF | 0.773 | 0.001 | 0.161 | 1.000 | 0.891 | 0.088 |
| | Qwen2.5-VL | RF | 0.822 | 0.017 | 0.384 | 0.994 | 0.768 | 0.082 |
| | | AF | 0.888 | 0.072 | 0.150 | 0.972 | 0.892 | 0.022 |
| | | RF+AF | **0.907** | 0.062 | 0.290 | 0.977 | 0.811 | **0.020** |
| | Qwen3-VL | RF | 0.879 | 0.004 | 0.290 | 0.999 | 0.811 | 0.035 |
| | | AF | 0.874 | 0.285 | 0.374 | 0.891 | 0.761 | 0.033 |

Table A.10: Original-image explanation SFT for Qwen3-teacher students. Companion to the explanation-SFT blocks of Tables 5.2–5.6. RF+AF is the mixed-order additional test of section 5.4. Acc.: accuracy. Ent.L / Conf.L: answer-letter entropy and confidence. Ent.X / Conf.X: explanation-span entropy and confidence. Bold Acc. and AURC are within-dataset extrema.

## A.10 Closed and open ablation splits

Tables A.11–A.13 report the closed and open splits of the stacked black-image and shuffled-image ablations deferred from section 5.1–section 5.3. Overall Acc., letter and explanation entropy and confidence, agreement, and AURC remain in Tables 5.2–5.6.

| | | | | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| Experiment 1 | Image | Model | Adaptation | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | Black images | LLaVA-1.5 | Base | 0.518 | 0.891 | 0.403 | 0.295 | 1.268 | 0.609 |
| | | | SFT | **0.713** | 0.091 | **0.196** | 0.590 | 0.133 | 0.230 |
| | | Qwen2.5-VL | Base | 0.562 | 0.600 | 0.372 | 0.355 | 1.226 | 0.517 |
| | | | SFT | 0.625 | 0.437 | 0.240 | **0.610** | 0.438 | 0.250 |
| | | Qwen3-VL | Base | 0.582 | 0.142 | 0.359 | 0.385 | 0.535 | 0.498 |
| | | | SFT | 0.594 | 0.624 | 0.302 | **0.610** | 0.775 | **0.195** |
| | Shuffled images | LLaVA-1.5 | Base | 0.470 | 0.797 | 0.480 | 0.280 | 1.200 | 0.574 |
| | | | SFT | **0.697** | 0.105 | **0.197** | 0.570 | 0.143 | **0.217** |
| | | Qwen2.5-VL | Base | 0.594 | 0.360 | 0.354 | 0.395 | 0.875 | 0.542 |
| | | | SFT | 0.653 | 0.239 | 0.289 | 0.575 | 0.307 | 0.313 |
| | | Qwen3-VL | Base | 0.582 | 0.128 | 0.367 | 0.390 | 0.395 | 0.553 |
| | | | SFT | 0.661 | 0.599 | 0.311 | **0.600** | 0.704 | 0.284 |

| Experiment 2 | Image | Model | Mode | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | Black images | LLaVA-1.5 | AF | 0.514 | 1.575 | 0.512 | 0.255 | 0.850 | 0.700 |
| | | | RF | 0.518 | 0.766 | 0.428 | 0.245 | 0.539 | 0.706 |
| | | | 2S | 0.610 | 0.413 | 0.356 | 0.240 | 0.381 | 0.781 |
| | | Qwen2.5-VL | AF | 0.582 | 0.670 | 0.397 | 0.385 | 1.091 | **0.466** |
| | | | RF | **0.614** | 0.201 | **0.300** | **0.410** | 0.128 | 0.495 |
| | | | 2S | 0.606 | 0.068 | 0.383 | 0.405 | 0.057 | 0.524 |
| | | Qwen3-VL | AF | 0.566 | 0.120 | 0.389 | 0.400 | 0.392 | 0.514 |
| | | | RF | 0.554 | 0.029 | 0.405 | 0.305 | 0.181 | 0.578 |
| | | | 2S | 0.542 | 0.044 | 0.402 | 0.330 | 0.338 | 0.635 |
| | Shuffled images | LLaVA-1.5 | AF | 0.482 | 1.398 | 0.498 | 0.320 | 0.851 | 0.566 |
| | | | RF | 0.482 | 0.661 | 0.509 | 0.310 | 0.502 | 0.641 |
| | | | 2S | 0.514 | 0.363 | 0.488 | 0.255 | 0.375 | 0.641 |
| | | Qwen2.5-VL | AF | 0.610 | 0.421 | **0.339** | **0.405** | 0.775 | **0.508** |
| | | | RF | **0.614** | 0.147 | 0.358 | 0.385 | 0.285 | 0.516 |
| | | | 2S | 0.582 | 0.075 | 0.378 | 0.350 | 0.285 | 0.546 |
| | | Qwen3-VL | AF | 0.606 | 0.111 | 0.378 | **0.405** | 0.275 | 0.547 |
| | | | RF | 0.582 | 0.019 | 0.377 | 0.370 | 0.129 | 0.604 |
| | | | 2S | 0.606 | 0.016 | 0.378 | 0.370 | 0.167 | 0.671 |

| Experiment 2 | Image | Model | Order | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.649 | 0.000 | 0.313 | 0.575 | 0.000 | 0.286 |
| | | | AF | **0.721** | 0.000 | 0.281 | 0.540 | 0.000 | 0.340 |
| | | Qwen2.5-VL | RF | 0.626 | 0.001 | 0.373 | 0.575 | 0.008 | 0.288 |
| | | | AF | 0.649 | 0.403 | 0.260 | **0.610** | 0.561 | **0.208** |
| | | | RF+AF | 0.677 | 0.068 | **0.247** | 0.575 | 0.084 | 0.242 |
| | | Qwen3-VL | RF | 0.518 | 0.017 | 0.428 | 0.565 | 0.043 | 0.303 |
| | | | AF | 0.562 | 0.660 | 0.358 | 0.530 | 0.977 | 0.290 |
| | Shuffled images | LLaVA-1.5 | RF | 0.693 | 0.000 | **0.250** | **0.600** | 0.000 | **0.272** |
| | | | AF | **0.697** | 0.000 | 0.277 | 0.535 | 0.000 | 0.415 |
| | | Qwen2.5-VL | RF | 0.641 | 0.000 | 0.318 | 0.515 | 0.020 | 0.407 |
| | | | AF | 0.657 | 0.242 | 0.335 | 0.525 | 0.447 | 0.383 |
| | | | RF+AF | 0.618 | 0.039 | 0.315 | 0.525 | 0.083 | 0.402 |
| | | Qwen3-VL | RF | 0.546 | 0.029 | 0.421 | 0.470 | 0.056 | 0.428 |
| | | | AF | 0.610 | 0.585 | 0.336 | 0.500 | 0.809 | 0.396 |

Table A.11: VQA-RAD stacked closed and open ablations, companion to the overall stack of Table 5.2. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

| Experiment 1 | Image | Model | Adaptation | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | Black images | LLaVA-1.5 | Base | 0.484 | 0.816 | 0.525 | 0.269 | 1.262 | 0.700 |
| | | | SFT | 0.547 | 0.208 | **0.242** | 0.805 | 0.176 | **0.042** |
| | | Qwen2.5-VL | Base | 0.547 | 0.565 | 0.514 | 0.317 | 1.300 | 0.574 |
| | | | SFT | 0.578 | 0.424 | 0.283 | **0.828** | 0.397 | 0.050 |
| | | Qwen3-VL | Base | 0.484 | 0.105 | 0.539 | 0.409 | 0.655 | 0.494 |
| | | | SFT | **0.609** | 0.497 | 0.243 | 0.823 | 0.447 | 0.043 |
| | Shuffled images | LLaVA-1.5 | Base | 0.375 | 0.684 | 0.529 | 0.303 | 1.038 | 0.679 |
| | | | SFT | 0.578 | 0.189 | **0.243** | **0.823** | 0.079 | **0.040** |
| | | Qwen2.5-VL | Base | 0.531 | 0.183 | 0.426 | 0.281 | 0.688 | 0.693 |
| | | | SFT | **0.594** | 0.332 | 0.287 | 0.761 | 0.299 | 0.088 |
| | | Qwen3-VL | Base | 0.500 | 0.104 | 0.499 | 0.354 | 0.293 | 0.674 |
| | | | SFT | 0.531 | 0.416 | 0.293 | 0.777 | 0.355 | 0.071 |

| Experiment 2 | Image | Model | Mode | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | Black images | LLaVA-1.5 | AF | 0.531 | 1.534 | 0.439 | 0.276 | 0.833 | 0.718 |
| | | | RF | 0.500 | 0.881 | 0.492 | 0.232 | 0.584 | 0.722 |
| | | | 2S | 0.469 | 0.460 | 0.519 | 0.251 | 0.398 | 0.736 |
| | | Qwen2.5-VL | AF | 0.531 | 0.611 | 0.472 | 0.326 | 1.201 | 0.549 |
| | | | RF | **0.578** | 0.325 | 0.400 | 0.375 | 0.126 | 0.590 |
| | | | 2S | 0.391 | 0.046 | 0.614 | 0.320 | 0.043 | 0.737 |
| | | Qwen3-VL | AF | 0.484 | 0.146 | **0.360** | **0.407** | 0.489 | **0.498** |
| | | | RF | 0.438 | 0.049 | 0.495 | 0.301 | 0.298 | 0.674 |
| | | | 2S | 0.484 | 0.065 | 0.513 | 0.287 | 0.343 | 0.631 |
| | Shuffled images | LLaVA-1.5 | AF | 0.500 | 1.588 | 0.414 | 0.310 | 0.736 | **0.662** |
| | | | RF | 0.484 | 0.594 | 0.561 | 0.267 | 0.482 | 0.730 |
| | | | 2S | 0.453 | 0.404 | 0.449 | 0.246 | 0.360 | 0.729 |
| | | Qwen2.5-VL | AF | 0.531 | 0.249 | 0.407 | 0.310 | 0.624 | 0.676 |
| | | | RF | 0.531 | 0.140 | 0.420 | 0.326 | 0.231 | 0.703 |
| | | | 2S | **0.547** | 0.058 | 0.449 | 0.303 | 0.211 | 0.723 |
| | | Qwen3-VL | AF | **0.547** | 0.101 | 0.524 | **0.352** | 0.222 | 0.669 |
| | | | RF | **0.547** | 0.003 | 0.360 | 0.336 | 0.117 | 0.689 |
| | | | 2S | **0.547** | 0.030 | **0.358** | 0.317 | 0.092 | 0.692 |

| Experiment 2 | Image | Model | Order | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.625 | 0.000 | 0.386 | 0.789 | 0.000 | 0.100 |
| | | | AF | 0.578 | 0.000 | 0.469 | 0.812 | 0.000 | 0.100 |
| | | Qwen2.5-VL | RF | **0.656** | 0.000 | 0.347 | 0.749 | 0.005 | 0.090 |
| | | | AF | 0.594 | 0.295 | 0.345 | 0.763 | 0.287 | 0.070 |
| | | | RF+AF | 0.484 | 0.236 | 0.398 | 0.660 | 0.401 | 0.150 |
| | | Qwen3-VL | RF | 0.531 | 0.000 | **0.335** | 0.766 | 0.005 | 0.083 |
| | | | AF | 0.500 | 0.570 | 0.357 | **0.814** | 0.454 | **0.052** |
| | Shuffled images | LLaVA-1.5 | RF | 0.500 | 0.000 | 0.447 | **0.754** | 0.000 | **0.090** |
| | | | AF | 0.609 | 0.000 | 0.425 | 0.669 | 0.000 | 0.302 |
| | | Qwen2.5-VL | RF | 0.547 | 0.000 | 0.495 | 0.628 | 0.017 | 0.280 |
| | | | AF | **0.625** | 0.108 | **0.379** | 0.720 | 0.194 | 0.142 |
| | | | RF+AF | 0.547 | 0.136 | 0.475 | 0.513 | 0.275 | 0.368 |
| | | Qwen3-VL | RF | 0.547 | 0.001 | 0.452 | 0.676 | 0.015 | 0.211 |
| | | | AF | 0.563 | 0.280 | 0.498 | 0.703 | 0.352 | 0.175 |

Table A.12: Med-VQA-2019 stacked closed and open ablations, companion to the overall stack of Table 5.4. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

| Experiment 1 | Image | Model | Adaptation | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Answer-only | Black images | LLaVA-1.5 | Base | 0.515 | 0.913 | **0.389** | 0.307 | 1.158 | 0.624 |
| | | | SFT | 0.493 | 0.060 | 0.443 | **0.782** | 0.079 | 0.143 |
| | | Qwen2.5-VL | Base | 0.501 | 0.455 | 0.537 | 0.381 | 1.140 | 0.470 |
| | | | SFT | **0.552** | 0.494 | 0.467 | 0.669 | 0.496 | 0.157 |
| | | Qwen3-VL | Base | 0.496 | 0.056 | 0.448 | 0.382 | 0.475 | 0.514 |
| | | | SFT | 0.527 | 0.629 | 0.390 | 0.775 | 0.392 | **0.102** |
| | Shuffled images | LLaVA-1.5 | Base | 0.549 | 0.851 | 0.391 | 0.322 | 0.997 | 0.643 |
| | | | SFT | **0.578** | 0.035 | 0.399 | **0.713** | 0.073 | **0.257** |
| | | Qwen2.5-VL | Base | 0.527 | 0.276 | 0.466 | 0.405 | 0.622 | 0.550 |
| | | | SFT | 0.572 | 0.247 | 0.405 | 0.569 | 0.323 | 0.435 |
| | | Qwen3-VL | Base | 0.521 | 0.095 | 0.494 | 0.397 | 0.256 | 0.559 |
| | | | SFT | 0.569 | 0.302 | **0.387** | 0.669 | 0.256 | 0.306 |

| Experiment 2 | Image | Model | Mode | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Zeroshot | Black images | LLaVA-1.5 | AF | **0.566** | 1.461 | **0.408** | 0.292 | 0.817 | 0.652 |
| | | | RF | 0.496 | 0.722 | 0.504 | 0.275 | 0.547 | 0.688 |
| | | | 2S | 0.501 | 0.441 | 0.520 | 0.293 | 0.405 | 0.688 |
| | | Qwen2.5-VL | AF | 0.507 | 0.520 | 0.522 | 0.360 | 1.063 | **0.490** |
| | | | RF | 0.482 | 0.202 | 0.513 | 0.341 | 0.160 | 0.557 |
| | | | 2S | 0.448 | 0.062 | 0.619 | 0.317 | 0.082 | 0.689 |
| | | Qwen3-VL | AF | 0.496 | 0.057 | 0.473 | **0.367** | 0.400 | 0.543 |
| | | | RF | 0.504 | 0.005 | 0.414 | 0.313 | 0.214 | 0.591 |
| | | | 2S | 0.510 | 0.026 | 0.476 | 0.309 | 0.253 | 0.551 |
| | Shuffled images | LLaVA-1.5 | AF | 0.515 | 1.444 | 0.446 | 0.282 | 0.796 | 0.658 |
| | | | RF | 0.501 | 0.634 | 0.487 | 0.227 | 0.480 | 0.739 |
| | | | 2S | 0.493 | 0.420 | 0.497 | 0.278 | 0.368 | 0.711 |
| | | Qwen2.5-VL | AF | **0.538** | 0.314 | **0.445** | **0.402** | 0.577 | 0.550 |
| | | | RF | 0.504 | 0.129 | 0.472 | 0.372 | 0.190 | 0.587 |
| | | | 2S | 0.515 | 0.050 | 0.482 | 0.365 | 0.177 | 0.598 |
| | | Qwen3-VL | AF | 0.535 | 0.087 | 0.491 | 0.394 | 0.177 | 0.553 |
| | | | RF | 0.515 | 0.014 | 0.540 | 0.387 | 0.079 | **0.547** |
| | | | 2S | 0.496 | 0.017 | 0.516 | 0.391 | 0.071 | 0.593 |

| Experiment 2 | Image | Model | Order | Closed | | | Open | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Acc. | Ent. | AURC | Acc. | Ent. | AURC |
| Explanation SFT | Black images | LLaVA-1.5 | RF | 0.442 | 0.000 | 0.539 | 0.150 | 0.000 | 0.844 |
| | | | AF | **0.597** | 0.001 | 0.370 | 0.688 | 0.001 | 0.206 |
| | | Qwen2.5-VL | RF | 0.549 | 0.001 | 0.386 | 0.584 | 0.022 | 0.291 |
| | | | AF | **0.597** | 0.298 | 0.409 | 0.686 | 0.190 | **0.148** |
| | | | RF+AF | 0.594 | 0.160 | **0.328** | **0.703** | 0.175 | **0.148** |
| | | Qwen3-VL | RF | 0.504 | 0.006 | 0.545 | 0.670 | 0.016 | 0.171 |
| | | | AF | 0.549 | 0.380 | 0.434 | 0.622 | 0.731 | 0.177 |
| | Shuffled images | LLaVA-1.5 | RF | 0.406 | 0.000 | 0.511 | 0.147 | 0.000 | 0.841 |
| | | | AF | **0.592** | 0.001 | **0.362** | 0.588 | 0.001 | 0.438 |
| | | Qwen2.5-VL | RF | 0.527 | 0.002 | 0.461 | 0.531 | 0.060 | 0.475 |
| | | | AF | 0.547 | 0.086 | 0.440 | 0.599 | 0.084 | 0.385 |
| | | | RF+AF | 0.563 | 0.064 | 0.467 | **0.603** | 0.103 | 0.439 |
| | | Qwen3-VL | RF | 0.544 | 0.008 | 0.424 | 0.594 | 0.030 | **0.374** |
| | | | AF | 0.583 | 0.204 | 0.403 | 0.582 | 0.409 | 0.414 |

Table A.13: SLAKE stacked closed and open ablations, companion to the overall stack of Table 5.6. Blocks from top to bottom: Experiment 1 (answer-only) and Experiment 2 (zeroshot explanation and explanation SFT). Solid rules separate protocols. A dashed rule splits black-image and shuffled-image rows inside each protocol. Columns are Acc., letter entropy, and AURC. RF+AF is the mixed-order additional test of section 5.4. Bold Acc. and AURC are the within-protocol, within-image-condition extrema.

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